GENERAL

PLATE	DRAWING NO.	CONTENT	DETAILS
1	1695-D-62526	DRAWING INDEX	
2	1695-529-60481	GENERAL MAP	
3	1695-529-60488	VICINITY MAP	SANOSTEE TURNOUT BUILDING
4	1695-529-60489	VICINITY MAP	NEWCOMB TURNOUT BUILDING
5	1695-529-60490	VICINITY MAP	NASCHITTI TURNOUT BUILDING
6	1695-529-60491	VICINITY MAP	BBN9 TURNOUT BUILDING
7	1695-529-60492	VICINITY MAP	COYOTE CANYON TURNOUT BUILDING
8	1695-529-60493	VICINITY MAP	ROCK SPRING TURNOUT BUILDING
9	1695-529-60494	VICINITY MAP	WINDOW ROCK TURNOUT BUILDING
10	1695-D-62527	SPECIAL INSPECTIONS	

CIVIL

PLATE	DRAWING NO.	CONTENT	DETAIL
11	1695-D-62528	SANOSTEE TURNOUT	SITE PLAN
12	1695-D-62529	SANOSTEE TURNOUT	GRADING PLAN
13	1695-D-62530	NEWCOMB TURNOUT	EXISTING CONDITIONS AND DEMOLITION PLAN
14	1695-D-62531	NEWCOMB TURNOUT	SITE PLAN
15	1695-D-62532	NEWCOMB TURNOUT	GRADING PLAN
16	1695-D-62533	NASCHITTI TURNOUT	SITE PLAN
17	1695-D-62534	NASCHITTI TURNOUT	GRADING PLAN
18	1695-D-62535	BBN9 TURNOUT	SITE PLAN
19	1695-D-62536	BBN9 TURNOUT	GRADING PLAN
20	1695-D-62537	COYOTE CANYON TURNOUT	SITE PLAN
21	1695-D-62538	COYOTE CANYON TURNOUT	GRADING PLAN
22	1695-D-62539	ROCK SPRINGS TURNOUT	SITE PLAN
23	1695-D-62540	ROCK SPRINGS TURNOUT	GRADING PLAN
24	1695-D-62541	WINDOW ROCK TURNOUT	SITE PLAN
25	1695-D-62542	WINDOW ROCK TURNOUT	GRADING PLAN
26	1695-D-62543	COYOTE CANYON ACCESS ROAD	PLAN & PROFILE
27	1695-D-62544	WINDOW ROCK ACCESS ROAD	PLAN & PROFILE
28	1695-D-62545	WINDOW ROCK ACCESS ROAD	PLAN & PROFILE
29	1695-D-62546	BBN9 TURNOUT	SITE PLAN - LOW WATER CROSSING
30	1695-D-62547	BBN9 TURNOUT	LOW WATER CROSSING - DETAIL AND SECTIONS

PIPELINE

PLATE	DRAWING NO.	CONTENT	DETAIL
31	1695-D-62548	BBN9 LINE PIPE TURNOUT	PLANS, SECTIONS AND DETAILS
32	1695-D-62549	BBN9 LINE PIPE - SECTIONALIZING VALVE INSTALLATION	PLANS, SECTIONS AND DETAILS
33	1695-D-62550	BBN9 LINE PIPE - COLLARS	DETAILS AND SECTIONS
34	1695-D-62551	BBN9 LINE PIPE - AIR VALVES WITH MANHOLES	SECTIONS AND DETAILS
35	1695-D-62552	BBN9 LINE PIPE - MANHOLE WITH BLOWOFF	SECTIONS AND DETAILS
36	1695-D-62553	BBN9 LINE PIPE - PRECAST CONCRETE MANHOLES	PLANS, SECTIONS AND DETAILS
37	1695-D-62554	PRESSURE PIPE TRENCH INSTALLATION	DETAILS AND SECTIONS

STRUCTURAL

PLATE	DRAWING NO.	CONTENT	DETAIL
38	1695-D-62555	STRUCTURAL DESIGN DATA & GENERAL ARRANGEMENT	PLAN AND SECTIONS
39	1695-D-62556	CONCRETE OUTLINE & REINFORCEMENT DESIGN	THRUST BLOCKS AND MISC. SITE CONCRETE

ARCHITECTURAL

PLATE	DRAWING NO.	CONTENT	DETAIL
40	1695-D-62557	TURNOUT BUILDING ARCHITECTURAL	CODE SUMMARY, PLANS, ELEVATIONS, SECTIONS

MECHANICAL - STEEL PIPE

PLATE	DRAWING NO.	CONTENT	DETAIL
41	1695-D-62558	COVER SHEET	DETAILS
42	1695-D-62559	6 INCH AND 8 INCH DIAMETER TURNOUT PIPING	PLAN AND SECTION
43	1695-D-62560	12 INCH DIAMETER TURNOUT PIPING	PLAN AND SECTION
44	1695-D-62561	SANOSTEE YARD PIPE	PLAN AND SECTION
45	1695-D-62562	NEWCOMB YARD PIPE	PLAN AND SECTION
46	1695-D-62563	NASCHITTI YARD PIPE	PLAN AND SECTION
47	1695-D-62564	BBN9 YARD PIPE	PLAN AND SECTION
48	1695-D-62565	COYOTE CANYON YARD PIPE	PLAN AND SECTION
49	1695-D-62566	ROCK SPRINGS YARD PIPE	PLAN AND SECTION
50	1695-D-62567	WINDOW ROCK YARD PIPE	PLAN AND SECTION
51	1695-D-62568	DETAILS	DETAILS

PIPING AND INSTRUMENTATION DIAGRAM

PLATE	DRAWING NO.	CONTENT	DETAIL
52	1695-D-62569	P&ID	LEGEND
53	1695-D-62570	P&ID	SYMBOLS AND ABBREVIATIONS
54	1695-D-62571	P&ID	SANOSTEE TURNOUT BUILDING
55	1695-D-62572	P&ID	NEWCOMB TURNOUT BUILDING
56	1695-D-62573	P&ID	NASCHITTI TURNOUT BUILDING
57	1695-D-62574	P&ID	BBN9 TURNOUT BUILDING
58	1695-D-62575	P&ID	COYOTE CANYON TURNOUT BUILDING
59	1695-D-62576	P&ID	ROCK SPRINGS TURNOUT BUILDING
60	1695-D-62577	P&ID	WINDOW ROCK TURNOUT BUILDING

ELECTRICAL

PLATE	DRAWING NO.	CONTENT	DETAIL
61	1695-D-62578	TYPICAL	SINGLE-LINE DIAGRAM
62	1695-D-62579	SANOSTEE TURNOUT	ELECTRICAL SITE PLAN
63	1695-D-62580	NEWCOMB	ELECTRICAL SITE PLAN
64	1695-D-62581	NASCHITTI	ELECTRICAL SITE PLAN
65	1695-D-62582	BBN9	ELECTRICAL SITE PLAN
66	1695-D-62583	COYOTE CANYON	ELECTRICAL SITE PLAN
67	1695-D-62584	ROCK SPRINGS	ELECTRICAL SITE PLAN
68	1695-D-62585	WINDOW ROCK	ELECTRICAL SITE PLAN
69	1695-D-62586	TYPICAL	GROUNDING PLAN
70	1695-D-62587	CABLE AND CONDUIT	SCHEDULE
71	1695-D-62588	NETWORK OVERVIEW	TURNOUT BUILDING
72	1695-D-62589	NOT INCLUDED IN PACKAGE	

CATHODIC PROTECTION

PLATE	DRAWING NO.	CONTENT	DETAIL
73	1695-D-62590	GALVANIC ANODE CATHODIC PROTECTION SYSTEM	JOINT BOND, CABLE TRENCH, IJFK, AND CABLE I.D. DETAILS
74	1695-D-62591	GALVANIC ANODE CATHODIC PROTECTION SYSTEM	TEST STATION AND ANODE BED DETAILS

INFORMATIONAL DRAWINGS

PLATE	DRAWING NO.	CONTENT	DETAIL
75	1695-D-60119	REACH 12.1 PLAN AND PROFILE	STA. 121300+00 TO STA. 121350+00
76	1695-D-60129	REACH 12.1 OUTLETS	PLAN, SECTIONS AND DETAILS
77	1695-D-60150	REACHES 12.2 & 12.2 2 OR 3-INCH AIR VALVE	PLAN, SECTION, AND DETAILS
78	1695-D-60157	REACHES 12.2 & 12.2 SECTIONIZING VALVE INSTALLATION	PLAN, SECTIONS, AND TABLE
79	1695-D-60253	REACHES 12.1 & 12.2 SECTIONALIZING VALVE INSTALLATIONS LAYOUT	PLAN, SECTION, AND TABLES
80	1695-D-60260	REACH 12.2 PLAN AND PROFILE	STA. 122280+00 TO END
81	1695-D-60297	REACH 9 PLAN AND PROFILE	STA. 90750+00 TO STA. 90800+00
82	1695-D-60309	REACH 11 PLAN AND PROFILE	BEGINNING TO STA. 11100+00
83	1695-D-60322	BLOCK 9-11 MANHOLE WITH BLOWOFF	SECTIONS, DETAILS, AND TABLES
84	1695-D-60324	BLOCK 9-11 SECTIONALIZING VALVE INSTALLATION	PLAN, SECTION, AND DETAILS
85	1695-D-60325	BLOCK 9-11 SECTIONALIZING VALVE INSTALLATION	TAPERS, ADAPTERS, AND COLLARS
86	1695-D-60326	BLOCK 9-11 SECTIONALIZING VALVE INSTALLATION	TYPICAL LAYOUT
87	1695-D-60327	BLOCK 9-11 PRECAST CONCRETE MANHOLES	DETAILS
88	1695-D-60684	REACH 4C PLAN AND PROFILE	STA. 42150+00 TO END
89	1695-D-60695	REACH 6 PLAN AND PROFILE	STA. 6500+00 TO END
90	1695-D-60713	REACH 8 PLAN AND PROFILE	STA. 8400+00 TO END
91	1695-D-60720	BLOCK 4C - 8 AIR VALVES	DATA TABLES
92	1695-D-60722	BLOCK 4C - 8 AIR VALVES WITH MANHOLES	SECTIONS AND DETAILS
93	1695-D-60724	BLOCK 4C - 8 MANHOLE WITH BLOWOFF	SECTIONS, DETAILS, AND TABLES
94	1695-D-60727	BLOCK 4C - 8 TAPERS, ADAPTERS AND COLLARS	DETAILS, SECTIONS, AND DATA TABLES
95	1695-D-60728	BLOCK 4C - 8 TYPICAL TURNOUT	PLAN, SECTIONS AND DETAILS
96	1695-D-60731	BLOCK 4C - 8 PRECAST CONCRETE MANHOLES	PLAN, SECTIONS AND DETAILS
97	085-N-1110	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	SANOSTEE TURNOUT NIP LAYOUT
98	085-N-1111	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	SANOSTEE TURNOUT NIP POWER DISTRIBUTION
99	085-N-1115	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	NEWCOMB TURNOUT NIP LAYOUT
100	085-N-1116	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	NEWCOMB TURNOUT NIP POWER DISTRIBUTION
101	085-N-1120	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	NASCHITTI TURNOUT NIP LAYOUT

INFORMATIONAL DRAWINGS

PLATE	DRAWING NO.	CONTENT	DETAIL
102	085-N-1121	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	NASCHITTI TURNOUT NIP POWER DISTRIBUTION
103	085-N-1125	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	BBN9 TURNOUT NIP LAYOUT
104	085-N-1126	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	BBN9 TURNOUT NIP POWER DISTRIBUTION
105	085-N-1130	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	COYOTE CANYON TURNOUT PP NIP LAYOUT
106	085-N-1131	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	COYOTE CANYON TURNOUT NIP POWER DISTRIBUTION
107	085-N-1135	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	ROCK SPRINGS TURNOUT NIP LAYOUT
108	085-N-1136	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	ROCK SPRINGS TURNOUT NIP POWER DISTRIBUTION
109	085-N-1150	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	WINDOW ROCK TURNOUT NIP LAYOUT
110	085-N-1151	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	WINDOW ROCK TURNOUT NIP POWER DISTRIBUTION
111	6928997 C-1	NACHITTI CHAPTER - SAN JUAN LATERAL CONNECTION	PLAN AND PROFILE STA 0+00 TO STA 28+00
112	6922268 C-1	NAVAJO GULLUP WATER SUPPLY PROJECT - BEACON BISTI N9 - REACH 10.1	PLAN AND PROFILE STA 0+35.5 TO STA 28+00
113	6928904 C-1	SANOSTEE CHAPTER - SAN JUAN LATERAL CONNECTION	PLAN AND PROFILE STA 1+03 TO STA 28+00
114	6922268 FIG-1	NAVAJO GULLUP WATER SUPPLY PROJECT - BEACON BISTI N9 - REACH 10.1.1	TURNOUT AS BUILT
114A	1695-D-60281	REACH 9 PLAN AND PROFILE	BEGINNING TO STA. 90005+00
114B	1695-D-60282	REACH 9 PLAN AND PROFILE	STA. 90005+00 TO STA. 90050+00
114C	1695-D-60283	REACH 9 PLAN AND PROFILE	STA. 90050+00 TO STA. 90099+00
114D	1695-D-60296	REACH 9 PLAN AND PROFILE	STA. 90700+00 TO STA. 90750+00
114E	1695-D-60298	REACH 9 PLAN AND PROFILE	STA. 90800+00 TO STA. 90850+00
114F	1695-D-60310	REACH 11 PLAN AND PROFILE	STA. 11100+00 TO STA. 11150+00
114G	1695-D-60311	REACH 11 PLAN AND PROFILE	STA. 11150+00 TO STA. 11200+00
114H	1695-D-60685	REACH 6 PLAN AND PROFILE	BEGINNING TO STA. 6050+00
114I	1695-D-60686	REACH 6 PLAN AND PROFILE	STA. 6050+00 TO STA. 6100+00
114J	1695-D-60696	REACH 7 PLAN AND PROFILE	BEGINNING TO STA. 7050+00
114K	1695-D-60697	REACH 7 PLAN AND PROFILE	STA. 7050+00 TO STA. 7100+00

STANDARD DRAWINGS

PLATE	DRAWING NO.	CONTENT	DETAIL
115	40-D-4334	Electrical Standards Electrical Standards - Typical Grounding Details	Typical Grounding Details
116	40-D-4335	Electrical Standards	Typical Grounding Details
117	40-D-4753	Electrical Installation - Typical Grounding Details - For Switchyards and Substations	
118	40-D-5246	Buildings and Other Structures	Method of Building Forms for Successive Lifts, Detail
119	40-D-5247	Buildings and Other Structures	Joints in Concrete Structures
120	40-D-6234	Standard Nameplates	
121	40-D-6551	Pressure Pipe Trench Installation	Select Material
122	40-D-6552	Pressure Pipe Trench Installation	Soil-Cement Slurry - 12-Inch and Larger Pipe
123	40-D-7102	Government Drawing Format	Drawing Border, Sample Title Block, and Signature Lines
124	40-D-60001	Electrical Equipment Support	Sections and Detail
125	40-D-60002	Typical Guard Posts	Non-Security Application
126	40-D-60003	General Notes for Concrete Outline and Reinforcement	
127	40-D-60004	Minimum Requirements for Detailing Reinforcement	
128	104-D-254	Box Termination	
129	104-D-757	Electrical Installations	Designations and Symbols
130	104-D-1150	Electrical Installation	Device Designations and Symbols to be Used on Single-Line and Schematic Diagrams
131	104-D-1165	Electrical Installation	Main Control Board CSA - Panel 7R - Wiring Diagram

NOTES

1. For the current drawing revision number, refer to Section 52 00 00 - Drawings.

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

As-built drawings

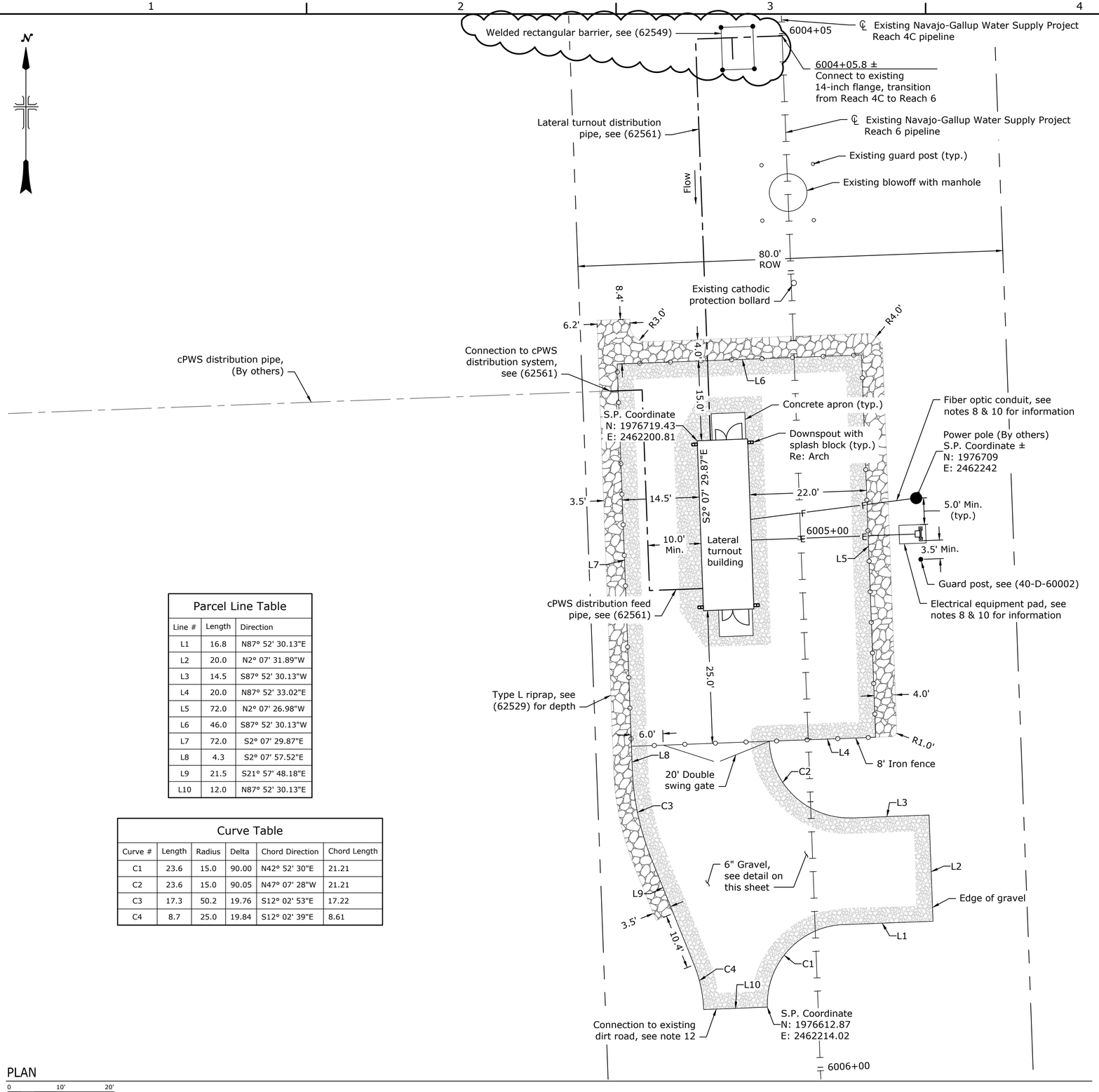
REV NO	STA.	DATE
1	D	2026-06-05

Asya Mcdaniel
DESIGNED
Asya Mcdaniel
DRAWN
Sara Putnam
CHECKED
Sara Putnam, P.E.
TECH. APPR. Sara Putnam, P.E.
Lan Nguyen, P.E.
APPROVED Lan Nguyen, P.E.
Peer Reviewer - Plant Structures Group

DENVER, CO 2026-06-19

DRAWING INDEX

1695-D-62526

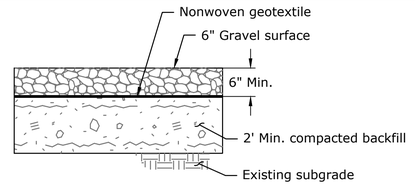


PLAN

0 10' 20'

Parcel Line Table		
Line #	Length	Direction
L1	16.8	N87° 52' 30.13"E
L2	20.0	N2° 07' 31.89"W
L3	14.5	S87° 52' 30.13"W
L4	20.0	N87° 52' 33.02"E
L5	72.0	N2° 07' 26.98"W
L6	46.0	S87° 52' 30.13"W
L7	72.0	S2° 07' 29.87"E
L8	4.3	S2° 07' 57.52"E
L9	21.5	S21° 57' 48.18"E
L10	12.0	N87° 52' 30.13"E

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	23.6	15.0	90.00	N42° 52' 30"E	21.21
C2	23.6	15.0	90.05	N47° 07' 28"W	21.21
C3	17.3	50.2	19.76	S12° 02' 53"E	17.22
C4	8.7	25.0	19.84	S12° 02' 39"E	8.61



DETAIL - YARD GRAVEL SURFACING SECTION

NOT TO SCALE

ABBREVIATIONS

S.P. - State Plane
℄ - Center line
ROW- Right-of-Way

LEGEND

- Permanent Right-of-Way
- ℄ Existing Navajo-Gallup Water Supply Project pipeline
- Lateral turnout distribution pipe/ cPWS distribution feed pipe
- cPWS distribution pipe (By others)
- E — E Electrical conduit
- F — F Fiber optic conduit
- Iron fence
- Concrete
- Riprap
- Gravel
- Compacted backfill
- Welded rectangular barrier
- Power pole (By others)
- Proposed air valve with manhole
- Proposed sectionalizing valve
- Proposed manhole with blowoff

GENERAL SITEWORK NOTES:

1. Reference Structural and Architectural drawings and specifications for concrete apron, downspout, and splash block information.
2. See 40-D-6551 and 40-D-6552 for pipeline trenching details.
3. Drawing numbers are 1695-D-XXXXX unless otherwise specified.
4. See specifications for sediment and erosion control information.
5. Reference Mechanical drawings and specifications for turnout yard piping information.
6. Clear and screen the turnout footprint and associated yard area for vegetation, large rocks, etc. before backfilling and grading. Protect in place existing infrastructure, see Specifications for additional information.
7. Power poles and overhead electric lines shall be installed by local utility provider.
8. Fiber and electrical conduits shown are for graphical purposes only and do not represent all conduits and vaults expected to be on site. See (62578-62588) for additional information. Coordinate with COR for final placement of electrical disconnect pad, see (62556) for pad information.
9. See specifications for fence and gate information.
10. See specifications for fiber and electrical conduit depths and minimum clearances at utility crossings.
11. See specifications for signage requirements and installation.
12. The location and elevation of the connection points to the existing dirt road are approximate and have not been field verified. Contractor to coordinate with the COR as needed for field adjustments for the road connection.
13. See (62558-62568) for yard pipe designs, (62555) for turnout building, and (62556) for thrust block information.
14. Existing infrastructure shall be protected in place in accordance with the specifications unless otherwise noted.
15. See (62556) for concrete apron details.
16. See specifications for cultural monitoring area requirements.

NOTES:

1. All control points are shown in New Mexico State Plan Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00026509926 (Reach 4) was used.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV.NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification and updated general sitework notes Marner, Andrew (Andy) J P.E.

B. ZOOK DESIGNED
L. VOGT DRAWN
C. HARMON CHECKED
Marner, Andrew (Andy) J P.E. TECH. APPR.
Steen, Timothy M APPROVED PEER REVIEWER
DENVER, CO 2026-06-01

SANOSTEE TURNOUT

SITE PLAN

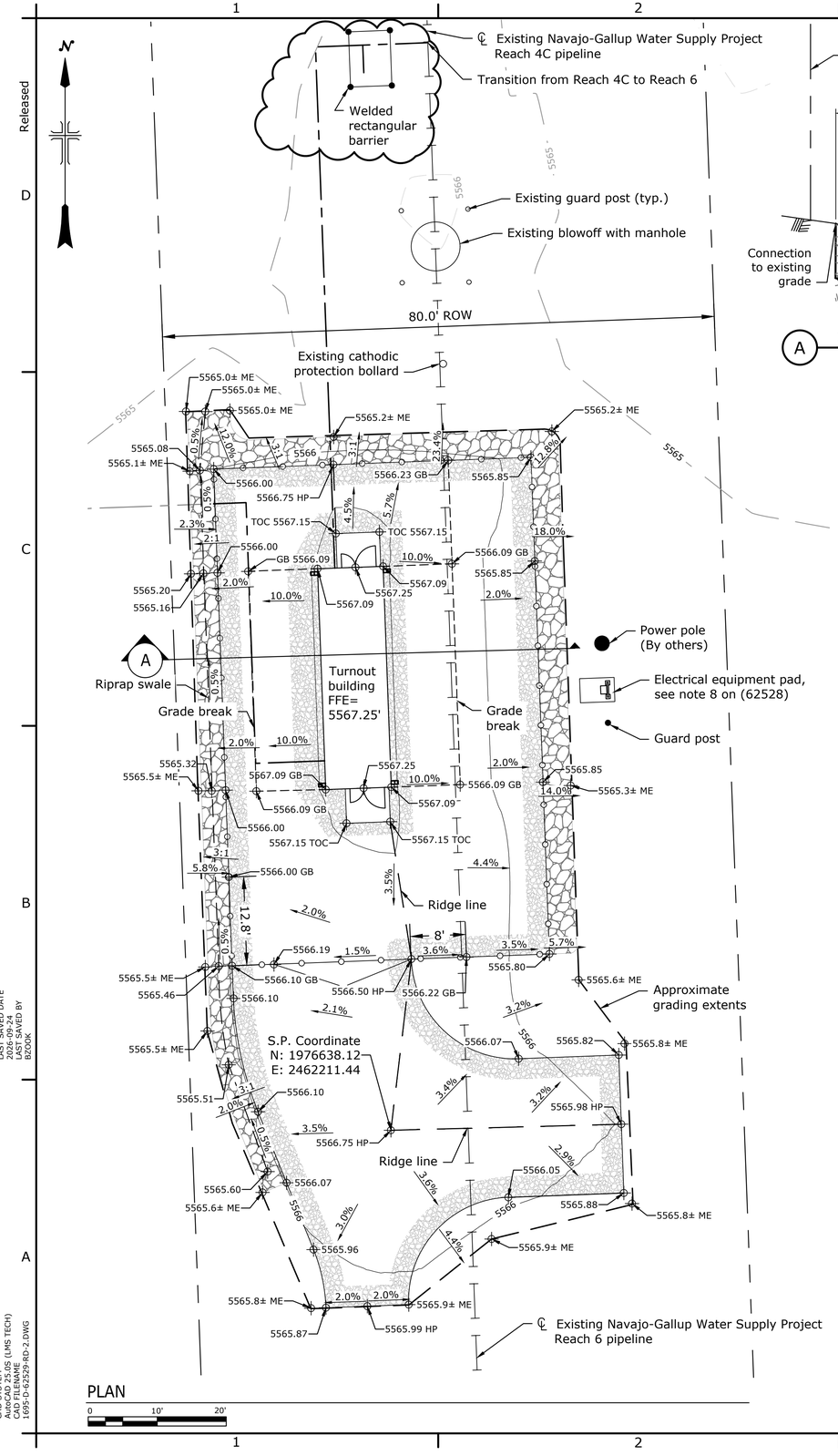
1695-D-62528

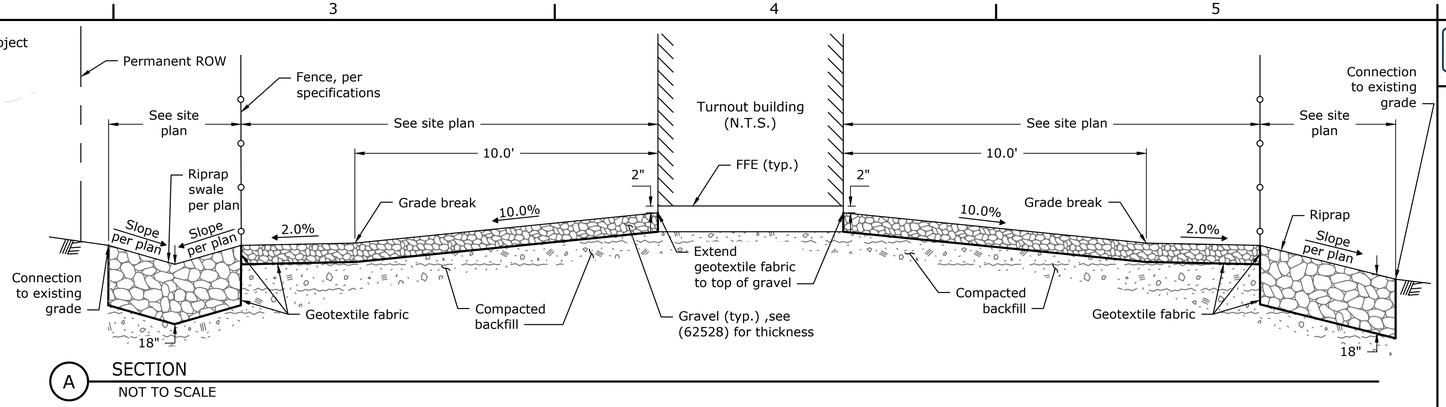
SHEET 1

Released

LAST SAVED DATE
2026-09-29
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD 2025 (LMS TECH)
CAD FILENAME
1695-D-62528-RD-1.DWG





A SECTION
NOT TO SCALE

ABBREVIATIONS

ME - Match existing (±)
HP - High point
GB - Grade break
FFE - Finished floor elevation
TOC - Top of concrete

GENERAL SITEWORK GRADING NOTES:

1. Contractor to verify existing grades and utilities prior to construction, notify COR of any conflicts or discrepancies.
2. Yard elevations are shown to top of gravel or concrete unless otherwise specified.
3. Provide a minimum slope of 10% away from the building in gravel areas, extending 10' minimum, or continuing to the edge of the fence where the gravel width is less than 10'.
4. Slope concrete stoops and aprons away from the building at 2.0% slope.
5. All vertical elevations are based on the NAVD88 vertical datum.
6. Contours shown are based off of survey data provided by the Four Corners Construction Office.
7. Contours are shown in 1' intervals.
8. Slope riprap as shown on drawings, maximum slope for riprap shall not exceed 1.5(H):1(V).
9. See (62528) for legend, abbreviations, and general sitework notes.

NOTES:

1. All control points are shown in New Mexico State Plan Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00026509926 (Reach 4) was used.

Released

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK — DESIGNED
L. VOGT — DRAWN
C. HARMON — CHECKED
Marner, Andrew (Andy) J P.E. — TECH. APPR.
Steen, Timothy M — APPROVED PEER REVIEWER
DENVER, CO — 2026-06-01

SANOSTEE TURNOUT

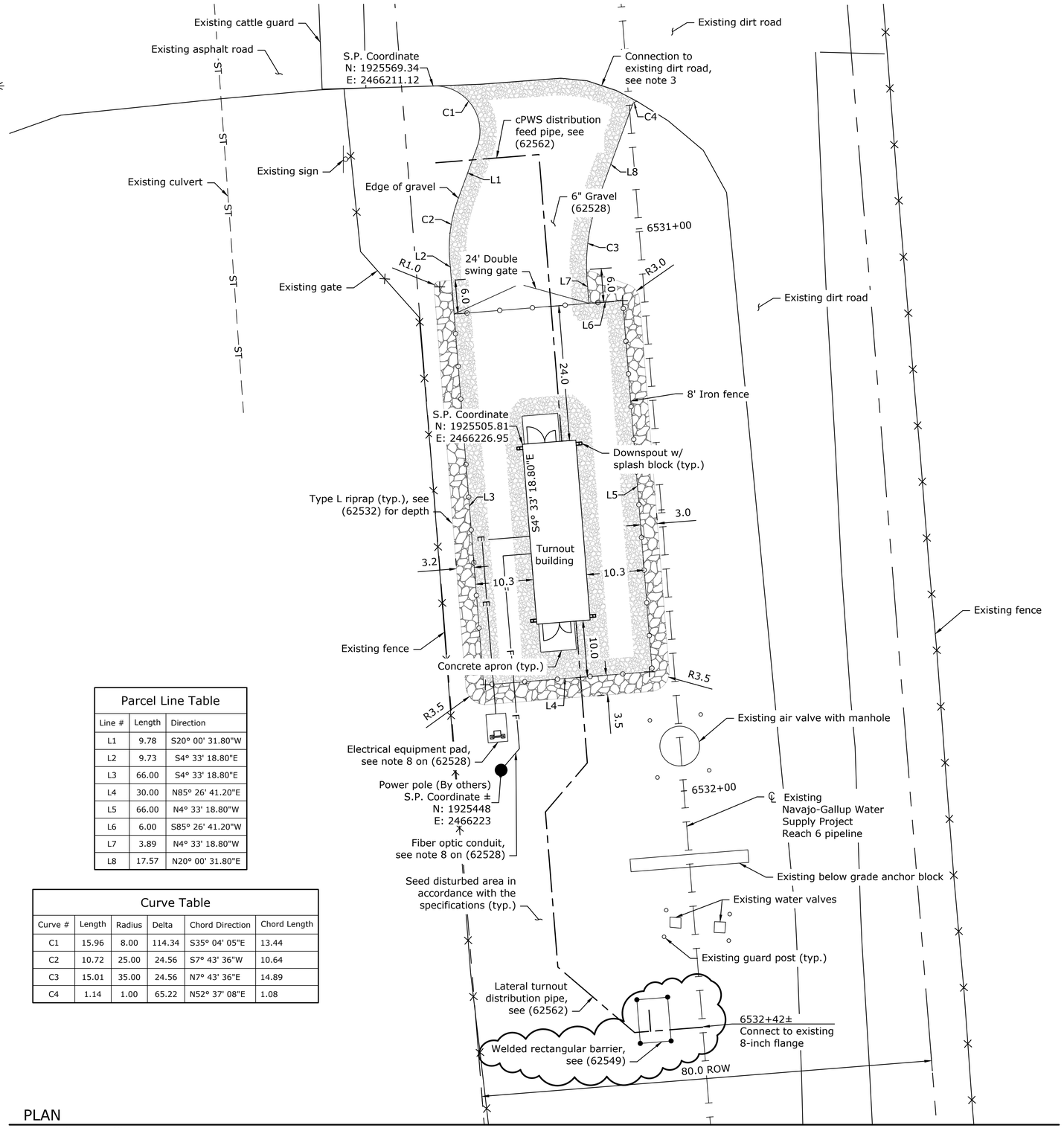
GRADING PLAN

1695-D-62529

SHEET 2

LAST SAVED DATE
2026-09-29
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62529-RD-2.DWG



PLAN

0 10' 20'

Parcel Line Table		
Line #	Length	Direction
L1	9.78	S20° 00' 31.80"W
L2	9.73	S4° 33' 18.80"E
L3	66.00	S4° 33' 18.80"E
L4	30.00	N85° 26' 41.20"E
L5	66.00	N4° 33' 18.80"W
L6	6.00	S85° 26' 41.20"W
L7	3.89	N4° 33' 18.80"W
L8	17.57	N20° 00' 31.80"E

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	15.96	8.00	114.34	S35° 04' 05"E	13.44
C2	10.72	25.00	24.56	S7° 43' 36"W	10.64
C3	15.01	35.00	24.56	N7° 43' 36"E	14.89
C4	1.14	1.00	65.22	N52° 37' 08"E	1.08

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027097141 (Reach 6) was used.
3. The location and elevation of the connection points to the existing dirt road are approximate and have not been field verified. Contractor to coordinate with the COR as needed for field adjustments for the road connection.

Released

LAST SAVED DATE
2026-09-25
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62531-RD-4.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO		
1	D	2026-09-29 Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED
PEER REVIEWER
DENVER, CO 2026-06-01

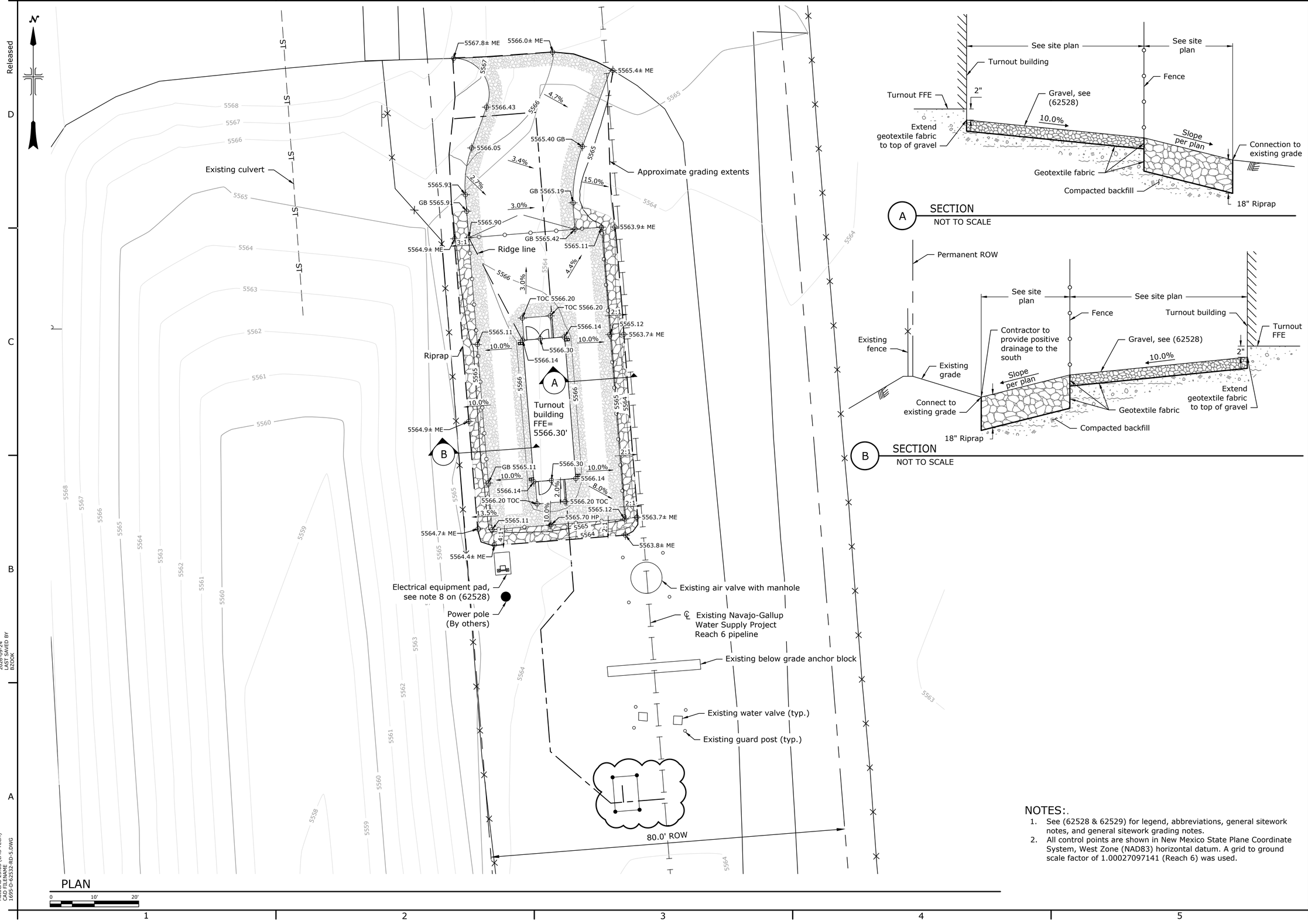
NEWCOMB TURNOUT

SITE PLAN

1695-D-62531

SHEET 4

Released



PLAN

0 10' 20'

A SECTION
NOT TO SCALE

B SECTION
NOT TO SCALE

NOTES:.

1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general sitework grading notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027097141 (Reach 6) was used.

BUREAU OF RECLAMATION

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED
PEER REVIEWER
DENVER, CO 2026-06-01

NEWCOMB TURNOUT

GRADING PLAN

1695-D-62532

SHEET 5

LAST SAVED DATE
2026-09-29
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD OpenDWG (LMS TECH)
CAD FILENAME
1695-D-62532-RD-5.DWG

Released

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	19.79	25.00	45.35	N39° 39' 51"W	19.28
C2	19.63	25.00	45.00	N39° 50' 24"W	19.13
C3	7.85	5.00	89.97	S62° 19' 25"E	7.07
C4	13.99	10.00	80.17	N32° 34' 23"E	12.88
C5	15.98	10.00	91.55	N61° 33' 56"W	14.33
C6	15.71	10.00	90.00	S27° 39' 36"W	14.14
C7	11.78	15.00	45.00	S39° 50' 24"E	11.48
C8	11.72	5.00	134.34	N50° 29' 20"E	9.22

Parcel Line Table		
Line #	Length	Direction
L1	67.53	N62° 20' 23.50"W
L2	84.78	N17° 20' 23.50"W
L3	72.00	N17° 20' 23.50"W
L4	48.00	N72° 39' 36.50"E
L5	72.00	S17° 20' 23.50"E
L6	24.00	S72° 39' 36.50"W
L7	59.79	S17° 20' 23.50"E
L8	25.22	N72° 41' 33.87"E
L9	34.57	S15° 52' 51.59"E
L10	16.19	S72° 39' 36.50"W
L11	2.49	S17° 20' 23.50"E
L12	32.30	S62° 20' 23.50"E
L13	45.11	S16° 51' 39.56"E

Existing fence

Existing cPWS distribution system

Type L riprap swale, see (62534) for depth

S.P. Coordinate
N: 1845579.33
E: 2471950.19

Seed disturbed areas in accordance with the specifications (typ.)

See (62563) for pipe connection information

cPWS distribution feed pipe, see (62563)

Lateral turnout distribution pipe, see (62563)

Flow

S.P. Coordinate
N: 1845491.32
E: 2471989.67

N17° 20' 23.50"W

Edge of gravel

6" Gravel, see (62528)

24' Double swing gate

Lateral turnout building

Existing Navajo-Gallup Water Supply Project Reach 8 pipeline

Welded rectangular barrier, see (62549)

Roof drain downspout w/ splashblock (typ.)

Concrete apron (typ.)

Existing cathodic protection bollard

Existing blowoff with manhole

Existing valve (typ.)

Existing blowoff with manhole

8' Iron fence

8414+88

8414+00

8413+00

8414+00.6±
Connect to existing 10-inch flange

Existing guard post (typ.)

Existing below grade anchor block

S.P. Coordinate
N: 1845322.48
E: 2472093.07

Approximate connection to existing road

Approximate connection to existing road

Existing dirt access road

Electrical equipment pad, see note 8 on (62528)

Guard post, see (40-D-60002)

Fiber optic cable, see note 8 on (62528)

Power pole (By others)

S.P. Coordinate ±
N: 1845523
E: 2472031

80.0' ROW

R3.0'

R10.0'

5.9'

3.5'

14.0'

34.4'

24.0'

16.0'

10.0'

4.0'

24.5'

10.0'

3.5' Min.

5.0' Min.

0 10' 20'

PLAN

NOTES:
1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00028975894 (Reach 8) was used.

LAST SAVED DATE
2026-09-29
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62533-RD-6.DWG

BUREAU OF RECLAMATION

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D | Welded rectangular barrier modification.
		Marner, Andrew (Andy) J P.E.

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED
PEER REVIEWER

DENVER, CO 2026-06-02

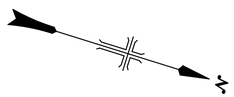
NASCHITTI TURNOUT

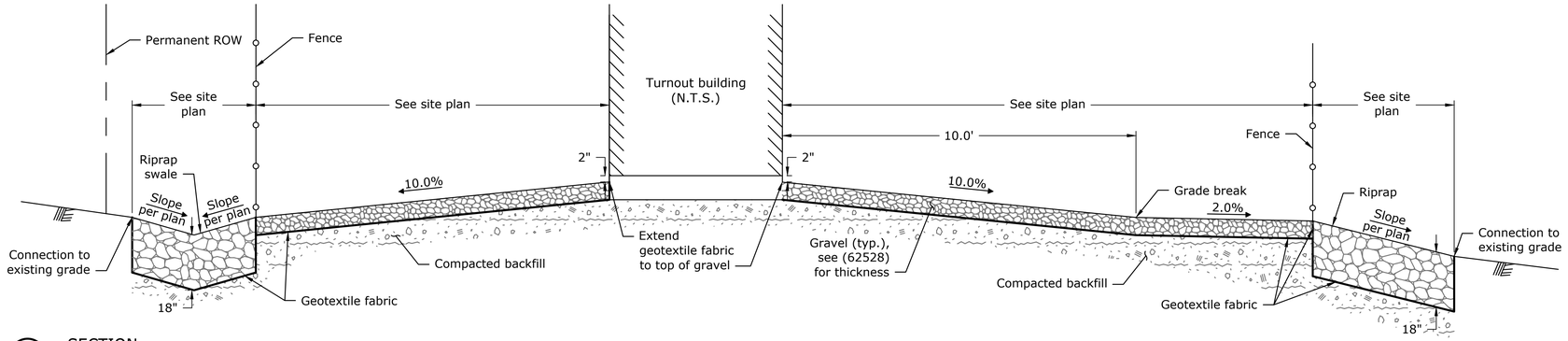
SITE PLAN

1695-D-62533

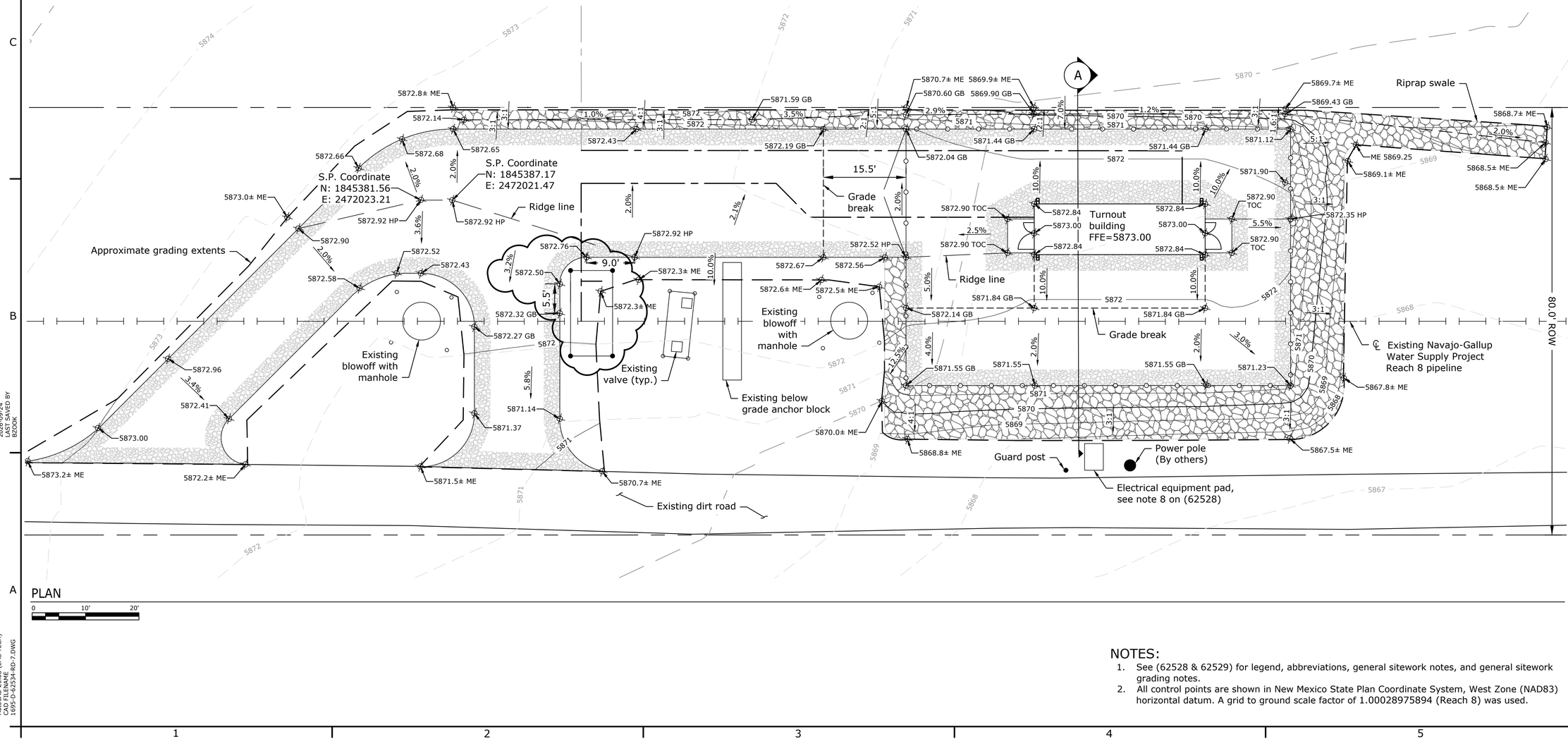
SHEET 6

Released





A SECTION
NOT TO SCALE



PLAN
0 10' 20'

NOTES:
1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general sitework grading notes.
2. All control points are shown in New Mexico State Plan Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00028975894 (Reach 8) was used.

LAST SAVED DATE 2026-05-29 LAST SAVED BY BZOOK

CAD SYSTEM AutoCAD Civil 3D (LMS TECH) CAD FILENAME 1695-D-62534-RD-7.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK DESIGNED
L. VOGT DRAWN
C. HARMON CHECKED
Marner, Andrew (Andy) J P.E. TECH. APPR.
Steen, Timothy M APPROVED ADMIN APPROVAL

DENVER, CO 2026-06-01

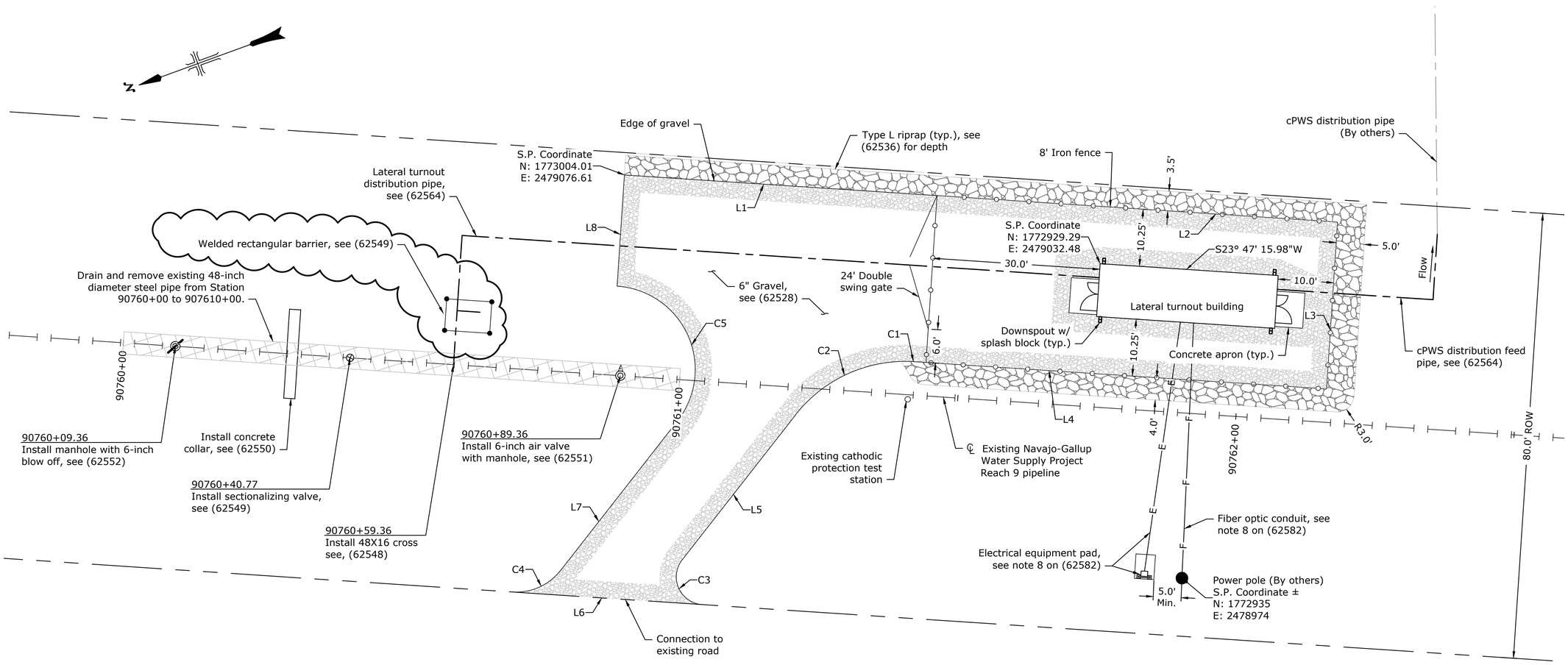
NASCHITTI TURNOUT

GRADING PLAN

1695-D-62534

SHEET 7

Released



PLAN

0 10' 20'

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	4.45	46.60	5.48	N22° 19' 10"E	4.45
C2	21.94	25.00	50.28	N6° 49' 31"W	21.24
C3	10.84	5.00	124.25	S85° 54' 48"W	8.84
C4	9.73	10.00	55.75	S4° 05' 18"E	9.35
C5	32.53	15.00	124.25	N85° 54' 45"E	26.52

Parcel Line Table		
Line #	Length	Direction
L1	56.19	S23° 47' 15.98"W
L2	72.00	S23° 47' 15.98"W
L3	30.00	N66° 12' 44.02"W
L4	72.00	N23° 47' 15.98"E
L5	32.97	N31° 57' 46.48"W
L6	34.10	N23° 47' 15.98"E
L7	34.10	S31° 57' 46.48"E
L8	20.00	S66° 12' 44.02"E

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.000308263000 (Reach 9) was used.

LAST SAVED DATE
2026-09-29
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD Plus 12 (LMS TECH)
CAD FILENAME
1695-D-62535-RD-8.DWG

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO		
1	2026-09-29	D Welded rectangular barrier modification.
	Marner, Andrew (Andy) J P.E.	

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED
PEER REVIEWER

DENVER, CO 2026-06-02

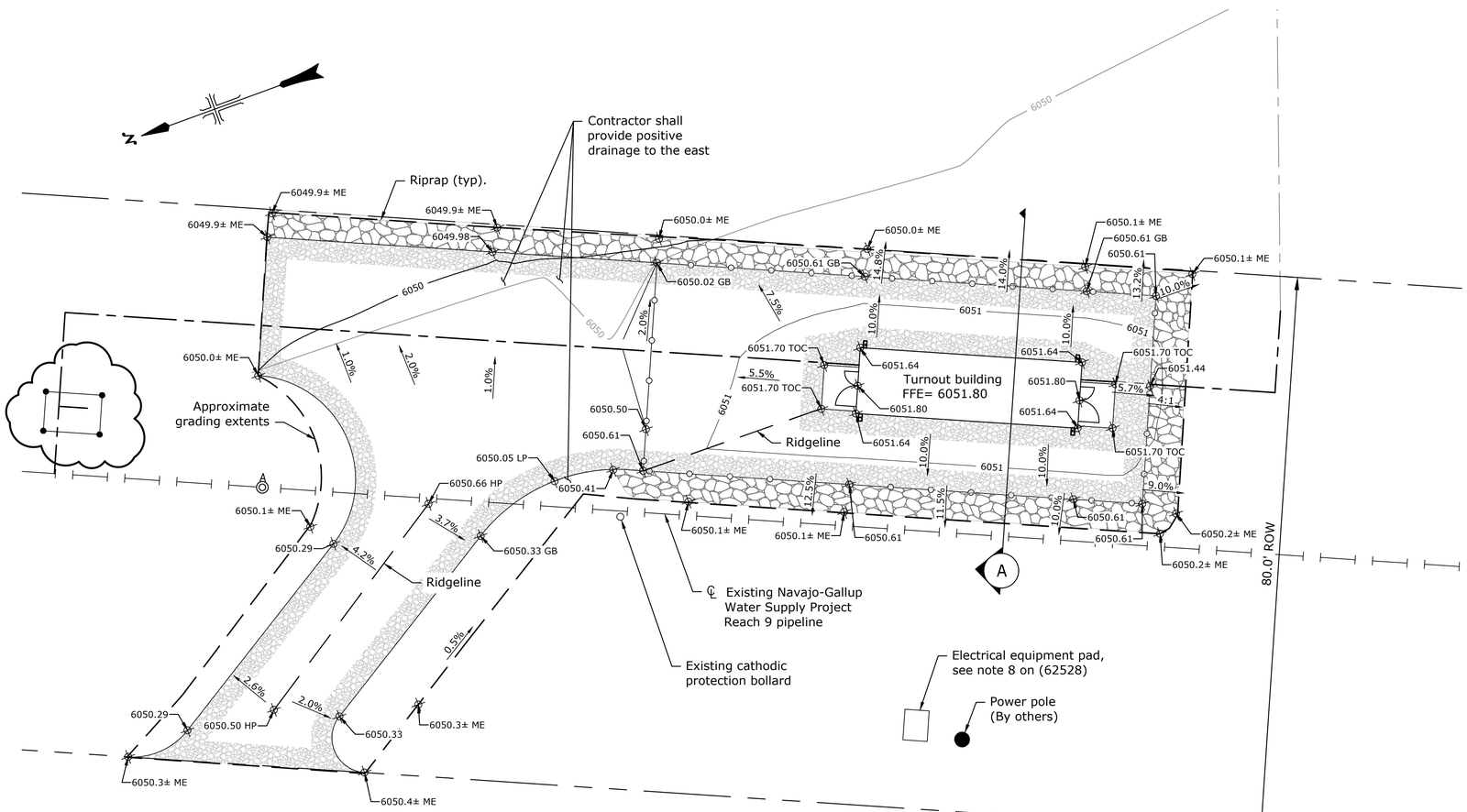
BBN9 TURNOUT

SITE PLAN

1695-D-62535

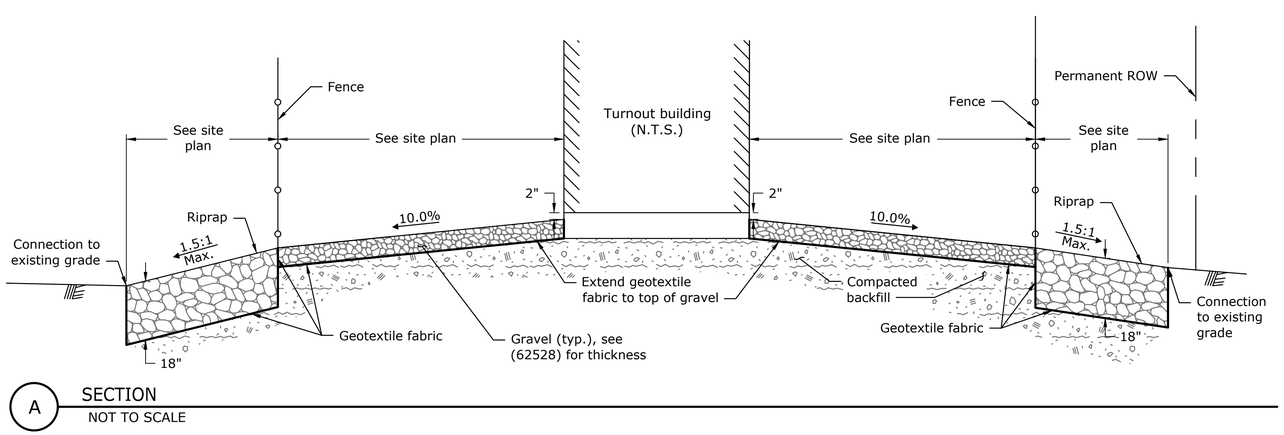
SHEET 8

Released



PLAN

0 10' 20'



A SECTION

NOT TO SCALE

NOTES:.

1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general sitework grading notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.000308263000 (Reach 9) was used.

BUREAU OF RECLAMATION

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED
PEER REVIEWER
DENVER, CO 2026-06-01

BBN9 TURNOUT

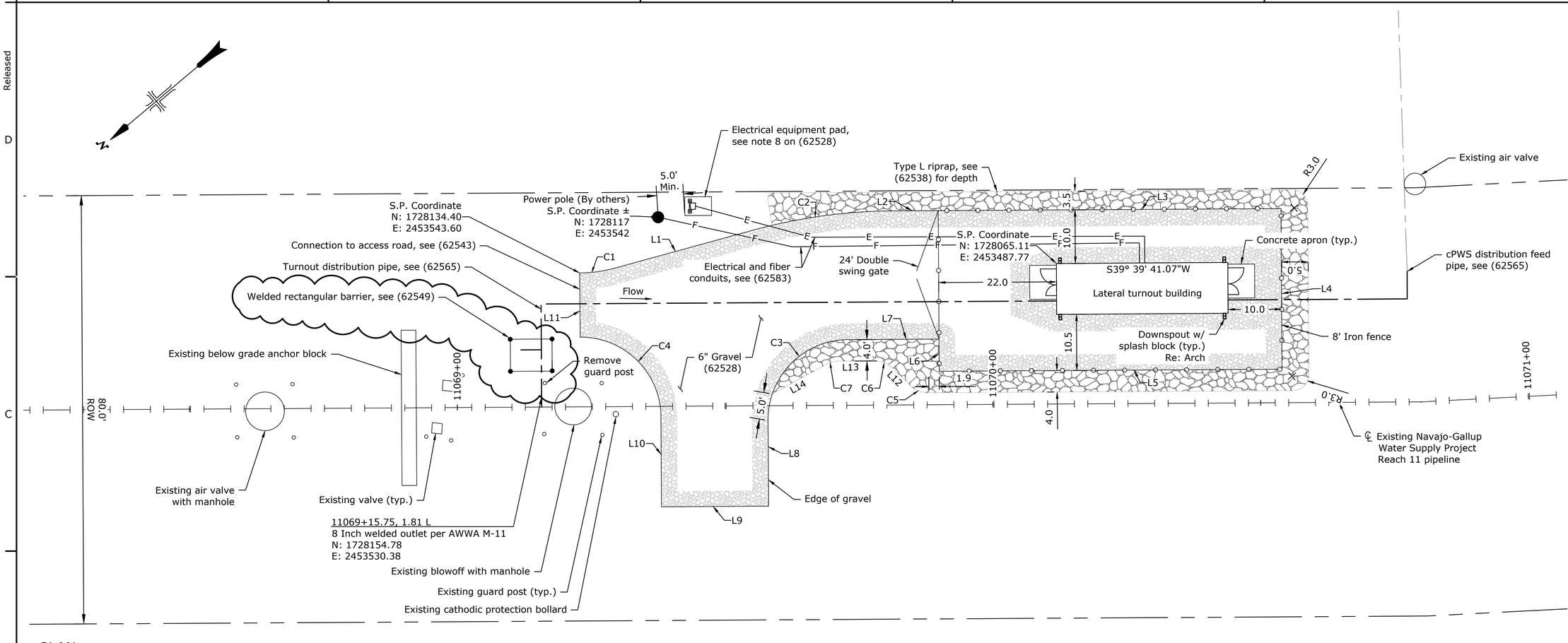
GRADING PLAN

1695-D-62536

SHEET 9

LAST SAVED DATE
2026-09-29 16:34
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62536-RD-9.DWG



PLAN

0 10' 20'

Parcel Line Table		
Line #	Length	Direction
L1	27.01	S24° 39' 41.23"W
L2	8.56	S39° 39' 41.07"W
L3	64.00	S39° 39' 41.07"W
L4	30.00	S50° 20' 18.93"E
L5	64.00	N39° 39' 41.07"E
L6	6.00	S50° 20' 18.93"E
L7	17.00	N39° 39' 41.07"E
L8	16.00	N50° 20' 18.77"W
L9	20.00	N39° 39' 41.23"E
L10	16.75	S50° 20' 18.77"E
L11	12.00	S50° 20' 18.77"E
L12	4.34	N84° 39' 41.07"E
L13	6.67	N39° 39' 41.07"E
L14	11.41	N11° 09' 08.19"E

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	6.54	25.00	15.00	S32° 09' 41"W	6.53
C2	26.18	100.00	15.00	S32° 09' 41"W	26.11
C3	23.57	14.97	90.21	N5° 20' 19"W	21.21
C4	23.56	15.00	90.00	N84° 39' 41"E	21.21
C5	3.93	5.00	45.00	N62° 09' 41"E	3.83
C6	3.93	5.00	45.00	N62° 09' 41"E	3.83
C7	2.49	5.00	28.51	N25° 24' 25"E	2.46

NOTES:
1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00029693715 (Reach 11) was used.

Released

LAST SAVED DATE
2026-09-15
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62537-RD-10.DWG

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO		
1	2026-09-29	D Welded rectangular barrier modification.
	Marner, Andrew (Andy) J P.E.	

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED
PEER REVIEWER
DENVER, CO 2026-06-01

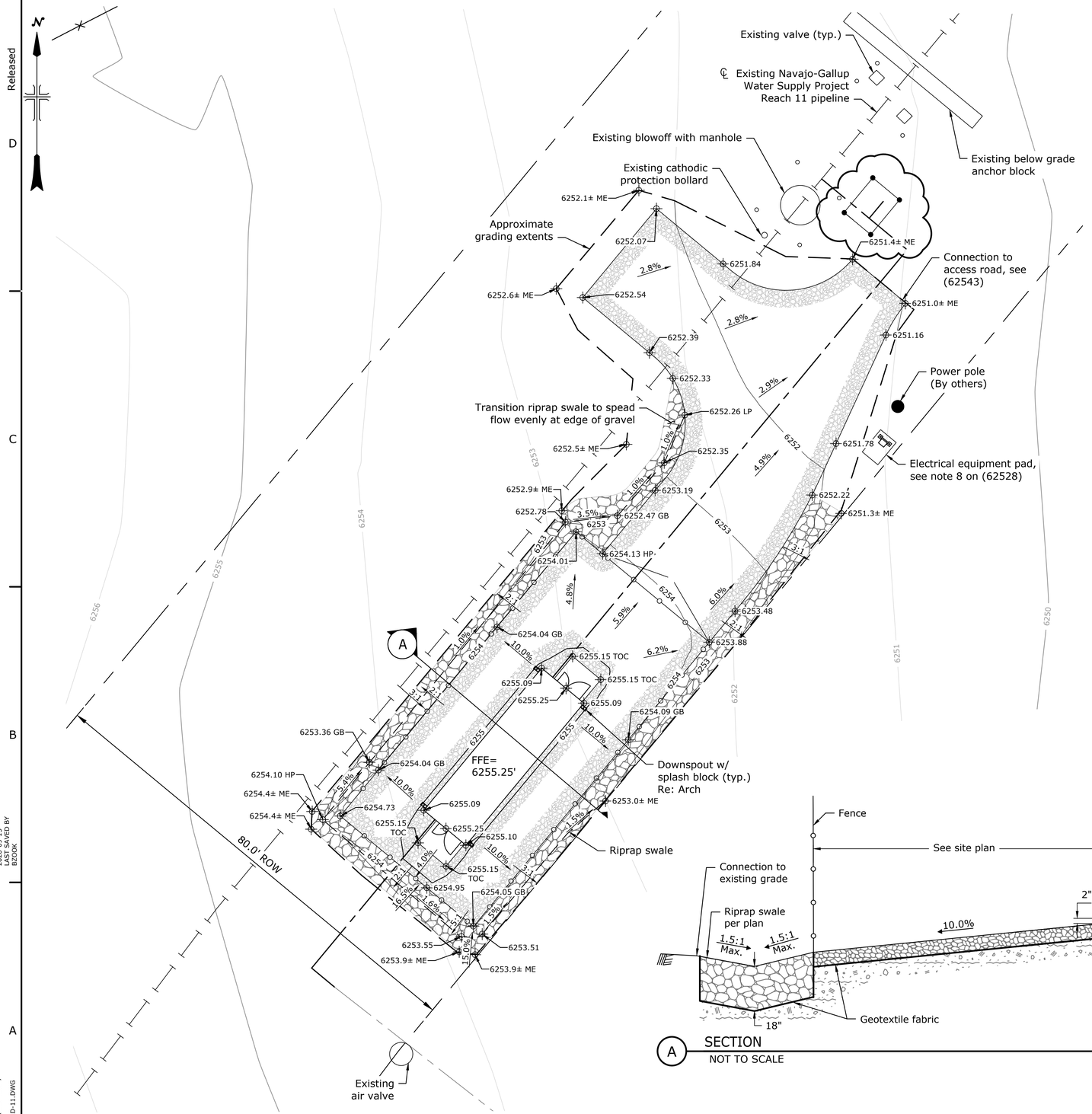
COYOTE CANYON TURNOUT

SITE PLAN

1695-D-62537

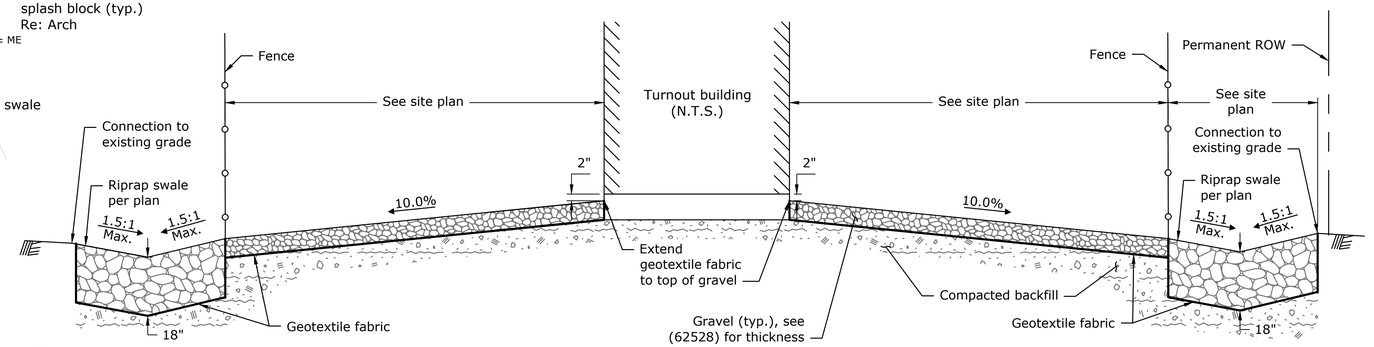
SHEET 10

Released



PLAN

0 10' 20'



A SECTION

NOT TO SCALE

NOTES:.

1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general grading notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00029693715 (Reach 11) was used.

— BUREAU OF — RECLAMATION

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK DESIGNED
L. VOGT DRAWN
C. HARMON CHECKED
Marner, Andrew (Andy) J P.E. TECH. APPR.
Steen, Timothy M APPROVED PEER REVIEWER
DENVER, CO 2026-06-01

COYOTE CANYON TURNOUT

GRADING PLAN

1695-D-62538

SHEET 11

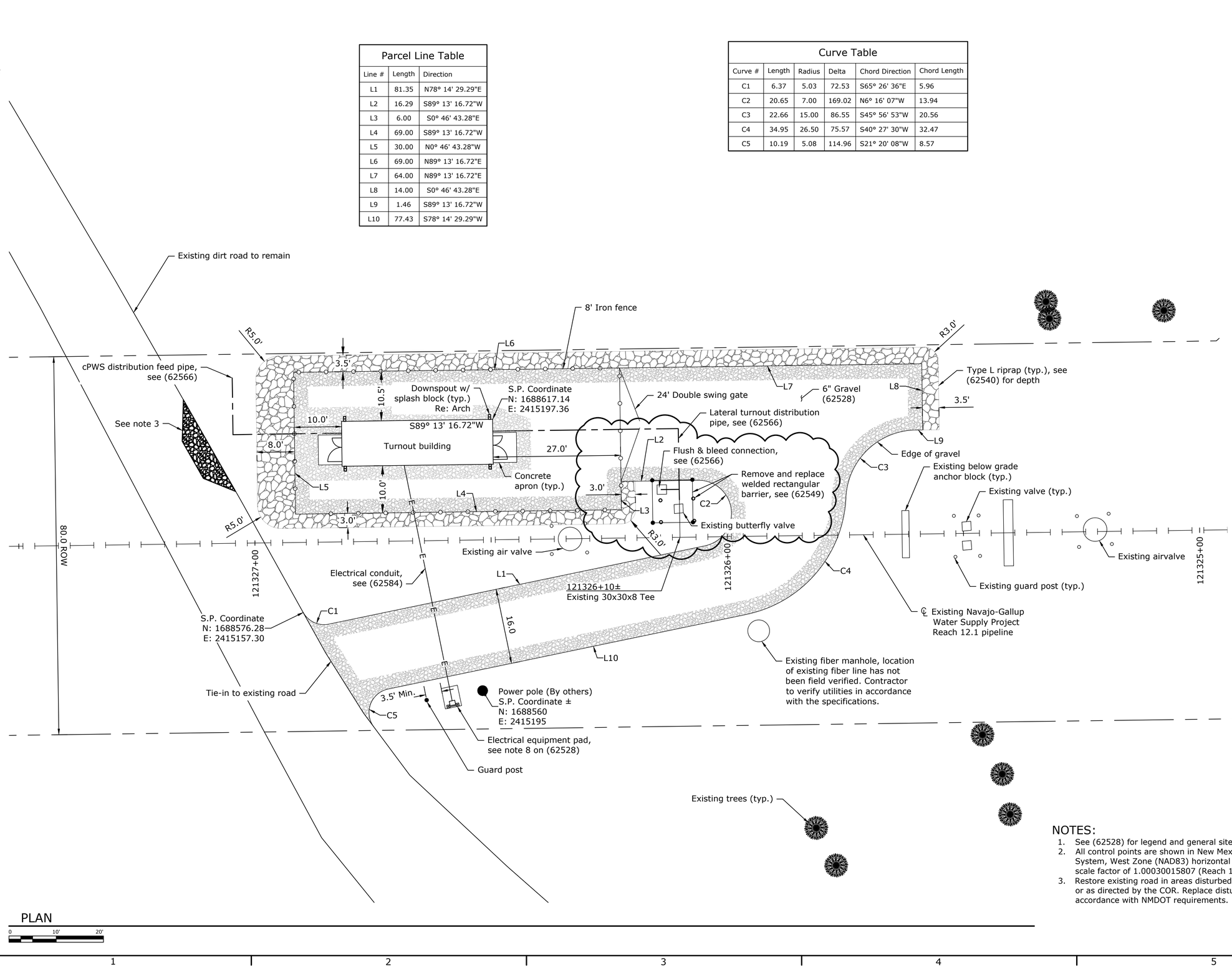
LAST SAVED DATE 2026-09-15
LAST SAVED BY BZOOK

CAD SYSTEM AUTOCAD CIVIL 3DS (LMS TECH)
CAD FILENAME 1695-D-62538-RD-11.DWG

Released

Parcel Line Table		
Line #	Length	Direction
L1	81.35	N78° 14' 29.29"E
L2	16.29	S89° 13' 16.72"W
L3	6.00	S0° 46' 43.28"E
L4	69.00	S89° 13' 16.72"W
L5	30.00	N0° 46' 43.28"W
L6	69.00	N89° 13' 16.72"E
L7	64.00	N89° 13' 16.72"E
L8	14.00	S0° 46' 43.28"E
L9	1.46	S89° 13' 16.72"W
L10	77.43	S78° 14' 29.29"W

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	6.37	5.03	72.53	S65° 26' 36"E	5.96
C2	20.65	7.00	169.02	N6° 16' 07"W	13.94
C3	22.66	15.00	86.55	S45° 56' 53"W	20.56
C4	34.95	26.50	75.57	S40° 27' 30"W	32.47
C5	10.19	5.08	114.96	S21° 20' 08"W	8.57



PLAN

0 10' 20'

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00030015807 (Reach 12.1) was used.
3. Restore existing road in areas disturbed by construction activities or as directed by the COR. Replace disturbed road areas in accordance with NMDOT requirements.

BUREAU OF RECLAMATION

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK DESIGNED
L. VOGT DRAWN
C. HARMON CHECKED
Marner, Andrew (Andy) J P.E. TECH. APPR.
Steen, Timothy M APPROVED PEER REVIEWER
DENVER, CO 2026-06-01

ROCK SPRINGS TURNOUT

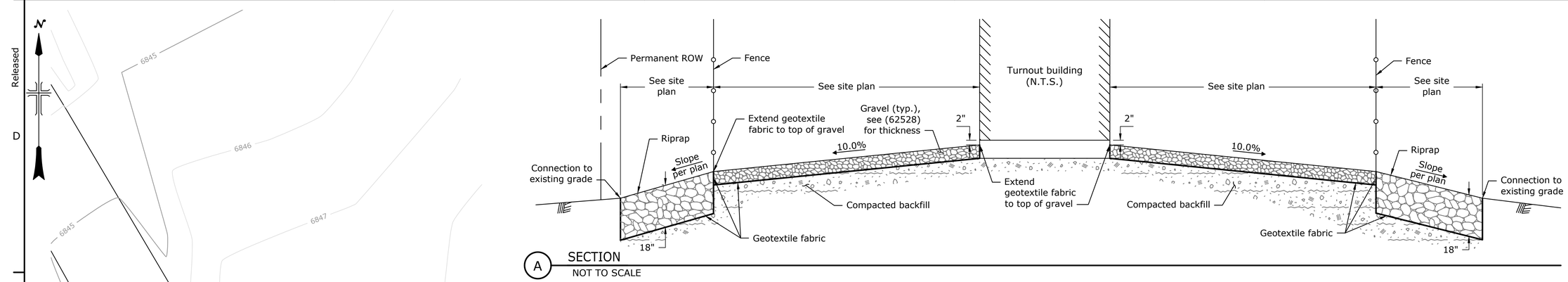
SITE PLAN

1695-D-62539

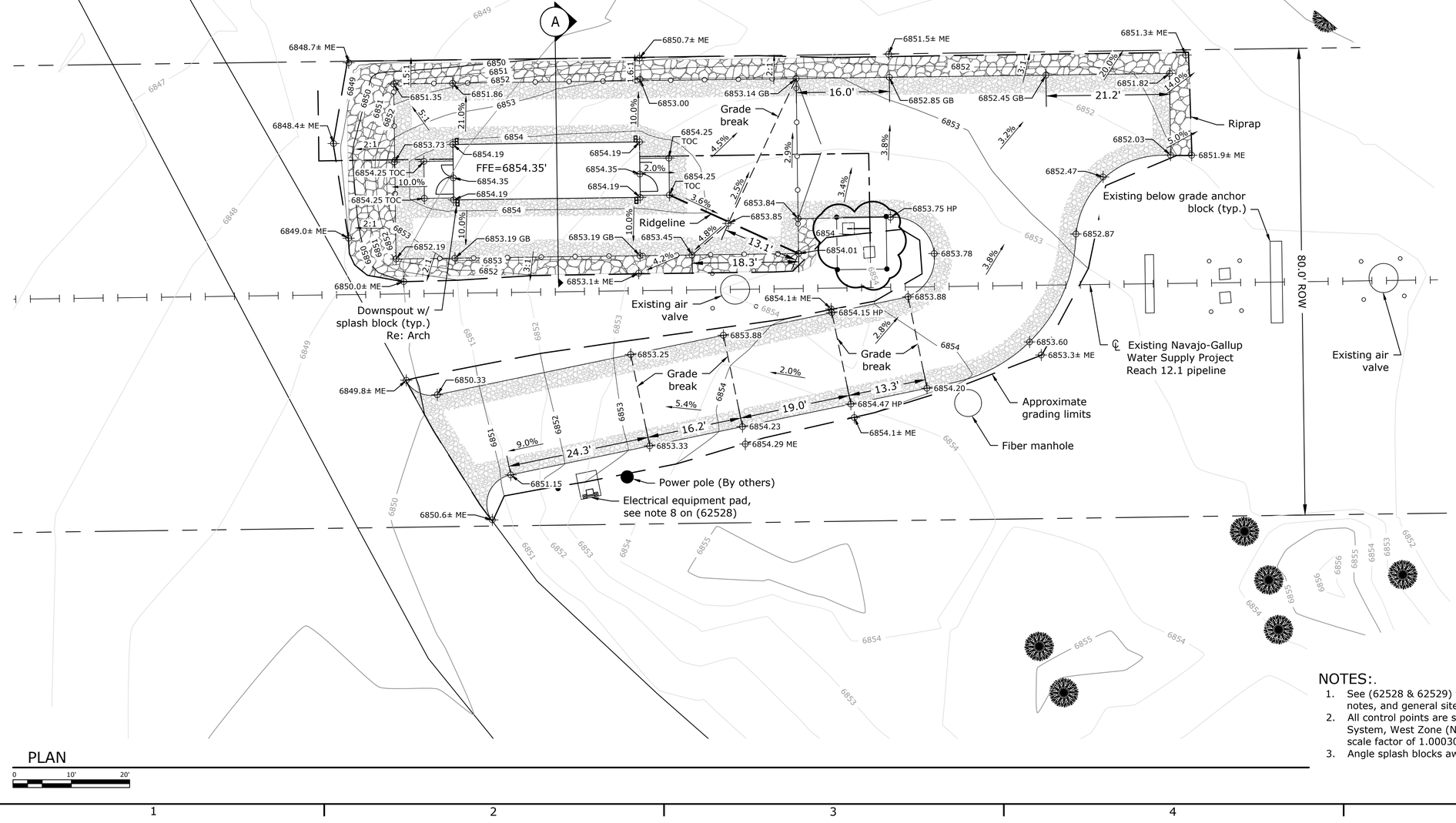
SHEET 12

LAST SAVED DATE
2026-09-29
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62539-RD-12.DWG



A SECTION
NOT TO SCALE



PLAN
0 10' 20'

NOTES:.

1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general sitework grading notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00030015807 (Reach 12.1) was used.
3. Angle splash blocks away from existing infrastructure.

Released

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK DESIGNED
L. VOGT DRAWN
C. HARMON CHECKED
Marner, Andrew (Andy) J P.E. TECH. APPR.
Steen, Timothy M APPROVED PEER REVIEWER
DENVER, CO 2026-06-01

ROCK SPRINGS TURNOUT

GRADING PLAN

1695-D-62540

SHEET 13

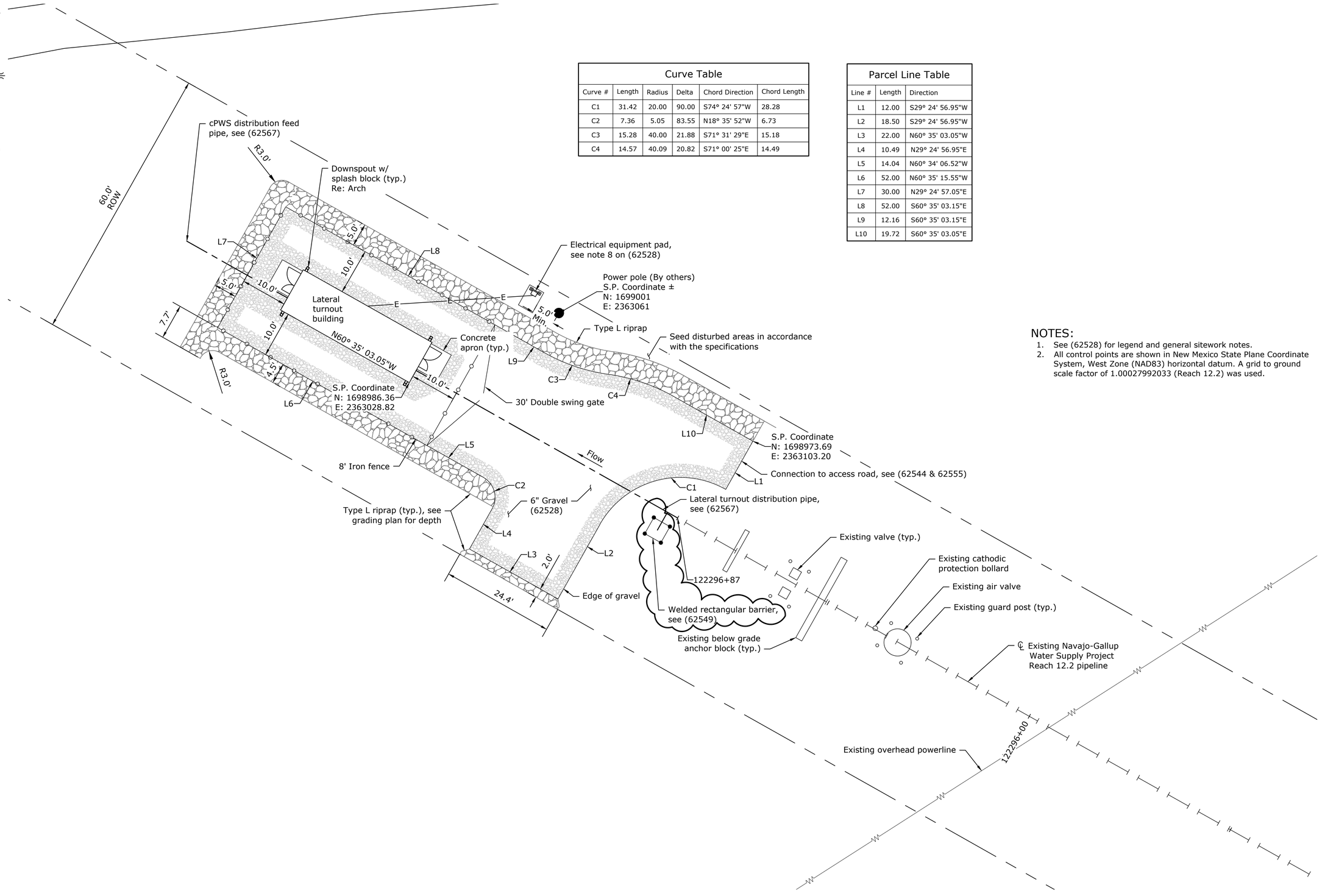
LAST SAVED DATE
2026-09-29
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD 2025 (LMS TECH)
CAD FILENAME
1695-D-62540-RD-13.DWG

Released

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	31.42	20.00	90.00	S74° 24' 57"W	28.28
C2	7.36	5.05	83.55	N18° 35' 52"W	6.73
C3	15.28	40.00	21.88	S71° 31' 29"E	15.18
C4	14.57	40.09	20.82	S71° 00' 25"E	14.49

Parcel Line Table		
Line #	Length	Direction
L1	12.00	S29° 24' 56.95"W
L2	18.50	S29° 24' 56.95"W
L3	22.00	N60° 35' 03.05"W
L4	10.49	N29° 24' 56.95"E
L5	14.04	N60° 34' 06.52"W
L6	52.00	N60° 35' 15.55"W
L7	30.00	N29° 24' 57.05"E
L8	52.00	S60° 35' 03.15"E
L9	12.16	S60° 35' 03.15"E
L10	19.72	S60° 35' 03.05"E



PLAN

0 10' 20'

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027992033 (Reach 12.2) was used.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REVISIONS			
REV NO	DATE	BY	DESCRIPTION
1	2026-09-29	Marner, Andrew (Andy) J P.E.	Welded rectangular barrier modification.

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED
PEER REVIEWER

DENVER, CO 2026-06-01

WINDOW ROCK TURNOUT

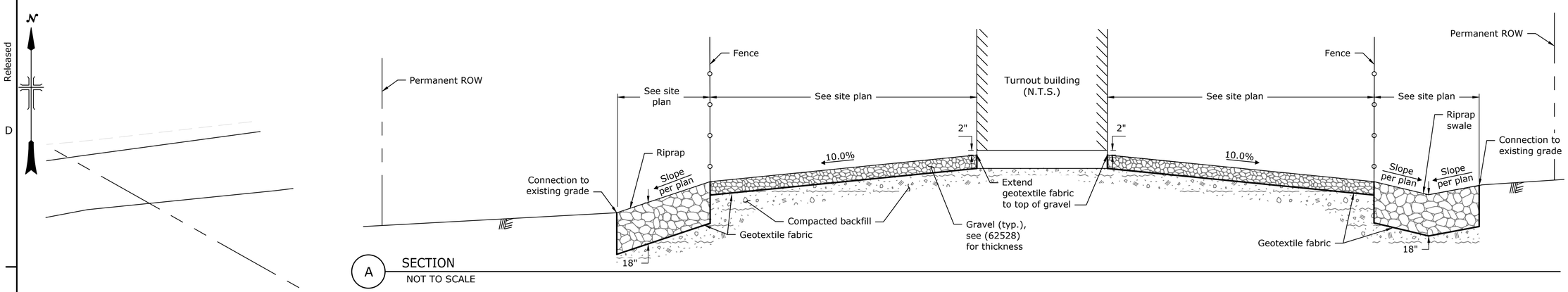
SITE PLAN

1695-D-62541

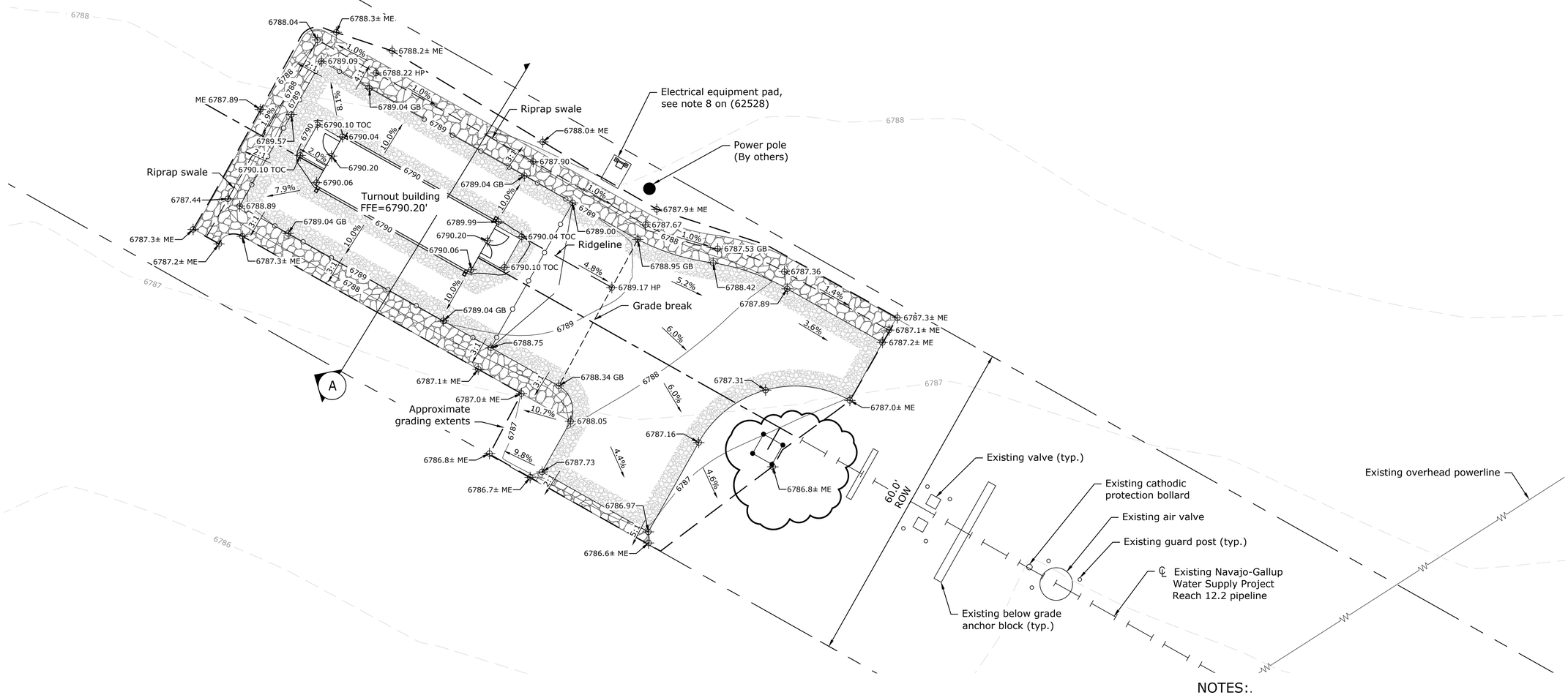
SHEET 14

LAST SAVED DATE
2026-09-29
LAST SAVED BY
BZOOK

CAD SYSTEM
AutoCAD 2025 (LMS TECH)
CAD FILENAME
1695-D-62541-RD-14.DWG



A SECTION
NOT TO SCALE



PLAN
0 10' 20'

NOTES:.
1. See (62528 & 62529) for legend, abbreviations, and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027992033 (Reach 12.2) was used.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-29	D Welded rectangular barrier modification. Marner, Andrew (Andy) J P.E.

B. ZOOK — DESIGNED
L. VOGT — DRAWN
C. HARMON — CHECKED
Marner, Andrew (Andy) J P.E. — TECH. APPR.
Steen, Timothy M — APPROVED PEER REVIEWER
DENVER, CO 2026-06-01

WINDOW ROCK TURNOUT

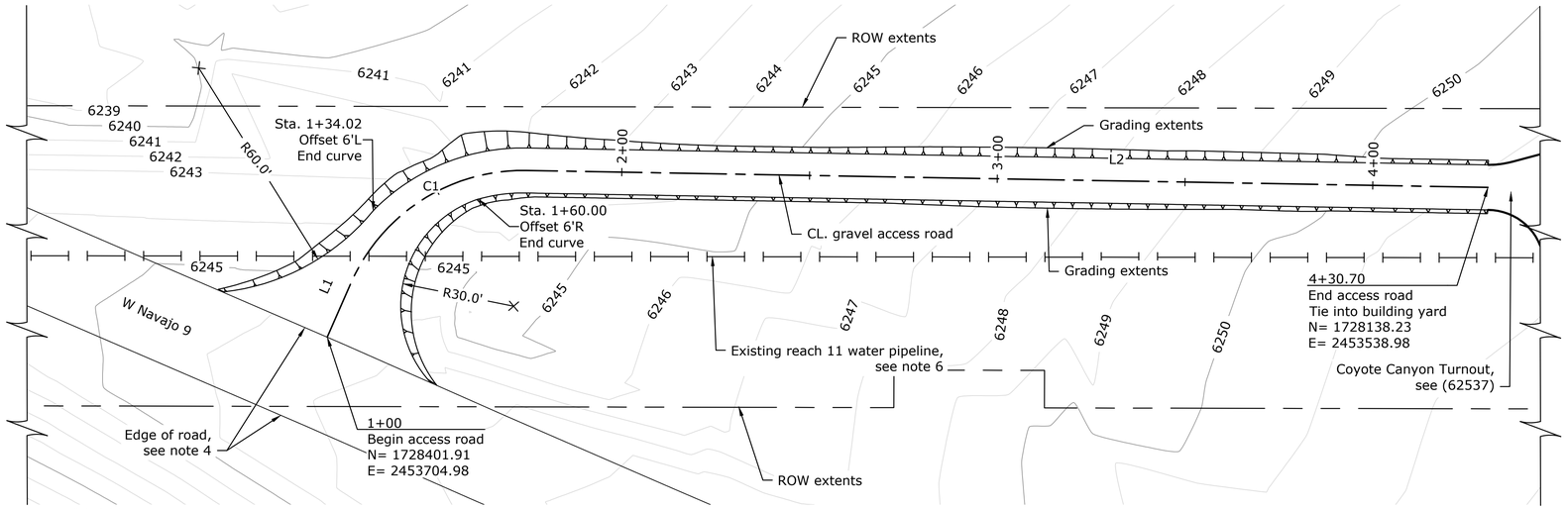
GRADING PLAN

1695-D-62542

SHEET 15

LAST SAVED DATE
2026-09-24
LAST SAVED BY
BZOOK

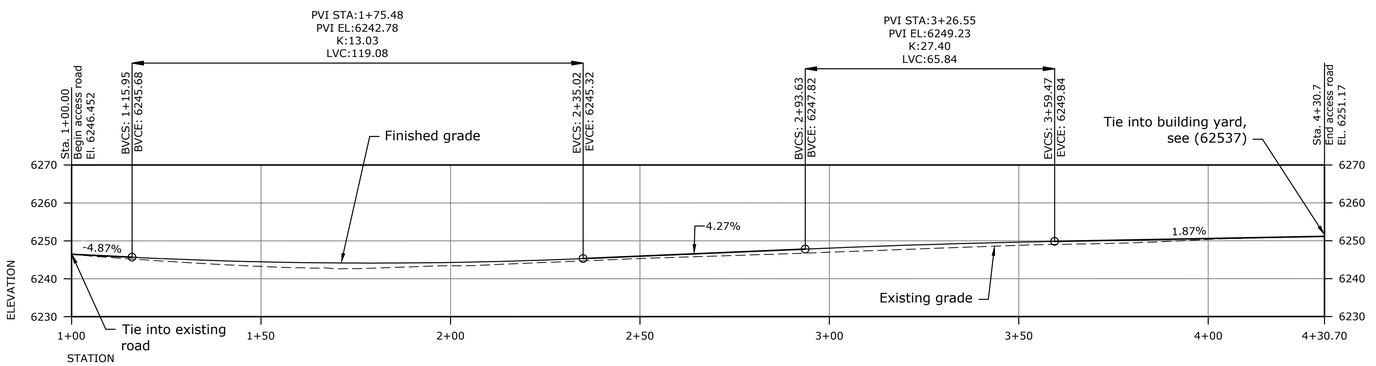
CAD SYSTEM
AUTOCAD OS (LMS TECH)
CAD FILENAME
1695-D-62542-RD-15.DWG



PLAN

0 10' 20' 40'

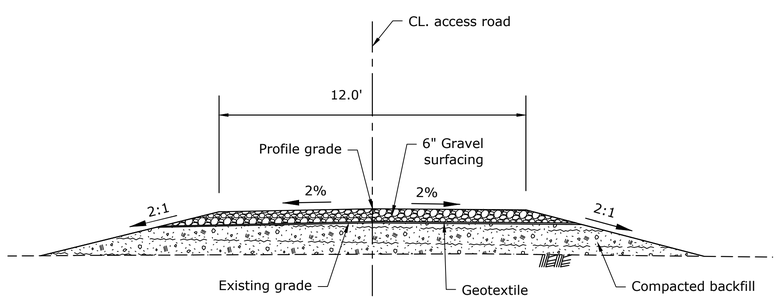
SCALE: 1" = 20'



PROFILE

0 10' 20' 40'

SCALE: 1" = 20'



TYPICAL CROSS SECTION

NOT TO SCALE

ALIGNMENT LINE DATA				
Line #	START STA	END STA	LENGTH	BEARING
L1	1+00.00	1+15.84	15.84	S26° 45' 40.95"E
L2	1+74.59	4+30.70	256.11	S40° 33' 22.53"W

ALIGNMENT CURVE DATA								
CURVE #	BC STA	PI STA	EC STA	LENGTH	RADIUS	DELTA	TANGENT	CHORD
C1	1+15.84	1+49.14	1+74.59	58.75	50.00	67.3176	33.29	55.42

NOTES

1. The design speed of the access road is 25 mph. The design vehicle is a passenger pickup truck.
2. Topography shown includes two sets of data. One prior to the pipe installation and the second post pipe installation construction completion.
3. Field verify the location and connection point from the access road to the existing road. If any major discrepancies exist, access road tie-in is to be constructed as directed by COR.
4. Boundaries shown for W. Navajo 9 are approximate.
5. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00029693715 (Reach 11) was used.
6. For Reach 11 pipeline crossing locations and elevation data, see 1695-D-60309.

ABBREVIATIONS

BVCE	Begin vertical curve elevation
BVCS	Begin vertical curve station
EVCE	End vertical curve elevation
EVCS	End vertical curve station
K	Rate of vertical curve curvature
LVC	Length of vertical curve
PI	Point of intersection
PVI	Point of vertical intersection
Sta.	Station
ROW	Right of way

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	BY	DESCRIPTION
1	2026-10-06	Hewett, Kyra M P.E.	There was a scaling issue within page setup. EL tiles been updated accordingly.

SEAN MILLER — DESIGNED
SEAN MILLER — DRAWN
K HEWETT, P.E. — CHECKED
Hewett, Kyra M P.E. — TECH. APPR.
Ledezma, Gabriel E P.E. — APPROVED
- PEER REVIEWER -
DENVER, CO — 2026-10-05

COYOTE CANYON ACCESS ROAD

PLAN AND PROFILE

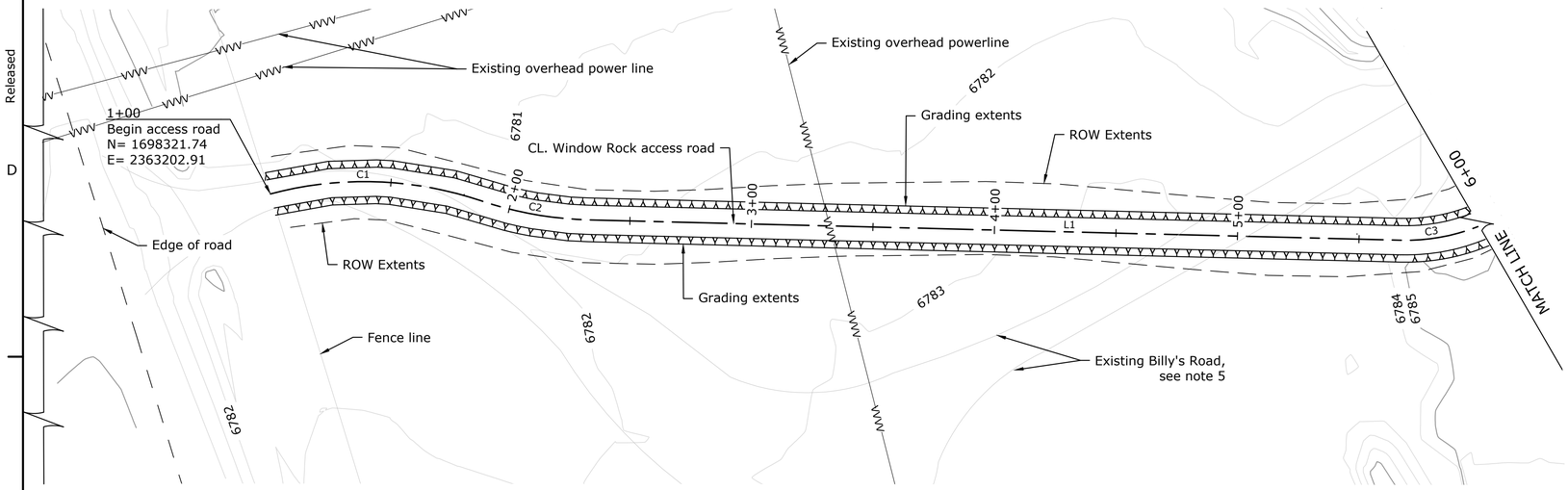
1695-D-62543

SHEET

LAST SAVED DATE
2026-10-05
LAST SAVED BY
MPERIGNY

CAD SYSTEM
AUTOCAD CIVIL (LMS TECH)
CAD FILENAME
1695-D-62543-RD-1.DWG

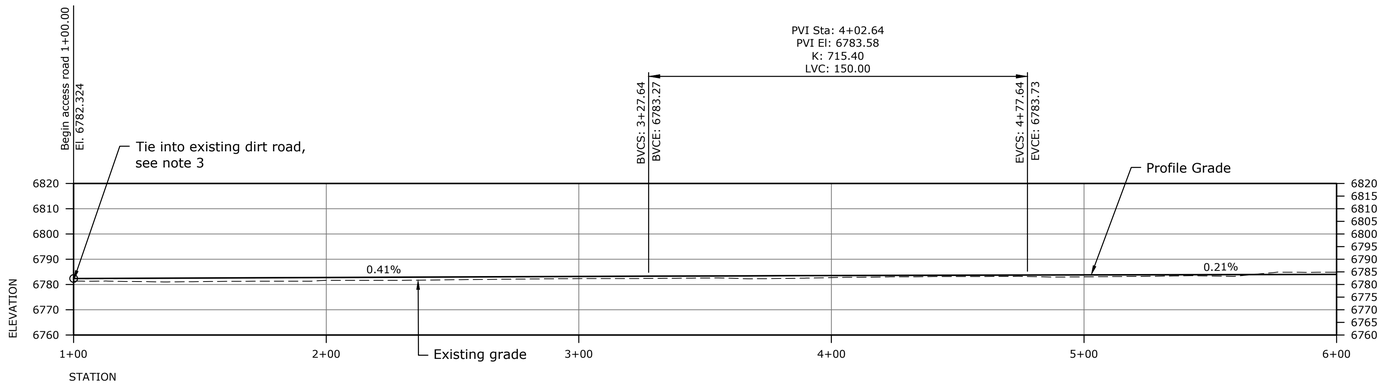
Released



PLAN

0 30' 60'

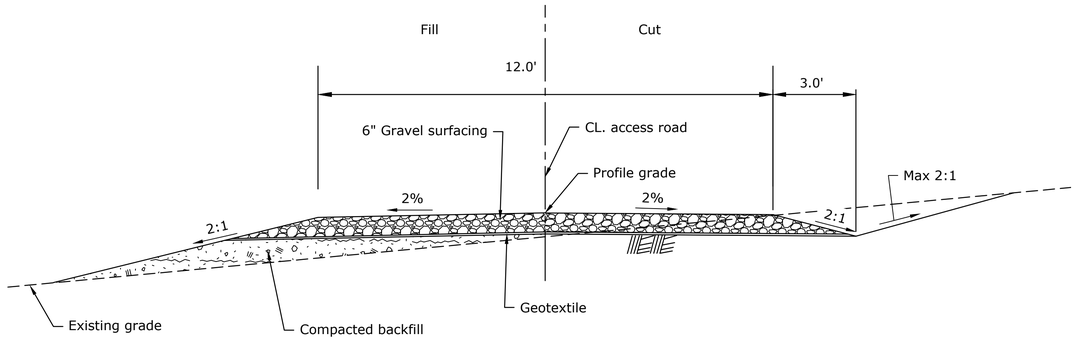
SCALE: 1" = 30'



PROFILE

0 30' 60'

SCALE: 1" = 30'



TYPICAL CROSS SECTION

NOT TO SCALE

ALIGNMENT LINE DATA

Line #	START STA	END STA	LENGTH (ft)	Bearing
L1	2+33.02	5+67.40	334.37	N61° 57' 31.71"E

ALIGNMENT CURVE DATA

Curve #	BC STA	PI STA	EC STA	LENGTH	RADIUS	DELTA	TANGENT	CHORD
C1	1+00.00	1+45.26	1+87.91	87.91	150.00	33.5800	45.26	86.66
C2	1+87.91	2+10.64	2+33.02	45.11	150.00	17.2310	22.73	44.94
C3	5+67.40	7+12.88	7+38.28	170.88	80.00	122.3864	145.48	140.20

ABBREVIATIONS

BVCE	Begin vertical curve elevation
BVCS	Begin vertical curve station
EVCE	End vertical curve elevation
EVCS	End vertical curve station
K	Rate of vertical curve curvature
LVC	Length of vertical curve
PVI	Point of vertical intersection
Sta.	Station

NOTES

1. The design speed of the access road is 25 mph. The design vehicle is a passenger pickup truck.
2. Topography shown includes two sets of data. One prior to the pipe installation and the second post pipe installation construction completion.
3. Field verify the location and connection point from the access road to the existing road. If any major discrepancies exist, access road tie-in is to be constructed as directed by COR.
4. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027992033 (Reach 12.2) was used.
5. Contractor shall field verify the boundaries and location of Billy's Road. Billy's Road intersection design is approximate and is to be constructed as directed by COR. The portions of Billy's Road inside of ROW to be abandoned in place.

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BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	BY	DESCRIPTION
1	2026-10-06	Hewett, Kyra M P.E.	D An incorrect scale factor was provided in the horizontal section and therefore was updated. Both have been updated accordingly.

SEAN MILLER
DESIGNED
SEAN MILLER
DRAWN
K HEWETT, P.E.
CHECKED
Hewett, Kyra M P.E.
TECH. APPR.
Ledezma, Gabriel E P.E.
APPROVED
PEER REVIEWER -

DENVER, CO 2026-10-05

WINDOW ROCK ACCESS ROAD

PLAN AND PROFILE

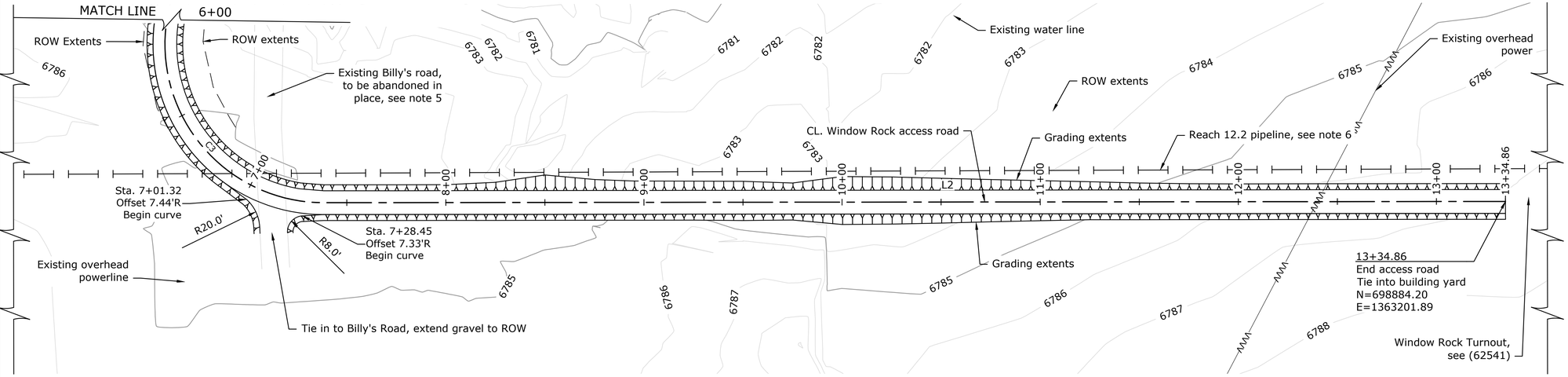
1695-D-62544

SHEET 1 OF 2

LAST SAVED DATE
2026-10-06
LAST SAVED BY
TRYAN

CAD SYSTEM
AUTOCAD CIVIL 3DS (LMS TECH)
CAD FILENAME
1695-D-62544-RD-1.DWG

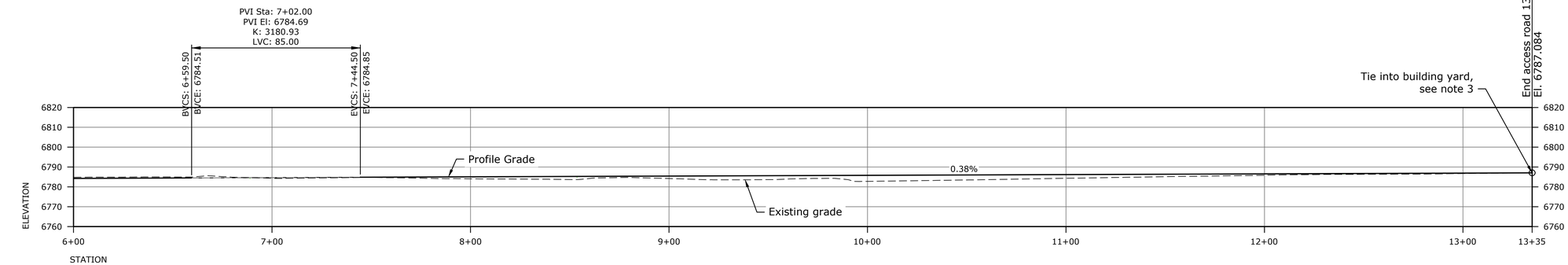
Released



PLAN

0 30' 60'

SCALE: 1" = 30'



PROFILE

0 30' 60'

SCALE: 1" = 30'

ALIGNMENT LINE DATA

LINE #	START STA	END STA	LENGTH (FT)	BEARING
L2	7+38.28	13+34.86	596.58	N60° 25' 39"W

ALIGNMENT CURVE DATA

Curve #	BC STA	PI STA	EC STA	LENGTH	RADIUS	DELTA	TANGENT	CHORD
C3	5+67.40	7+12.88	7+38.28	170.88	80.00	122.3864	145.48	140.20

NOTES

1. The design speed of the access road is 25 mph. The design vehicle is a passenger pickup truck.
2. Topography shown includes two sets of data. One prior to the pipe installation and the second post pipe installation construction completion.
3. Field verify the location and connection point from the access road to the existing road. If any major discrepancies exist, access road tie-in is to be constructed as directed by COR.
4. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027992033 (Reach 12.2) was used.
5. Contractor shall field verify the boundaries and location of Billy's road. Billy's Road intersection design is approximate and is to be constructed as directed by COR. The portions of Billy's road inside of ROW to be abandoned in place.
6. For Reach 12.2 pipeline crossing locations and elevation data, see 1695-D-60260.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	BY	DESCRIPTION
D	2026-10-06	Hewett, Kyra M P.E.	An incorrect scale factor was provided in the project control and was causing issues. Both have been updated accordingly.
1			

SEAN MILLER
DESIGNED
SEAN MILLER
DRAWN
K HEWETT, P.E.
CHECKED
Hewett, Kyra M P.E.
TECH. APPR.
Ledezma, Gabriel E P.E.
APPROVED
PEER REVIEWER -

DENVER, CO 2026-10-05

WINDOW ROCK ACCESS ROAD

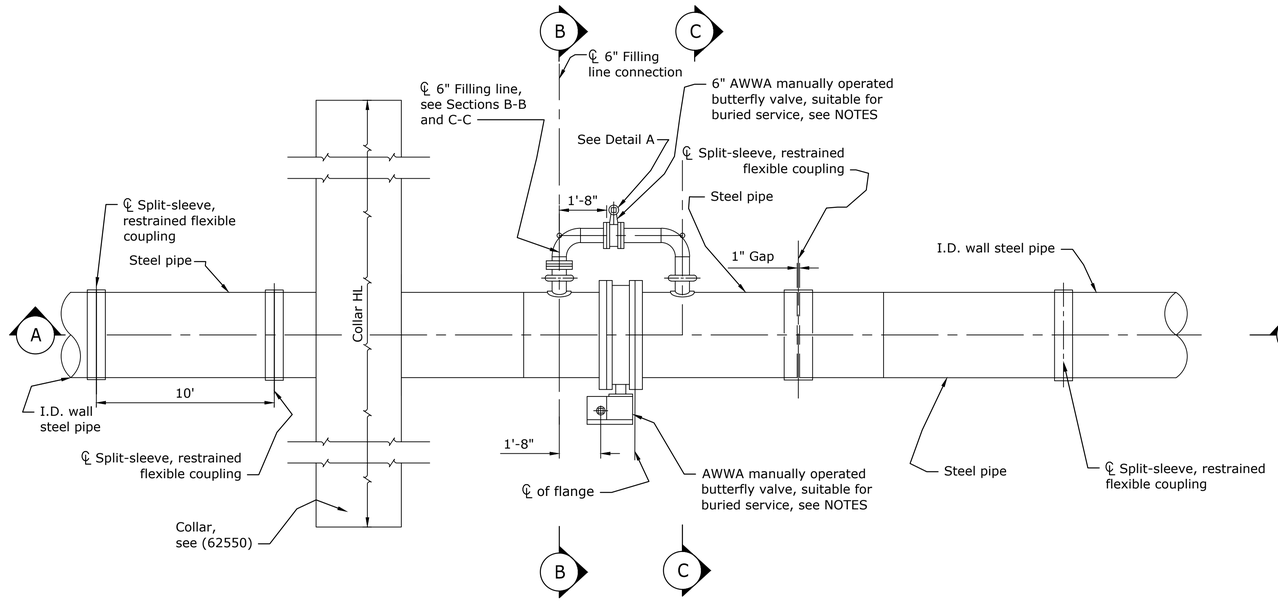
PLAN AND PROFILE

1695-D-62545

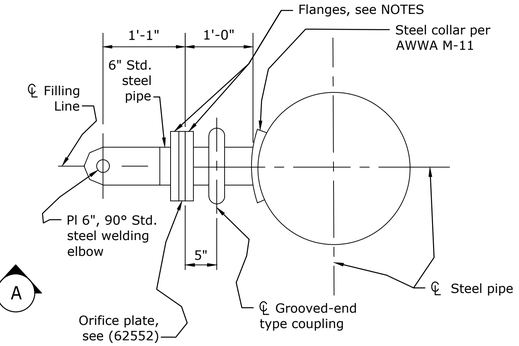
SHEET 2 OF 2

LAST SAVED DATE
2026-10-06
LAST SAVED BY
TRYAN

CAD SYSTEM
AUTOCAD CIVIL (LMS TECH)
CAD FILENAME
1695-D-62545-RD-1.DWG



PLAN



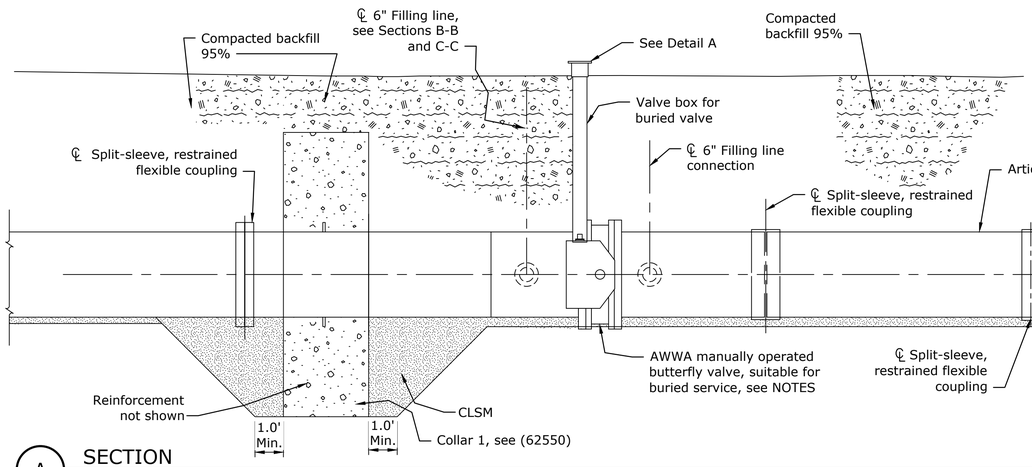
B SECTION

OPERATING CRITERIA

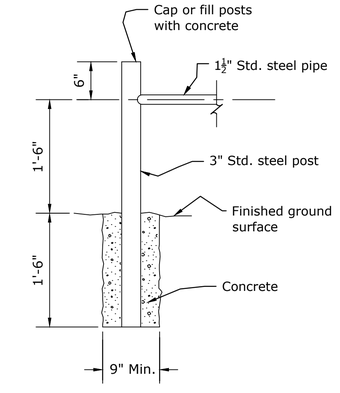
1. High performance Bypass butterfly valve shall be fully open when operated.
2. Excavation around blocking shall not occur when the pipeline is under hydrostatic pressure.

NOTES

1. Drawing not to scale.
2. See Specifications on exact label text required for each valve.
3. All sectionalizing valves will require a bypass.
4. For pipe trench earthwork details, see (62554).
5. The minimum steel cylinder thickness for the section of line pipe with the valve outlet shall be the thickness of the steel line pipe. For the steel line pipe thickness see the specification.
6. Piping next to butterfly valves may require 1" oversizing so that the applied interior lining will not reduce the inside dimension to a smaller size which could prevent the butterfly valve from opening. Valve bypass piping shall be epoxy lined.
7. For collar details, see (62550).
8. Flange class shall meet pressure requirement designated by adjacent line pipe. Flange drilling shall match mating flange.
9. For Flange and Valve ratings, see notes (62551).
10. Dissimilar metals require a dielectric coupling for electrical isolation.

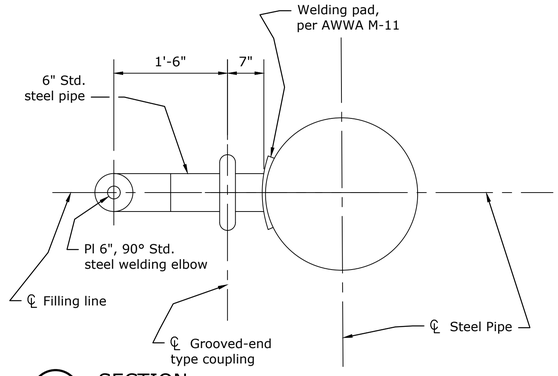


A SECTION

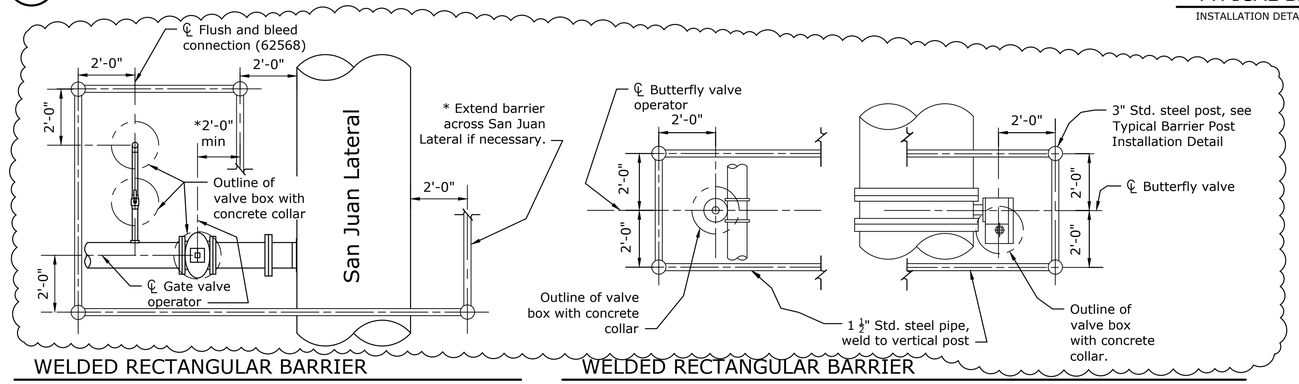


TYPICAL BARRIER POST

INSTALLATION DETAIL



C SECTION

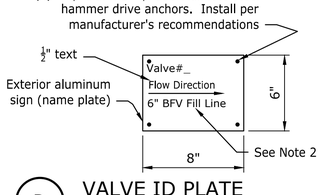


WELDED RECTANGULAR BARRIER

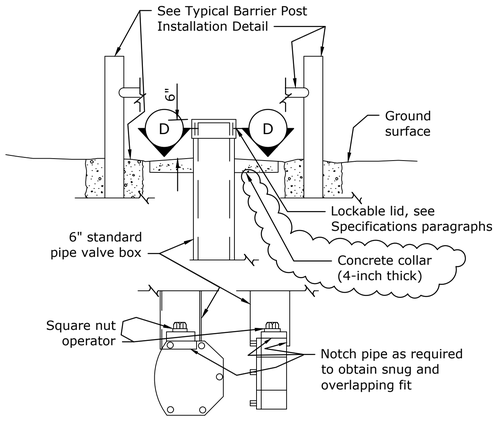
TURNOUT LOCATION

WELDED RECTANGULAR BARRIER

48-INCH BUTTERFLY VALVE



D VALVE ID PLATE



A DETAIL

(PROFILE VIEW)

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	DESCRIPTION
1	2026-09-30	D Revised to show chlorine injection port with gasketed... Marner, Andrew (Andy) J P.E.

A. MARNER
DESIGNED
A. MARNER
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Pelzer, Kylie C P.E.
APPROVED
PEER REVIEWER
DENVER, CO 2026-06-02

BBN9 LINE PIPE - SECTIONALIZING VALVE INSTALLATION

PLAN, SECTIONS AND DETAILS

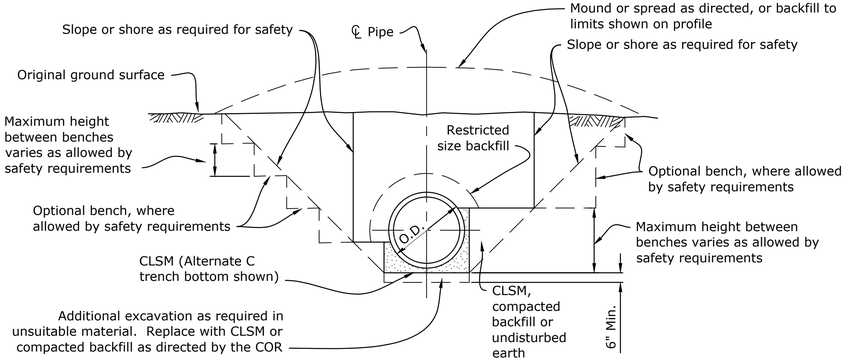
1695-D-62549

SHEET 2

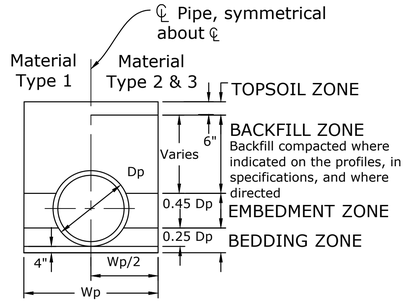
Released

LAST SAVED DATE
2026-09-30
LAST SAVED BY
AMARNER

CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62549-RD-2.DWG

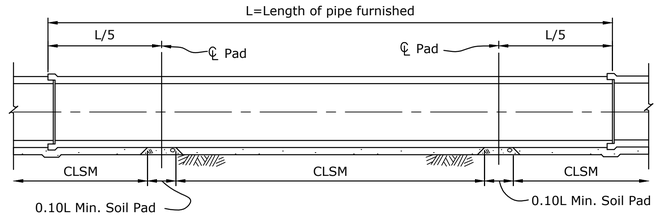


TYPICAL TRENCH DETAILS



TRENCH DEFINITION

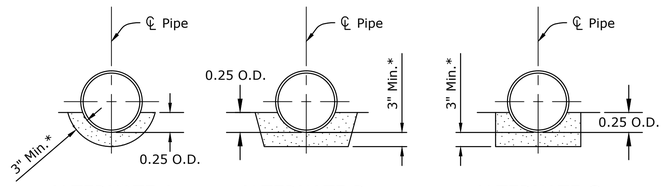
ALL TYPES OF PIPE



NOTE: Soil pads allow observation of the CLSM flowing from one side of the pipe to the other. See Note 8.

CLSM INSTALLATION WITH SOIL PADS

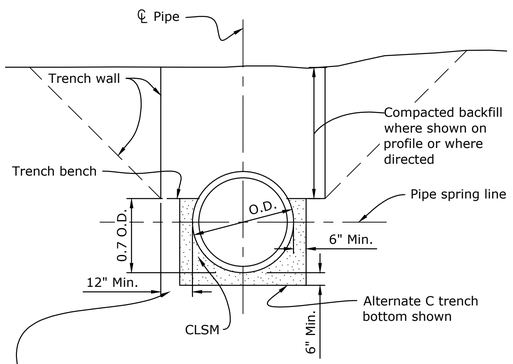
TYPICAL FOR ALL PIPE TYPES



ALTERNATE A ALTERNATE B ALTERNATE C

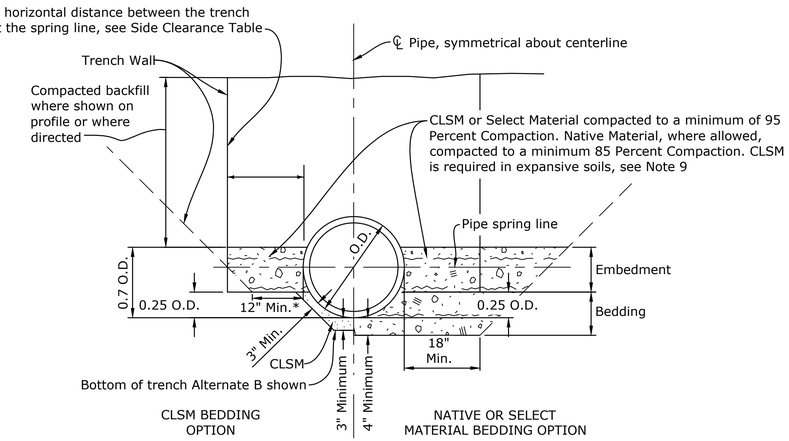
ACCEPTABLE TRENCH BOTTOMS FOR CLSM

* 6" Min required between bottom of pipe and rock surfaces



Minimum working/inspection clearance is the horizontal distance between the intersection of the trench wall with the trench bench and the pipe at spring line

MATERIAL TYPE 1 TRENCH



*12" Minimum bench not required using CLSM embedment, minimum side clearance still required

MATERIAL TYPE 2 AND 3 TRENCH

SELECT MATERIAL GRADATION LIMITS

Size *	Percent by Weight
Passing No. 200 sieve	12 or less
Passing No. 50 sieve	25 or less

* Maximum size shall not exceed ¾-inch

CLSM/SELECT EMBEDMENT SIDE CLEARANCE

Height of Cover	STEEL					
	<=16"		24"		48"	
	Material Type 2	Material Type 3	Material Type 2	Material Type 3	Material Type 2	Material Type 3
<=10'	6	6	12	15	15	18
<=15'	8	8	12	15	21	27
<=20'	10	10	12	18	30	33

NATIVE MATERIAL PROPERTIES

MATERIAL TYPE 1 - Typical trench wall materials would be rock; materials described as claystone, mudstone, or siltstone; highly cemented soils even if low density; sands and gravels; and silty or clayey material with inplace densities 95% Standard Proctor or higher.

MATERIAL TYPE 2 - Typical trench wall materials would include silty or clayey material or sands and gravels with inplace densities between 85% Standard Proctor or higher, but less than 95%.

MATERIAL TYPE 3 - Typical trench wall materials would include elastic silts, low density silty or clayey soils, or low-density cohesionless soils with a density below 85% Standard Proctor.

Trench Material Type locations, based on the above Native Material Properties and geology investigation, have been identified and correspond to minimum side clearance widths listed in the Side Clearance Table. Material Type 1's inferred bedrock surface is shown in the line pipe profile on drawings. Material Types 2 and 3 locations are listed in the specifications.

1. Side clearance measurement is at pipe springline for all materials.
2. Backfill can not exceed 5' until CLSM embedment obtains an unconfined compressive strength of 30 psi.
3. Any dimension less than 12" must use CLSM embedment. Select and native material embedment requires minimum of 18" side clearance.

Steel - Mortar lined, flexible coated.
Ductile Iron - Special thickness class.
HDPE - Only DR ratios 21, 17, 13.5, 11, and 9 allowed.

NATIVE MATERIAL EMBEDMENT

PIPE MATERIAL TYPE	MAX COVER DEPTH
Steel Pipe	Not Allowed

Screened excavated native material is allowed for the embedment zone for Material Types 2 and 3 based on the following conditions:

1. Particle sizes greater than ¾-inch are removed.
2. Compacted backfill to 95% is not required above pipe.
3. Native material embedment is compacted to 85%.
4. Side clearance is 18" minimum.
5. Restraint joints are not required for thrust.

NOTES

1. Drawing not to scale.
2. Dp and Wp are used for calculating pay quantities for all pipe and trench types. Calculations are based on vertical walls.
3. O.D is the outside diameter, in inches, of the pipe actually installed.
4. CLSM is Controlled Low Strength Material. See Specifications for details.
5. Sloping, shoring, and benching shall be in accordance with Reclamation Safety and Health Standards.
6. CLSM is required in the bedding for all pipe types.
7. Warning tape and tracer wire (for plastic pipes) not shown, but are required. See specifications for details.
8. Other material may be submitted in lieu of soil pads for approval. Soil pad materials must not exceed the compressive strength of the CLSM.
9. CLSM is required for the embedment when there is expansive soils and the native trench material's plastic index exceeds 25 as listed in Table 1.2 of Reclamation's Earth Manual.
10. CLSM is required for the embedment when saturated soil is encountered in the pipe trench.
11. Install pipe based on these Specifications or pipe manufacturer's, whichever is more stringent.

Released

LAST SAVED DATE 2026-06-02
LAST SAVED BY AMARNER

CAD SYSTEM AutoCAD 2025 (LMS TECH)
CAD FILENAME 1695-D-62554-RD-7.DWG

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV	DATE	DESCRIPTION
D	2026-09-30	Amend side clearance table. Marner, Andrew (Andy) J P.E.

A. MARNER DESIGNED
A. MARNER DRAWN
C. HARMON CHECKED
Marner, Andrew (Andy) J P.E. TECH. APPR.
Pelzer, Kylie C P.E. APPROVED PEER REVIEWER
DENVER, CO 2026-06-02

PRESSURE PIPE TRENCH INSTALLATION

SECTIONS AND DETAILS

1695-D-62554

SHEET 7

Flow Control Valve Sizes

Turnout Location	Flow Control Valve Size	Maximum Operating Pressure	Maximum Operating Flows	Yard Pipe Drawing	Interior Pipe Centerline Elevation	Site Plan
Sanostee	6 inch	233 psi	2.27 cfs	1695-D-62561	5571.75	1695-D-62528
Newcomb	8 inch	184 psi	1.52 cfs	1695-D-62562	5570.80	1695-D-62531
Naschitti	6 inch	210 psi	1.77 cfs	1695-D-62563	5877.50	1695-D-62533
BBN9	8 inch	123 psi	3.80 cfs	1695-D-62564	6056.30	1695-D-62535
Coyote Canyon	6 inch	230 psi	1.25 cfs	1695-D-62565	6259.75	1695-D-62537
Rock Springs	6 inch	148 psi	1.71 cfs	1695-D-62566	6858.85	1695-D-62539
Window Rock[1]	12 inch	138 psi	11.51 cfs	1695-D-62567	6795.70	1695-D-62541

Notes:

1. Future phase, not for construction

Location of Lateral Turnout

Lateral Turnout	Lateral Turnout Tie-In Location		Reference Drawing	Consecutive Distribution Entry Point	
	Northing[1]	Easting[1]		Northing[1]	Easting[1]
Sanostee	1976795.6044	2462216.8104	1695-D-60728	1976728.7800	2462184.4600
Newcomb	1925402.5175	2466258.7219	1695-D-60728	1925555.7300	2466211.6000
Naschitti	1845416.9037	2472035.7886	1695-D-60728	1845407.4200	2472005.3900
BBN9[2]	1773044.4500	2479055.1100	1695-D-62535	1772870.6700	2479016.7600
Coyote Canyon	1728154.7800	2453530.3800	1695-D-62537	1728015.9400	2453444.5400
Rock Springs	1688592.1811	2415237.0131	1695-D-60119	1688625.8900	2415142.2500
Window Rock	1698957.0600	2363086.8899	1695-D-60260	1699016.9700	2362980.6500

Notes:

1. Coordinate system code NM83-WF
2. Crownpoint is now BBN9

REFERENCE DRAWINGS

1695-D-60119 ------ STA. 121300+00 TO STA. 121350+00
1695-D-60260 ------ STA. 122280+00 TO END
1695-D-60326 ------ Sectionalizing Valve Installation
1695-D-60328 ------ Crownpoint Turnout
1695-D-60728 ------ Typical Turnout
1695-D-62535 ------ BBN9 Site Plan
1695-D-62537 ------ Coyote Canyon Site Plan
1695-D-62556 ------ Thrust Blocks and Misc. Site Concrete
1695-D-62590 ------ Galvanic Anode Cathodic Protection

GENERAL NOTES

1. Provide nuts, bolts and gaskets suitable for cold water service.
2. All flanges shall be AWWA Class F as shown unless otherwise noted.
3. Additional fittings, flanges, or welded joints added by the contractor to facilitate installation of the piping are subject to Government approval.
4. Set grooved coupling gap to half of manufacturer's maximum allowable pipe end separation.
5. Thead-O-Let type fittings may be substituted for the 3000 lb. steel threaded couplings and threaded half-couplings. Weld Thread-O-Let fittings to pipe as recommended by Thread-O-Let manufacturer.
6. Welding symbols apply to the joints of all members of similar identification.
7. Hydrostatic test pressure for the turnout building steel pipe: 1.5 times the maximum operating pressure, see table.
8. Dimensions marked with an asterisk (*) to be field verified.
9. All valves and equipment to be rated for a minimum 300 psi working pressure.
10. Provide control valve flange supports to match control valve flanges, round or flat bottomed.
11. All elbows shall be long radius unless otherwise noted.
12. At locations where fittings have a greater wall thickness than steel piping, machine to match lighter pipe wall and taper excess thickness internally to form transition zone.
13. All fillet welds shall be returned.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	STA.	2026-09-30	ADDED NOTE TO FLOW CONTROL VALVE SIZES TABLE
1	D		

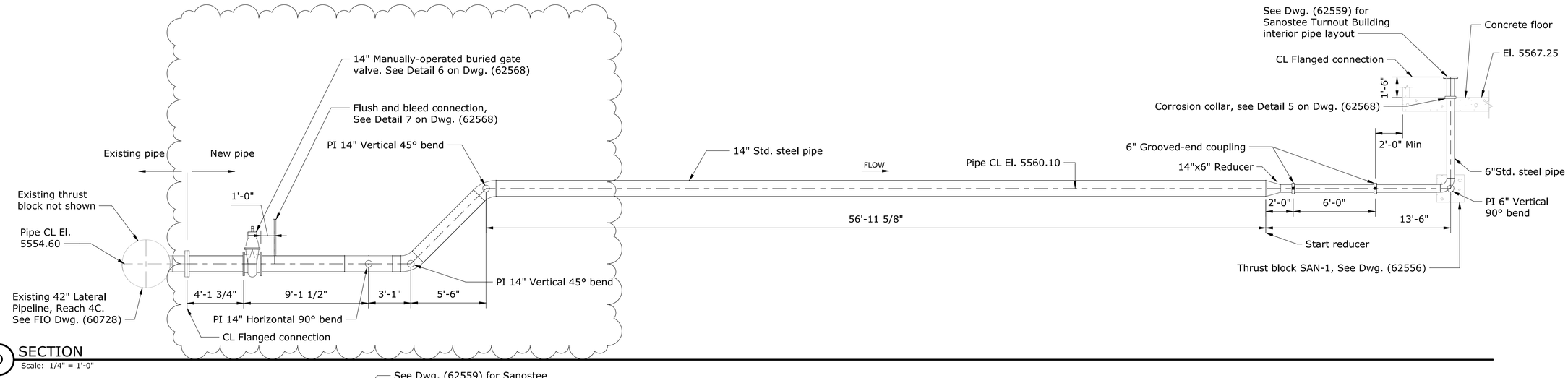
CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems

DENVER, CO 2026-06-19

COVERSHEET

DETAILS

1695-D-62558



D SECTION
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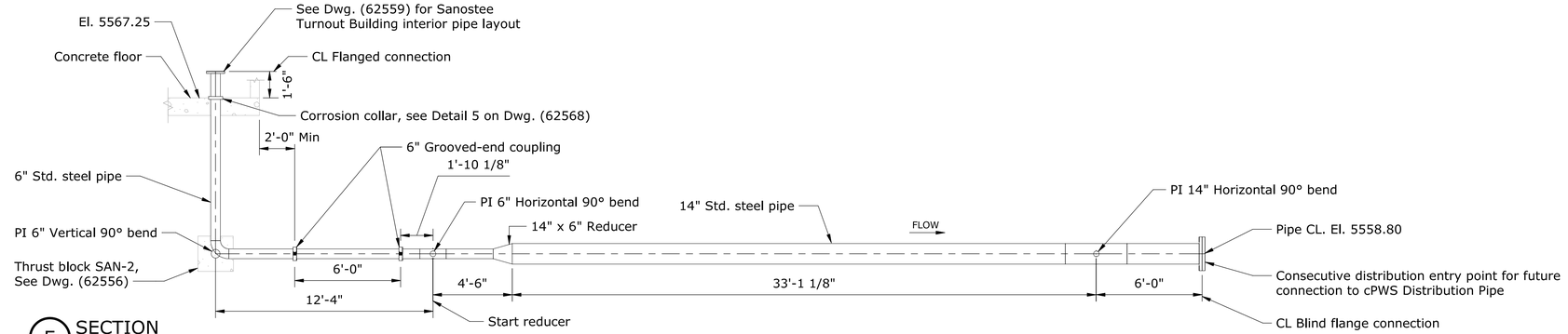
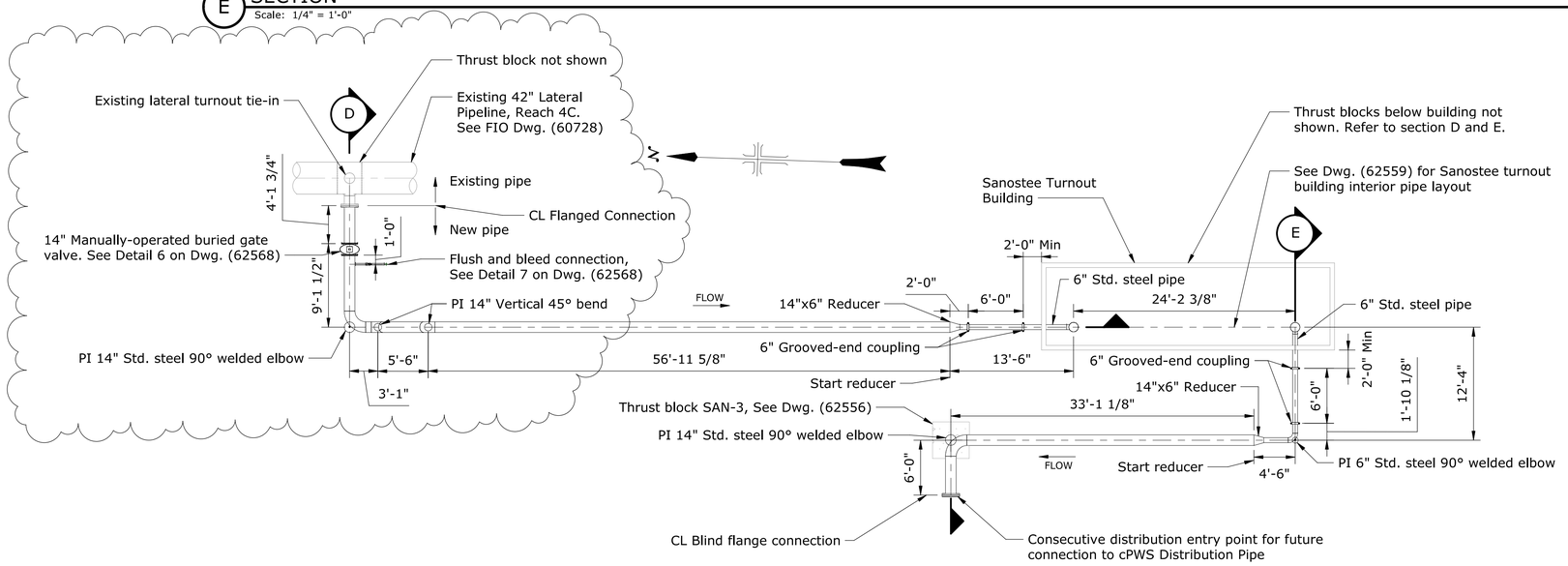


E SECTION
Scale: 1/4" = 1'-0"



SANOSTEE YARD PIPE PLAN
Scale: 1/8" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 8'
Scale: 1/4" = 1'-0"

0 4' 8' 16'
Scale: 1/8" = 1'-0"

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

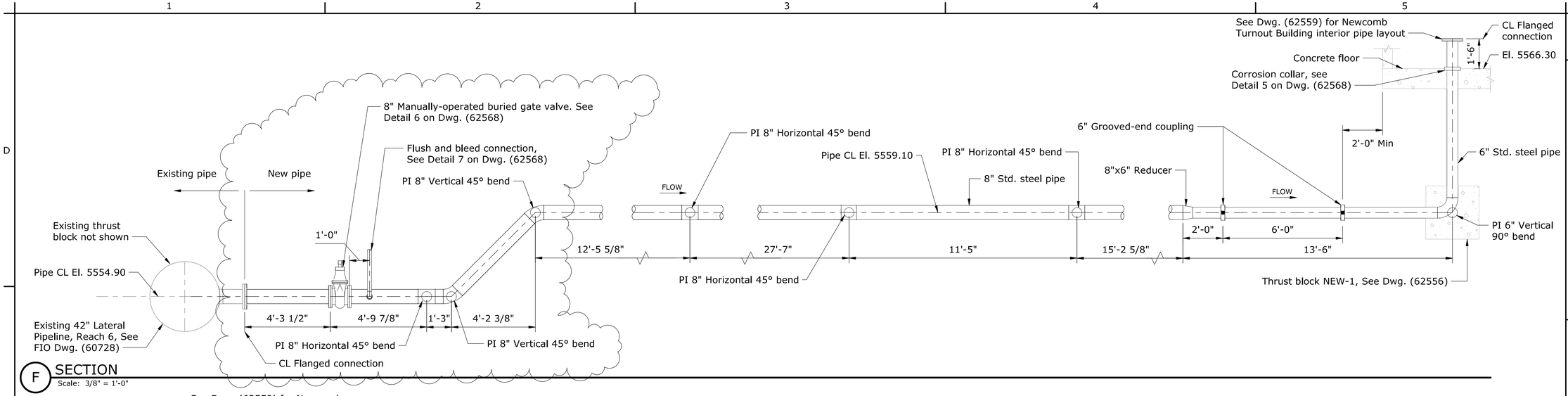
REV NO	STA	DATE	
1	D	2026-09-30	ADDED FLUSH AND BLEED CONNECTION

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E. (JEREMY HAMMOND, P.E.) Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

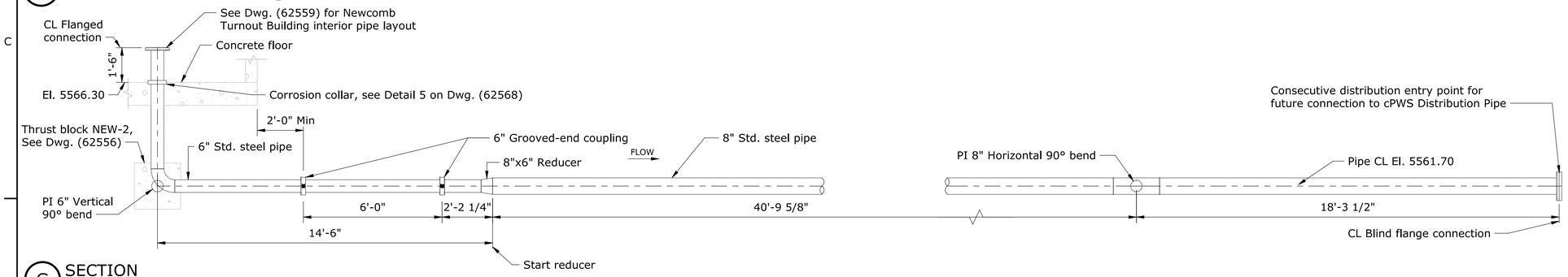
SANOSTEE YARD PIPE

PLAN AND SECTION

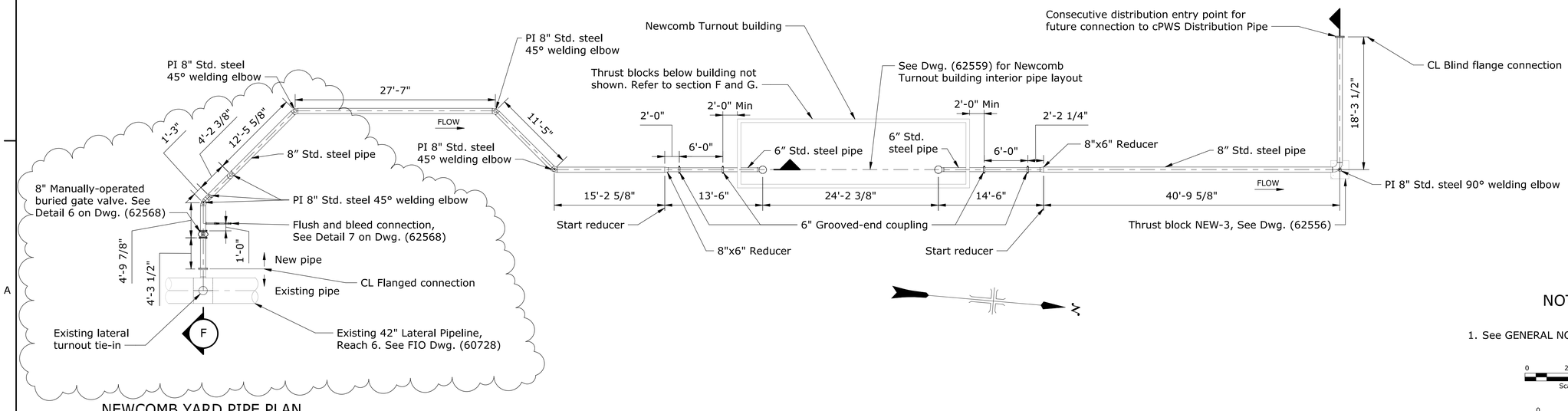
1695-D-62561



F SECTION
Scale: 3/8" = 1'-0"



G SECTION
Scale: 3/8" = 1'-0"



NEWCOMB YARD PIPE PLAN
Scale: 1/8" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

Scale: 3/8" = 1'-0"

Scale: 3/16" = 1'-0"

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NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

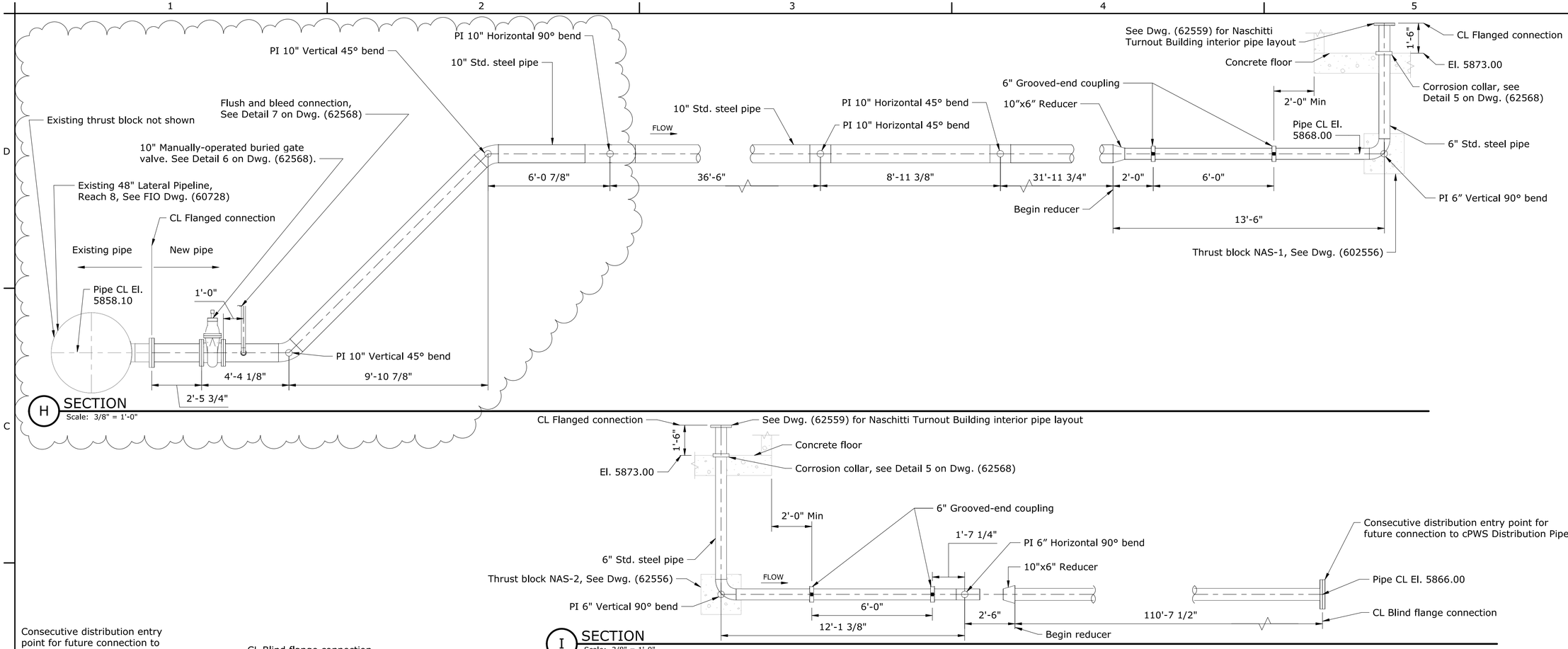
REV NO	STA.	
1	D	ADDED FLUSH AND BLEED CONNECTION

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
	Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

NEWCOMB YARD PIPE

PLAN AND SECTION

1695-D-62562



NASCHITTI YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

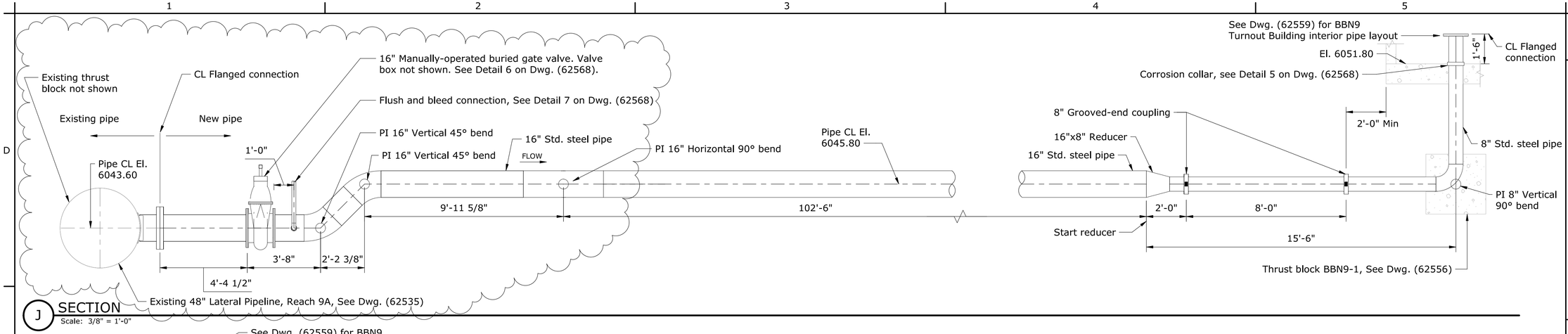
REV. NO	STA.		
1	D	2026-09-30	ADDED FLUSH AND BLEED CONNECTION

CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO 2026-06-19

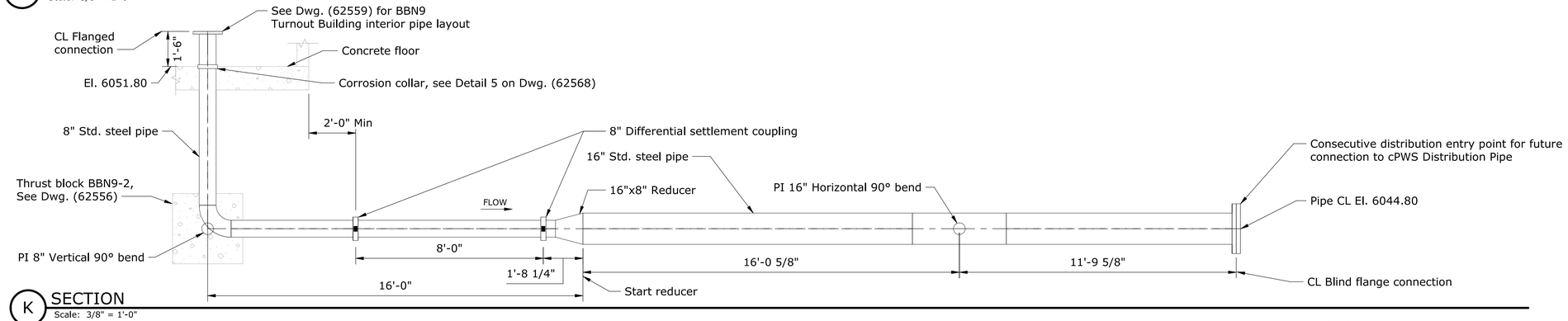
NASCHITTI YARD PIPE

PLAN AND SECTION

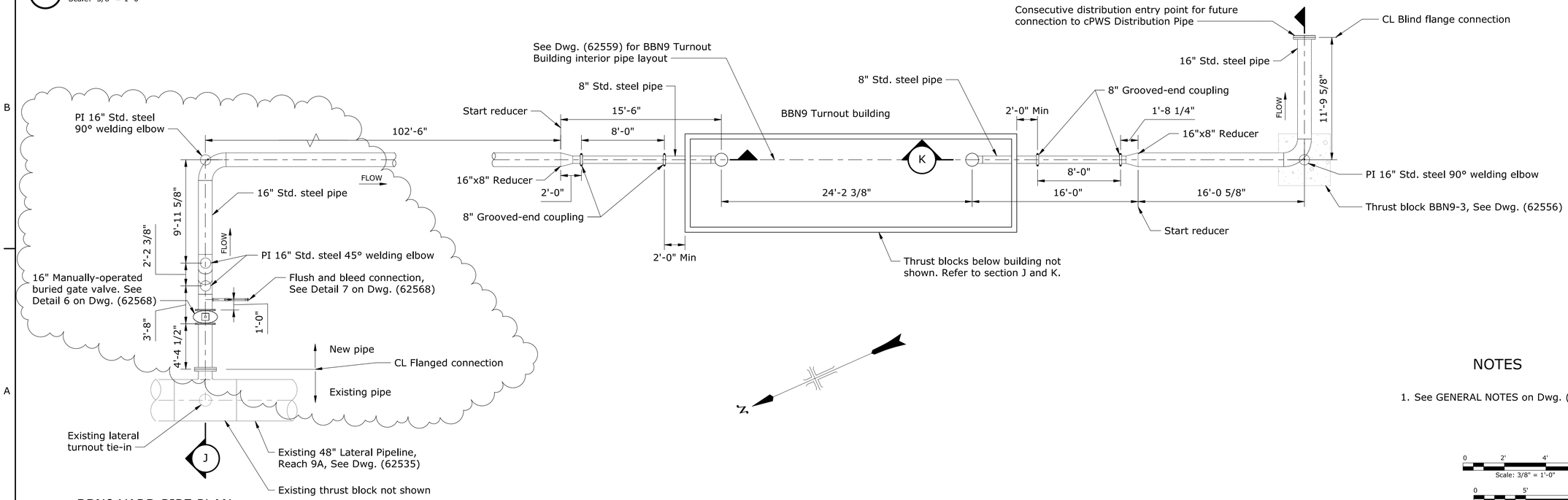
1695-D-62563



J SECTION
Scale: 3/8" = 1'-0"



K SECTION
Scale: 3/8" = 1'-0"



BBN9 YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

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NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

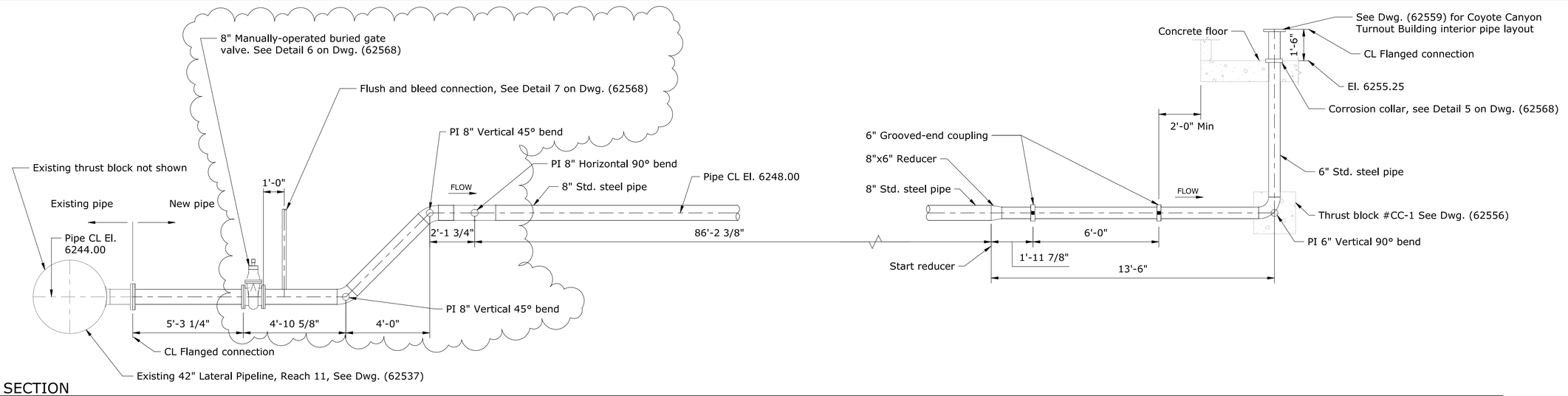
REV NO	STA.	
1	D	ADDED FLUSH AND BLEED CONNECTION

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
	Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

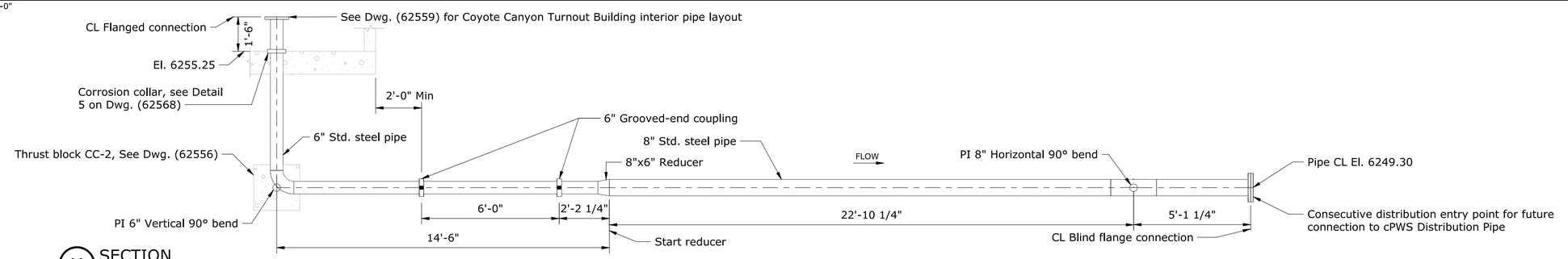
BBN9 YARD PIPE

PLAN AND SECTION

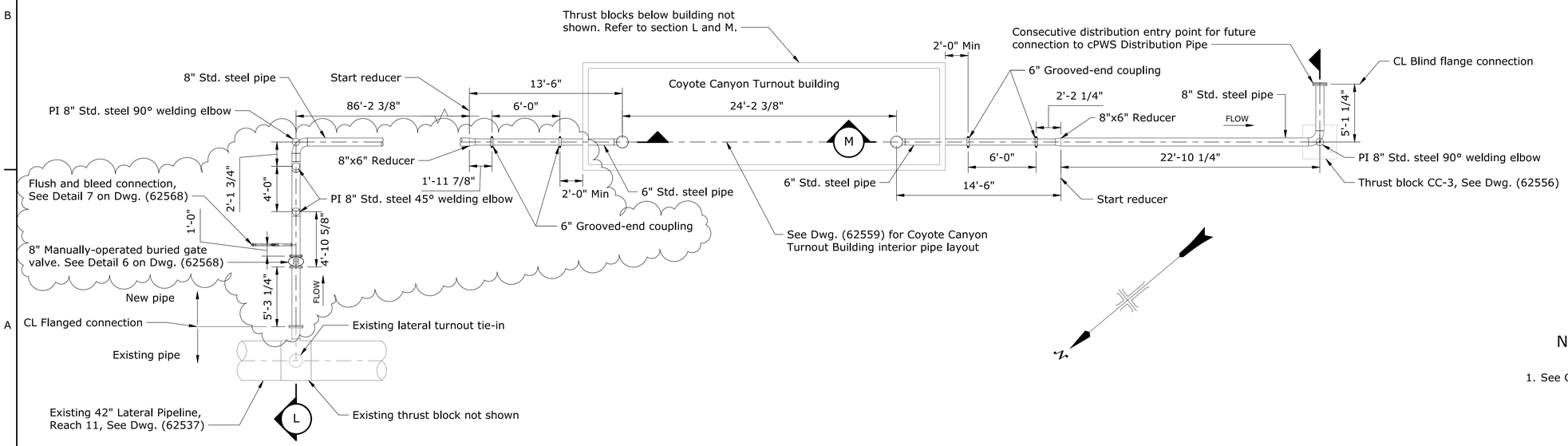
1695-D-62564



L SECTION
Scale: 3/8" = 1'-0"



M SECTION
Scale: 3/8" = 1'-0"



COYOTE CANYON YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

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BUREAU OF RECLAMATION
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NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

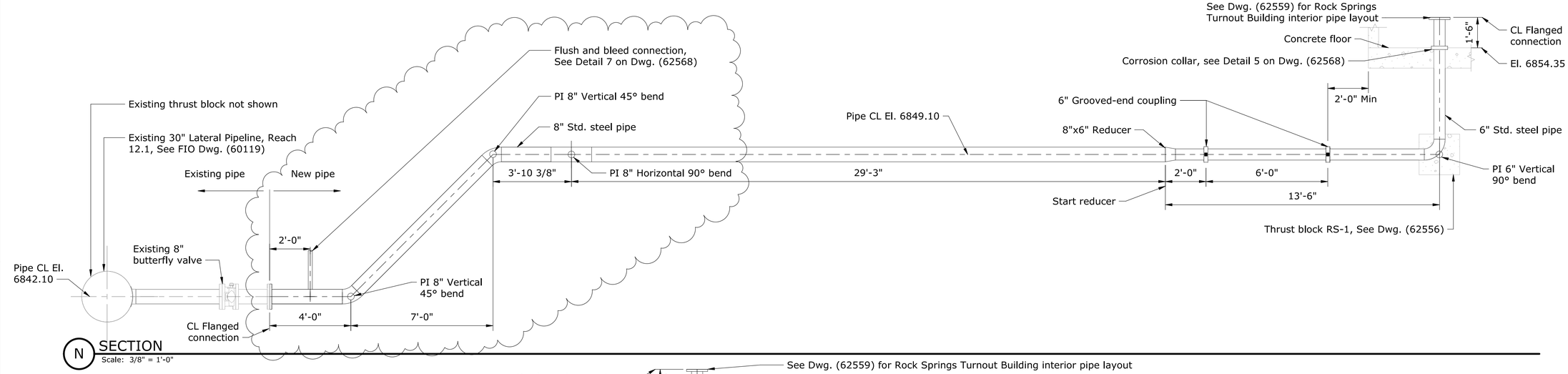
REV NO	STA.		
1	D	2026-09-30	ADDED FLUSH AND BLEED CONNECTION

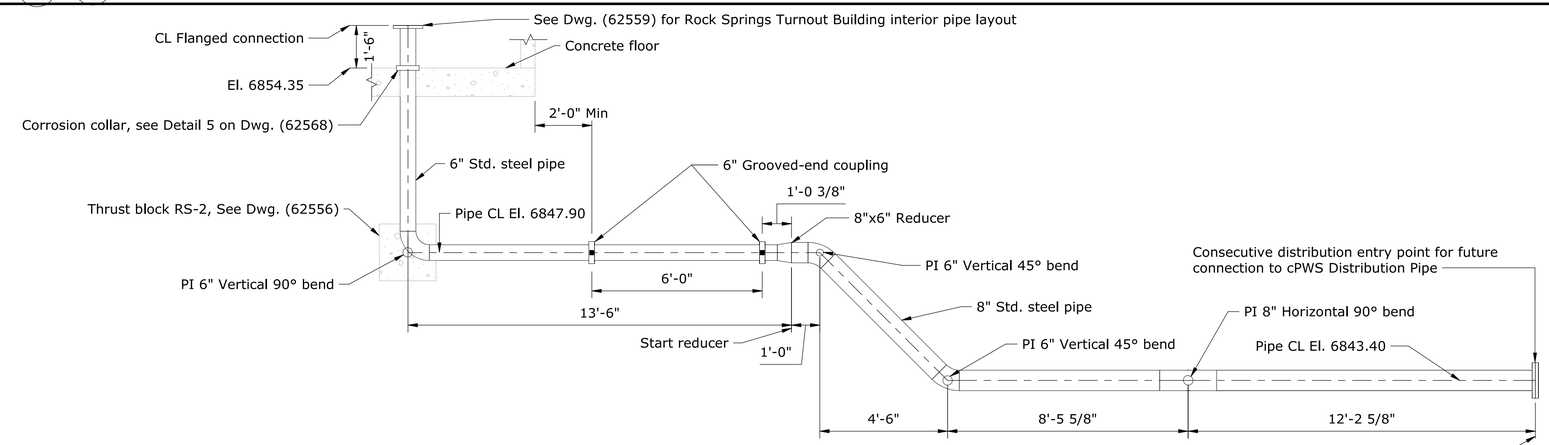
CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO 2026-06-19

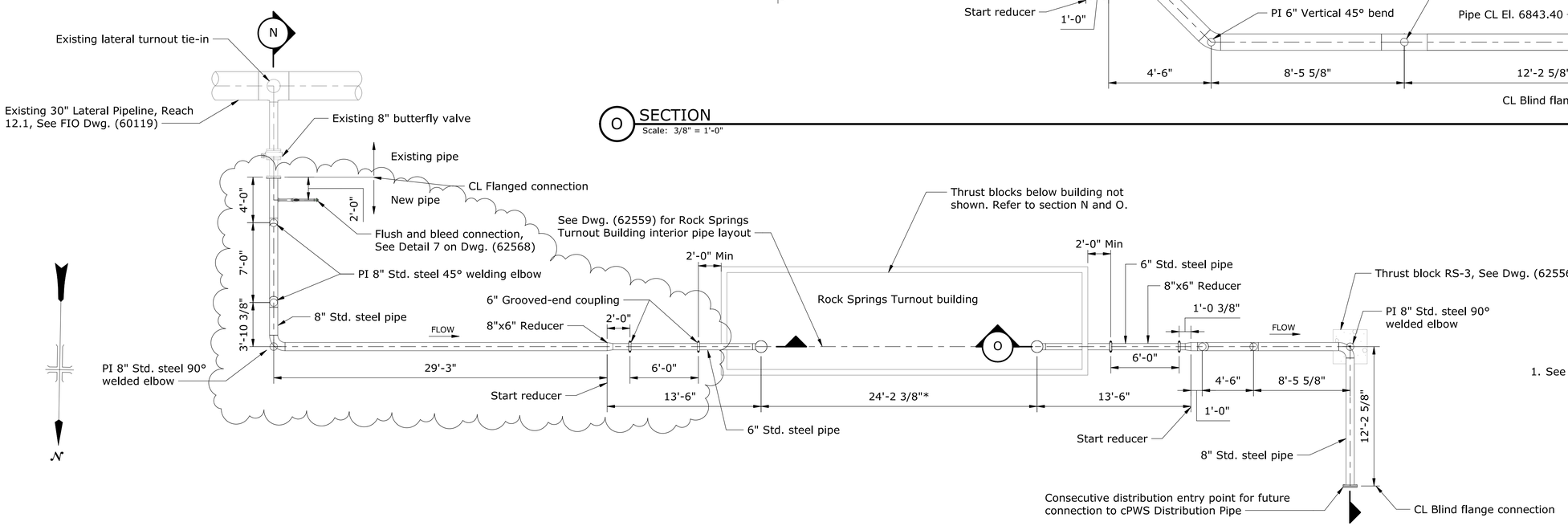
COYOTE CANYON YARD PIPE

PLAN AND SECTION

1695-D-62565







ROCK SPRINGS YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

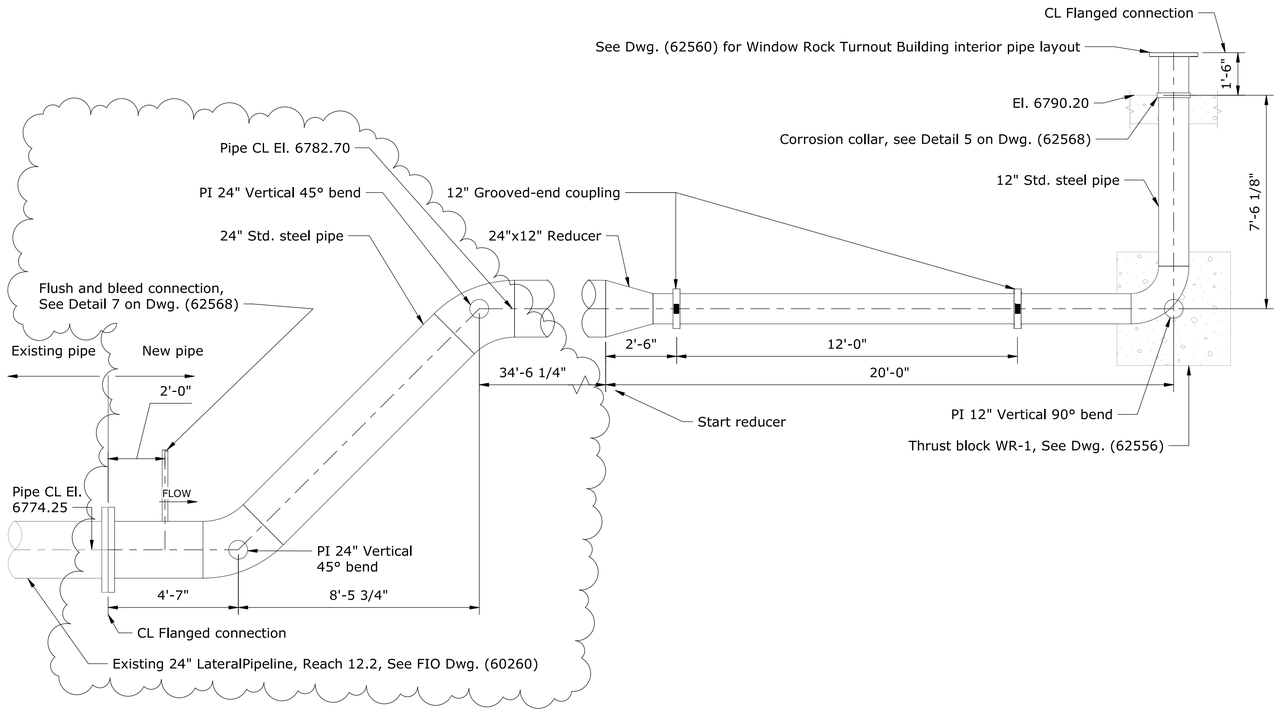
REV. NO	STA.	DATE	
1	D	2026-09-30	ADDED FLUSH AND BLEED CONNECTION

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
	Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

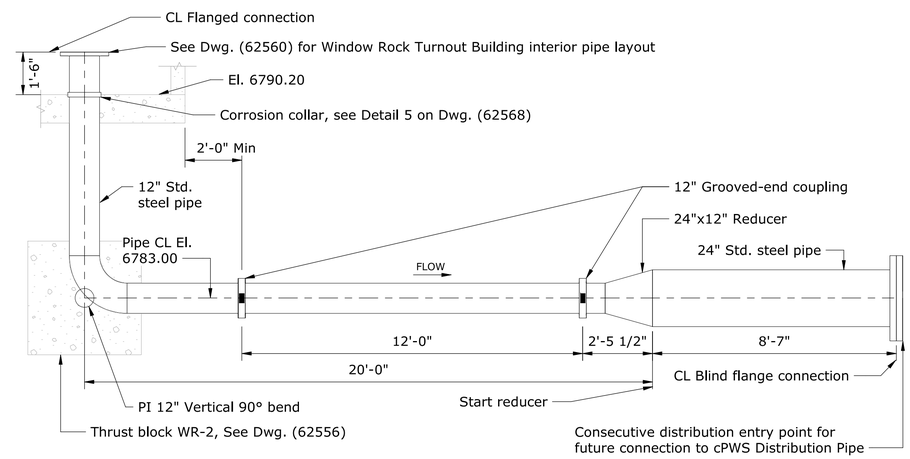
ROCK SPRINGS YARD PIPE

PLAN AND SECTION

1695-D-62566



P SECTION
Scale: 3/8" = 1'-0"



Q SECTION
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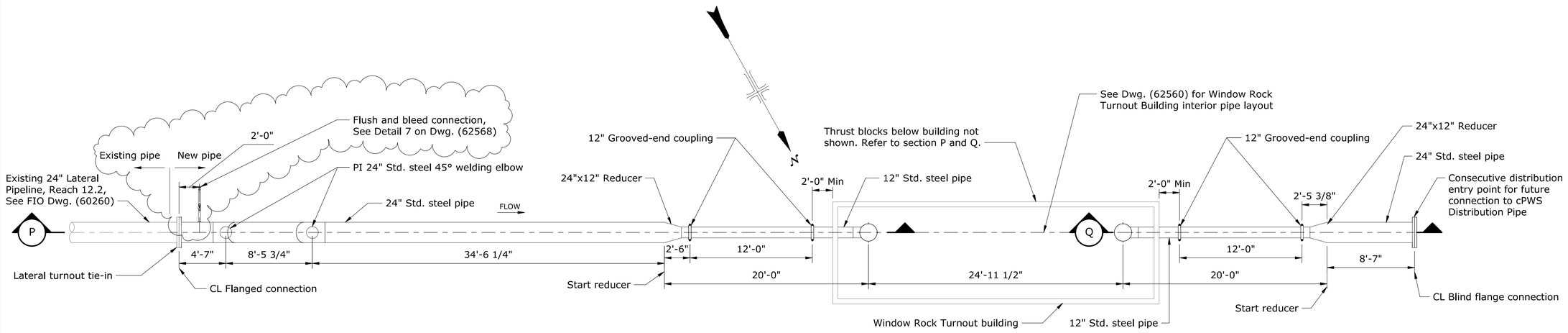

WINDOW ROCK YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

BUREAU OF RECLAMATION

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

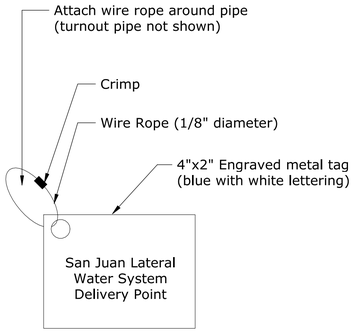
REV NO	STA.	
1	D	ADDED FLUSH AND BLEED CONNECTION

CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO 2026-06-19

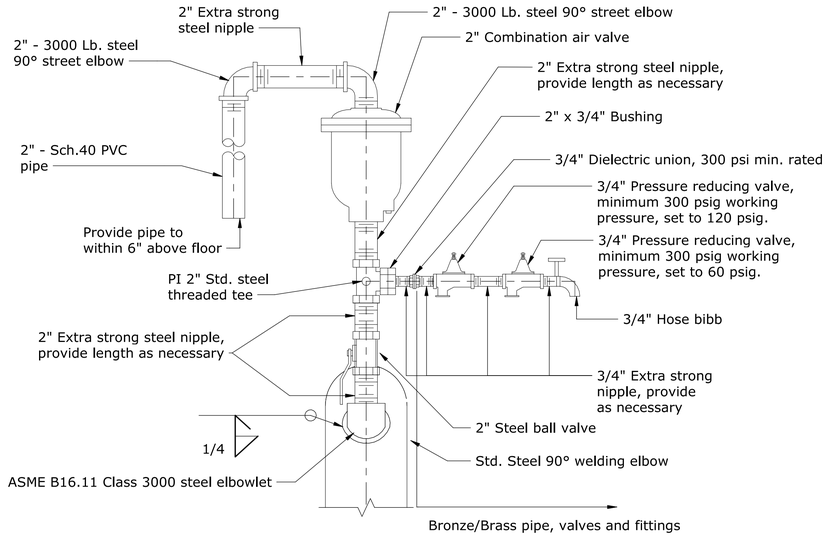
WINDOW ROCK YARD PIPE

PLAN AND SECTION

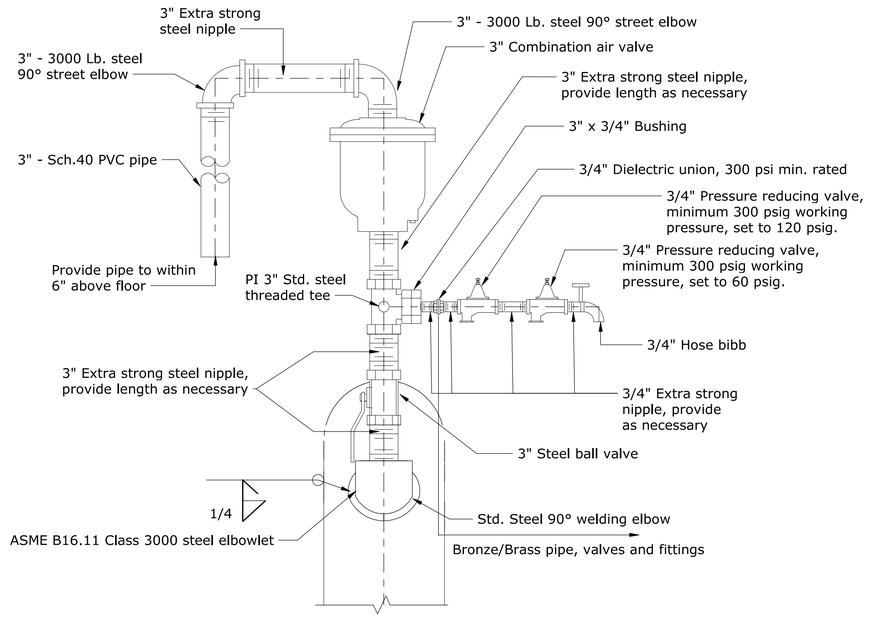
1695-D-62567



1 DELIVERY POINT SIGNAGE DETAIL
Scale: 6" = 1'-0"

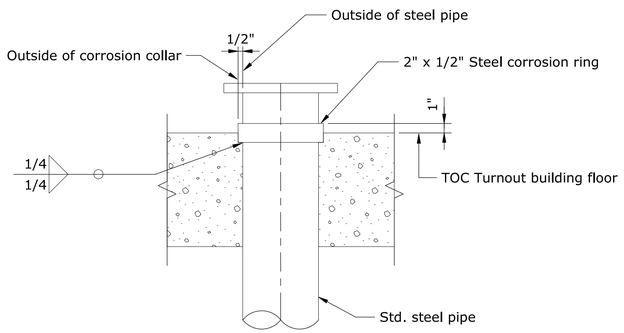


2 2-INCH COMBINATION AIR VALVE WITH SAMPLING PORT DETAIL
Scale: 1 1/2" = 1'-0"



3 3 INCH COMBINATION AIR VALVE WITH SAMPLING PORT DETAIL
Scale: 1 1/2" = 1'-0"

4 INTERIOR PIPE DRAIN DETAIL
Scale: 3" = 1'-0"



5 Corrosion Collar
Scale: 1 1/2" = 1'-0"

6 BURIED VALVE BOX DETAIL
Scale: 1" = 1'-0"

7 FLUSH AND BLEED DETAIL
Scale: 1" = 1'-0"

0 1/2' 1' 1 1/2'
Scale: 1 1/2" = 1'-0"

0 3" 6" 9"
Scale: 3" = 1'-0"

0 1" 2" 3" 4"
Scale: 6" = 1'-0"

0 1' 2'
Scale: 1" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

REV NO	STA.	2026-09-30	ADDED FLUSH AND BLEED DETAIL
1	D		

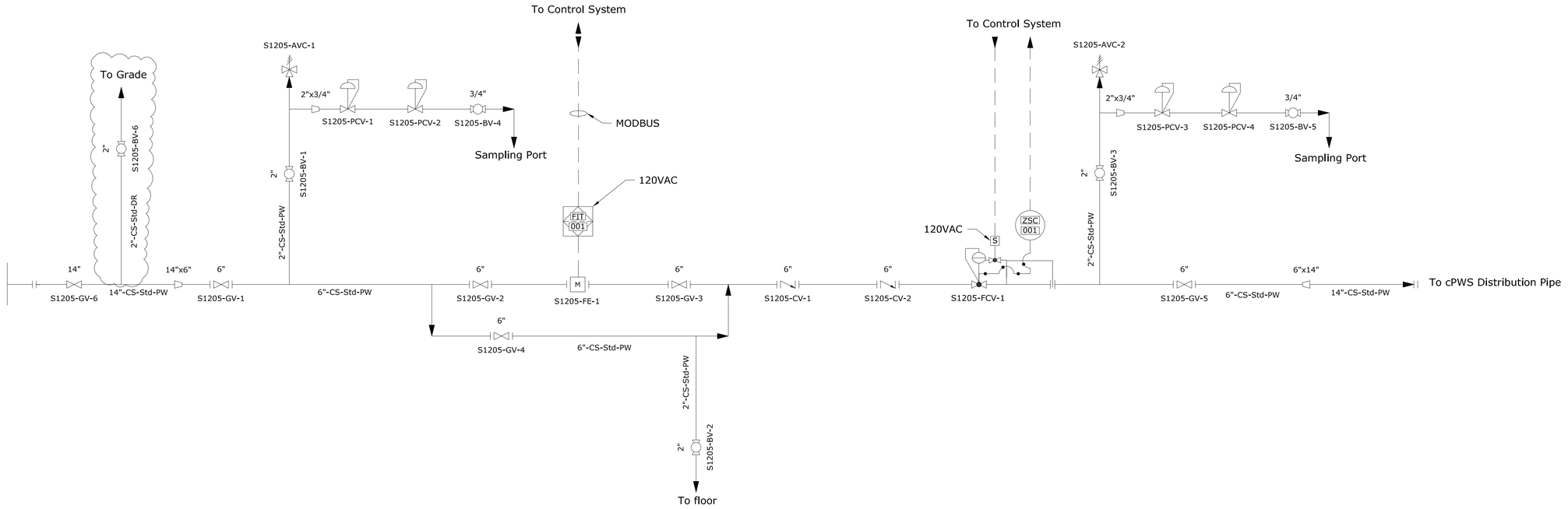
CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems

DENVER, CO 2026-06-19

DETAILS

DETAILS

1695-D-62568



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CAD SYSTEM
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CAD FILENAME
SANOSTEE TURNOUT UPDATED.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	STA.		
1	D	2026-09-30	ADDED FLUSH AND BLEED CONNECTION

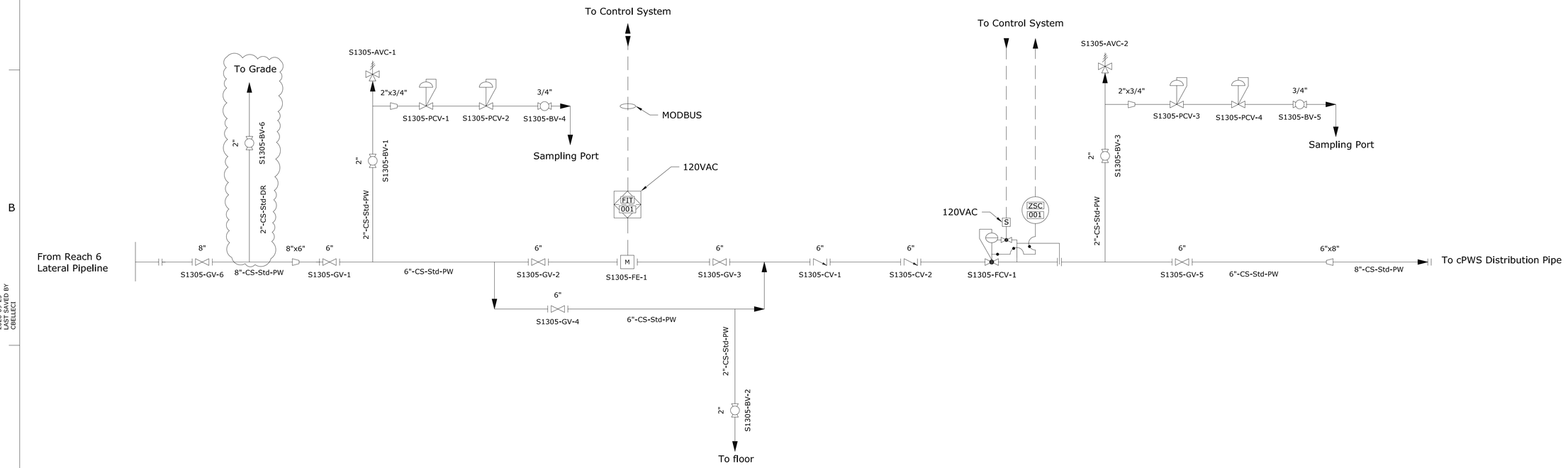
CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
DENVER, CO 2026-06-19

P&ID

SANOSTEE TURNOUT BUILDING

1695-D-62571

SHEET 1



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2026-06-05
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CAD FILENAME
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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	STA.		
1	D	2026-09-30	ADDED FLUSH AND BLEED CONNECTION

CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS

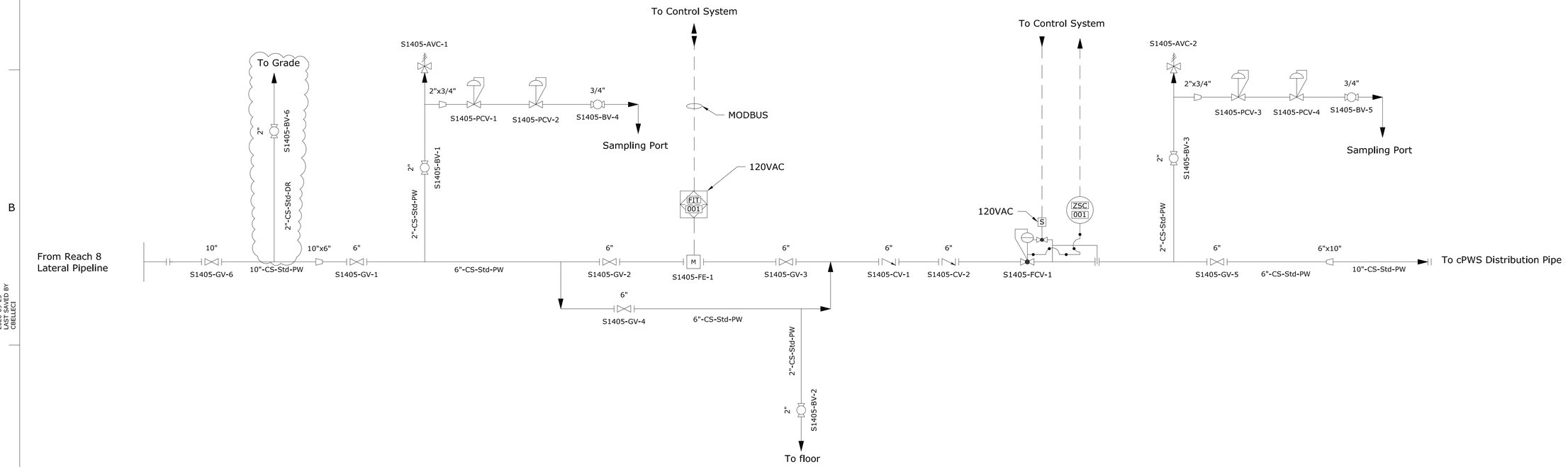
DENVER, CO 2026-06-19

P&ID

NEWCOMB TURNOUT BUILDING

1695-D-62572

SHEET 1



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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	STA.		
1	D	2026-09-30	ADDED FLUSH AND BLEED CONNECTION

CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
DENVER, CO 2026-06-19

P&ID

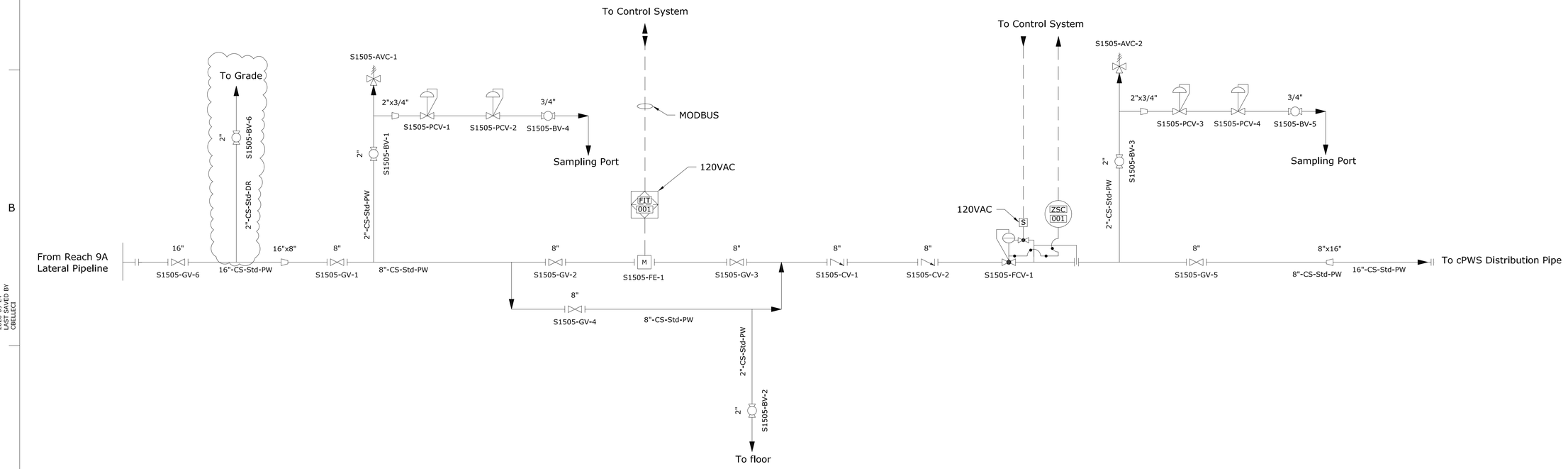
NASCHITTI TURNOUT BUILDING

1695-D-62573

SHEET 1

LAST SAVED DATE
2026-09-30
LAST SAVED BY
CBELLECI

CAD SYSTEM
AutoCAD 2025 (LMS TECH)
CAD FILENAME
NASCHITTI TURNOUT.DWG



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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	STA.		
1	D	2026-06-30	ADDED FLUSH AND BLEED CONNECTION

CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
DENVER, CO 2026-06-19

P&ID

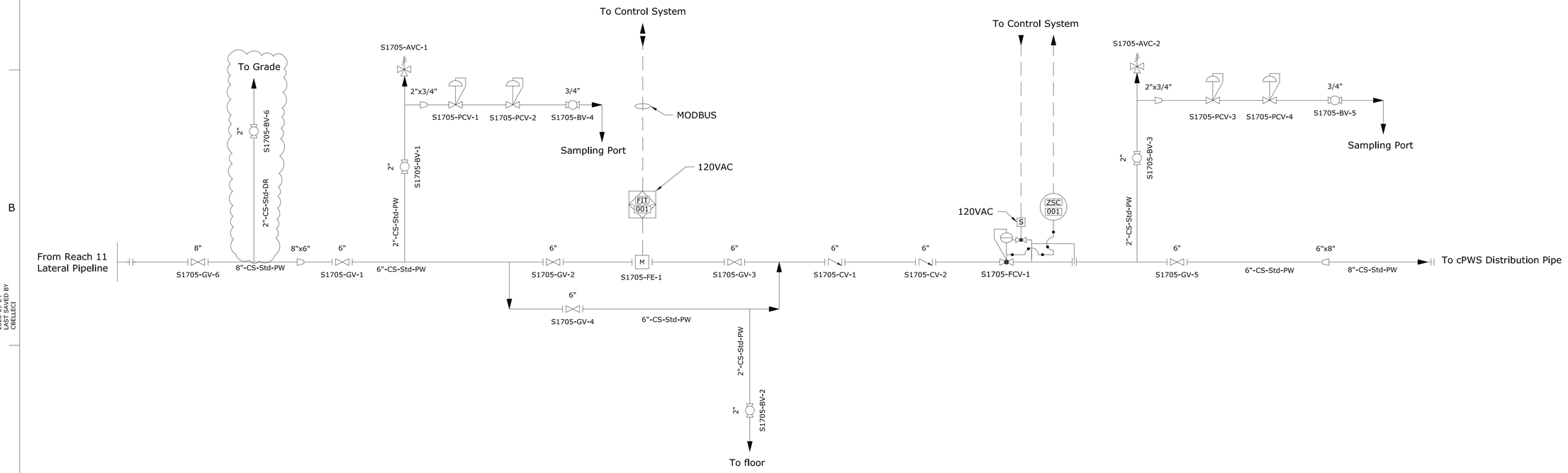
BBN9 TURNOUT BUILDING

1695-D-62574

SHEET 1

LAST SAVED DATE
2026-06-30
LAST SAVED BY
CBELLECI

CAD SYSTEM
AutoCAD 2025 (LMS TECH)
CAD FILENAME
BBN9 TURNOUT.DWG



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BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	STA.		
1	D	2026-06-30	ADDED FLUSH AND BLEED CONNECTION

CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
DENVER, CO 2026-06-19

P&ID

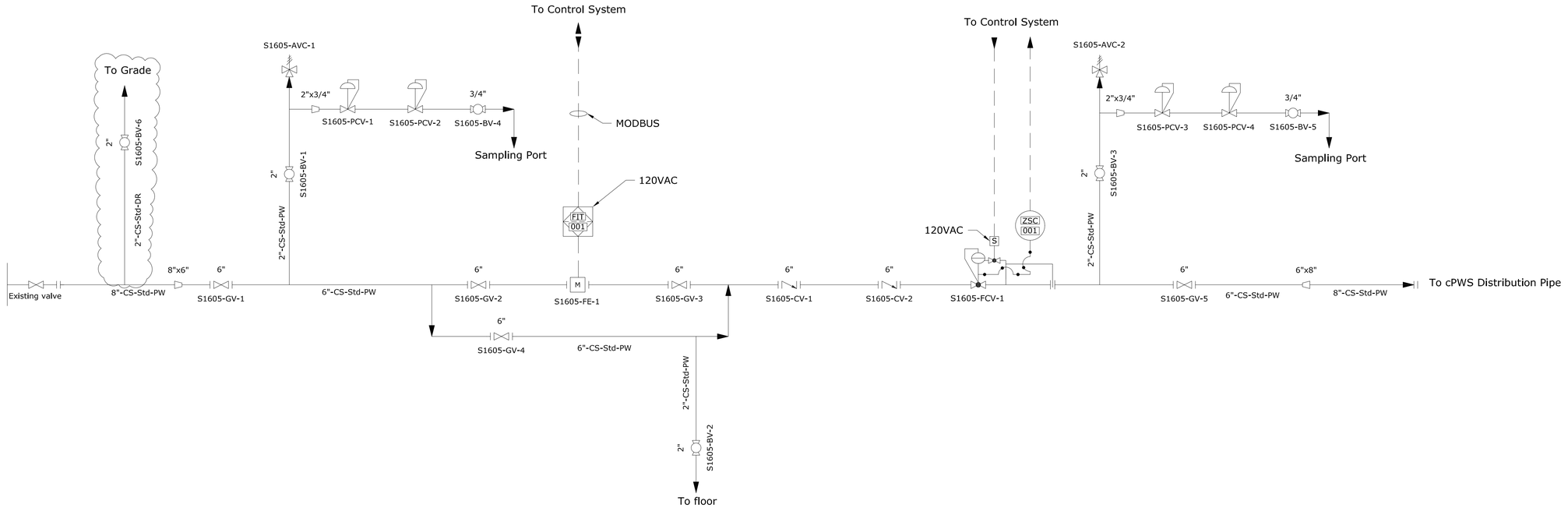
COYOTE CANYON
TURNOUT BUILDING

1695-D-62575

SHEET 1

LAST SAVED DATE
2026-06-30
LAST SAVED BY
CBELLECI

CAD SYSTEM
AutoCAD 2025 (LMS TECH)
CAD FILENAME
COYOTE CANYON TURNOUT.DWG



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CBELLECI

CAD SYSTEM
AutoCAD 2025 (LMS TECH)
CAD FILENAME
ROCK SPRINGS TURNOUT.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	STA.	
1	D	2026-09-30 ADDED FLUSH AND BLEED CONNECTION

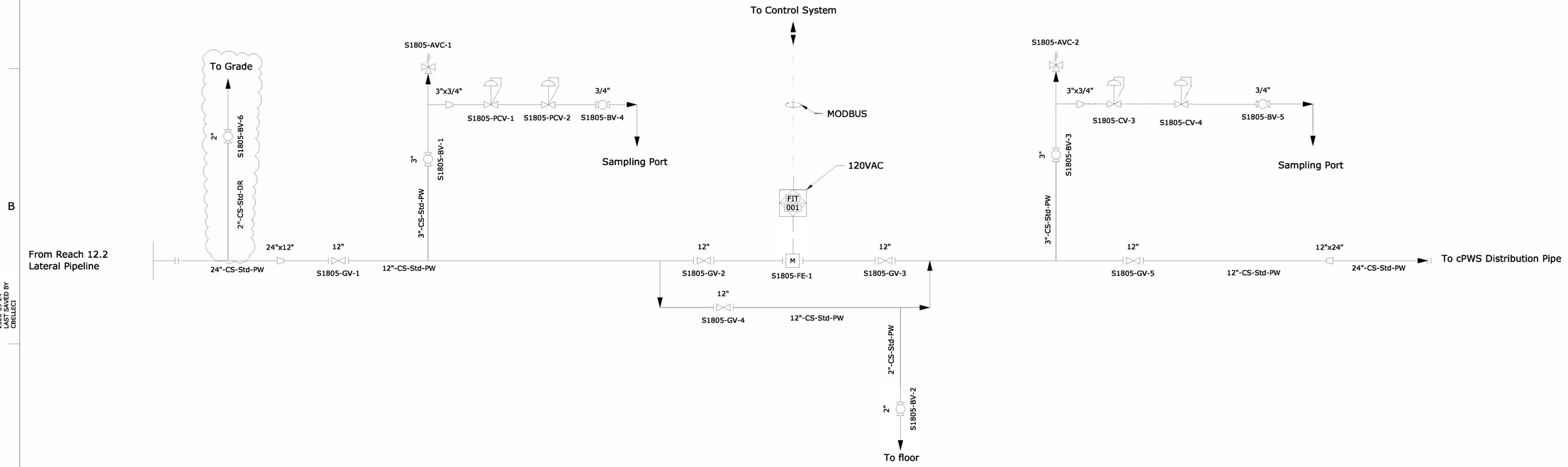
CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
DENVER, CO 2026-06-19

P&ID

ROCK SPRINGS
TURNOUT BUILDING

1695-D-62576

SHEET 1



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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	STA.		
1	D	2026-09-30	ADDED FLUSH AND BLEED CONNECTION

CHRISTOPHER BELLECI
DESIGNED
CHRISTOPHER BELLECI
DRAWN
KYLE TAYLOR
CHECKED
JEREMY HAMMOND, P.E.
TECH. APPR. JEREMY HAMMOND, P.E.
CODY YOUNG, P.E.
APPROVED CODY YOUNG, P.E.
PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
DENVER, CO 2026-06-19

P&ID

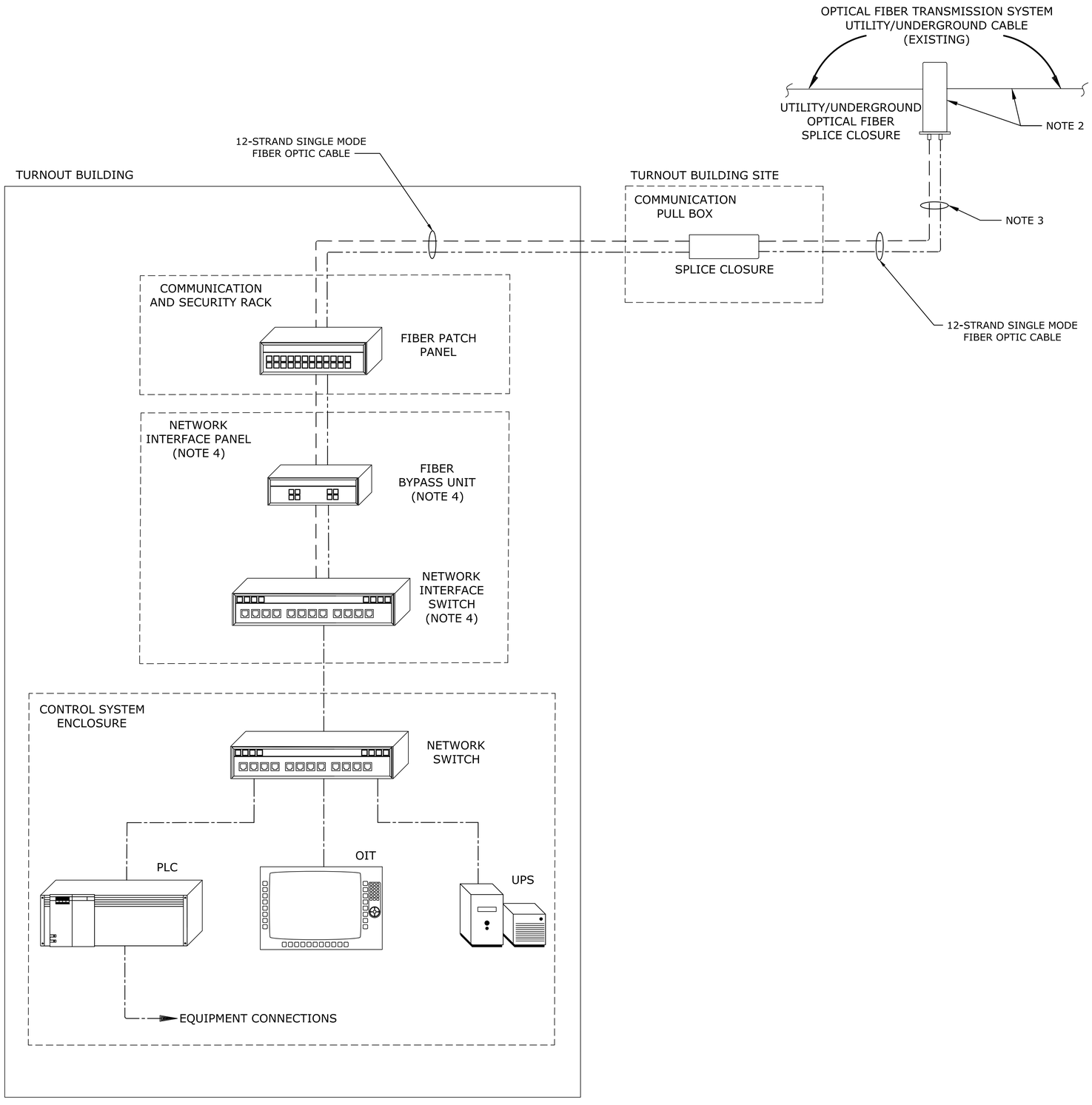
WINDOW ROCK
TURNOUT BUILDING

1695-D-62577

SHEET 1

LAST SAVED DATE
2026-09-30
LAST SAVED BY
CBELLECI

CAD SYSTEM
AutoCAD 2024
CAD FILENAME
WINDOW ROCK TURNOUT.DWG



NOTES

1. NETWORK OVERVIEW IS TYPICAL FOR ALL TURNOUT BUILDING SITES EXCEPT AS NOTED.
2. UTILITY/UNDERGROUND OPTICAL FIBER CABLE AND UTILITY/UNDERGROUND SPLICE ENCLOSURE PROVIDED BY OTHERS.
3. OPTICAL FIBER UTILITY CONNECTIONS FOR EACH OF THE FOLLOWING SITES ARE AS FOLLOWS:
   - 3.1. SANOSTEE, NASCHITTI, NEWCOMB, BBN9, COYOTE CANYON: CONNECT TO UTILITY FIBER OPTIC TRANSMISSION SYSTEM INTERFACE AT THE UNDERGROUND CONTRACTOR SUPPLIED COMMUNICATION PULL BOX LOCATED CLOSEST TO UTILITY POLE.
   - 3.2. ROCK SPRINGS AND WINDOW ROCK: CONNECT TO EXISTING UNDERGROUND FIBER OPTIC TRANSMISSION SYSTEM INTERFACE AT THE SPLICE CLOSURE IN THE UNDERGROUND FIBER OPTIC PULL BOX LOCATED AT THE SITE.
4. NETWORK INTERFACE PANEL TO BE PROVIDED BY OTHERS. CONTRACTOR TO COORDINATE WITH GOVERNMENT TO FACILITATE INSTALLATION AND INTEGRATION.

ABBREVIATIONS

OIT	OPERATOR INTERFACE TERMINAL
PLC	PROGRAMMABLE LOGIC CONTROLLER
SM	SINGLE MODE (FIBER OPTIC CABLE)
UPS	UNINTERRUPTIBLE POWER SUPPLY

CABLE LEGEND

- CAT 6 ETHERNET CABLE
- SM OPTICAL FIBER CABLE - PRIMARY PAIR
- SM OPTICAL FIBER CABLE - BACKUP PAIR

REFERENCE DRAWINGS

SINGLE LINE DIAGRAM - TYPICAL	1695-D-62578
ELECTRICAL SITE PLAN - SANOSTEE	1695-D-62579
ELECTRICAL SITE PLAN - NASCHITTI	1695-D-62581
ELECTRICAL SITE PLAN - NEWCOMB	1695-D-62580
ELECTRICAL SITE PLAN - BBN9	1695-D-62582
ELECTRICAL SITE PLAN - COYOTE CANYON	1695-D-62583
ELECTRICAL SITE PLAN - ROCK SPRINGS	1695-D-62584
ELECTRICAL SITE PLAN - WINDOW ROCK	1695-D-62585

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1695, NAVAJO-GALLUP WATER SUPPLY PROJECT, NM

SAN JUAN LATERAL

TURNOUT BUILDINGS

REV NO	DATE	BY	DESCRIPTION
1	2026-09-30	L. Gentry	D UPDATED NETWORK INTERFACE PANEL NOTE AND REMOVED GFE ABBREVIATION.

Christian Underwood — DESIGNED
Christian Underwood — DRAWN
Lisa Gentry — CHECKED
Lisa Gentry, P.E. — TECH. APPR.
Sean Kyer, F.P.E. — APPROVED ADMIN APPROVAL
DENVER, CO 2026-05-29

NETWORK OVERVIEW

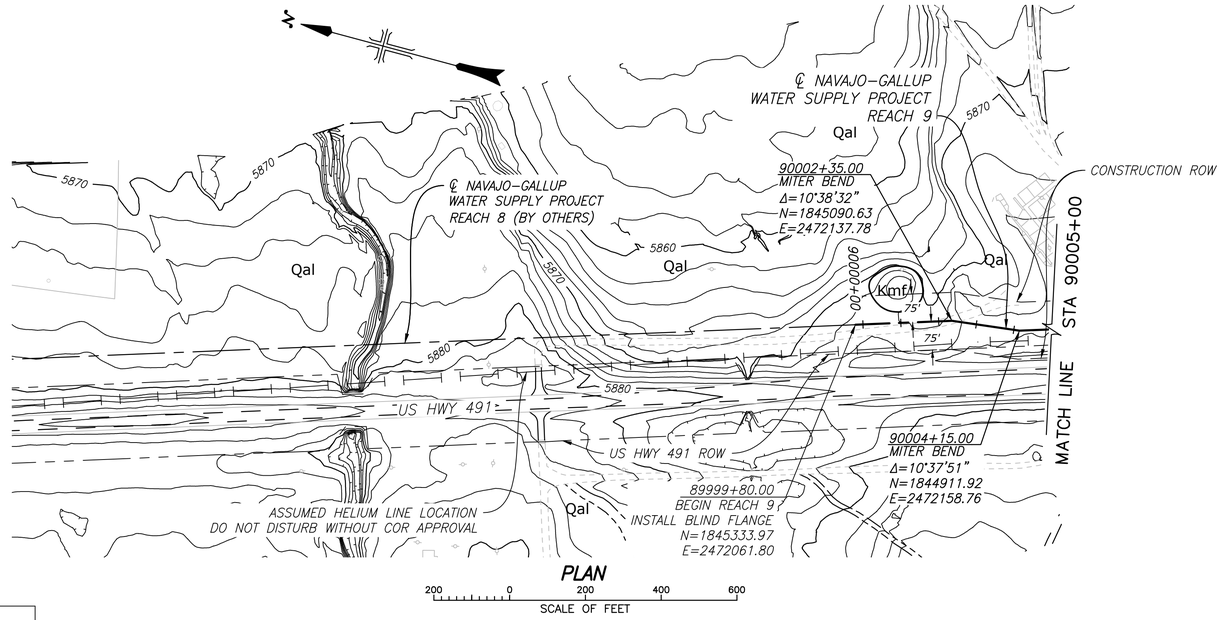
TURNOUT BUILDING

1695-D-62588

SHEET 1

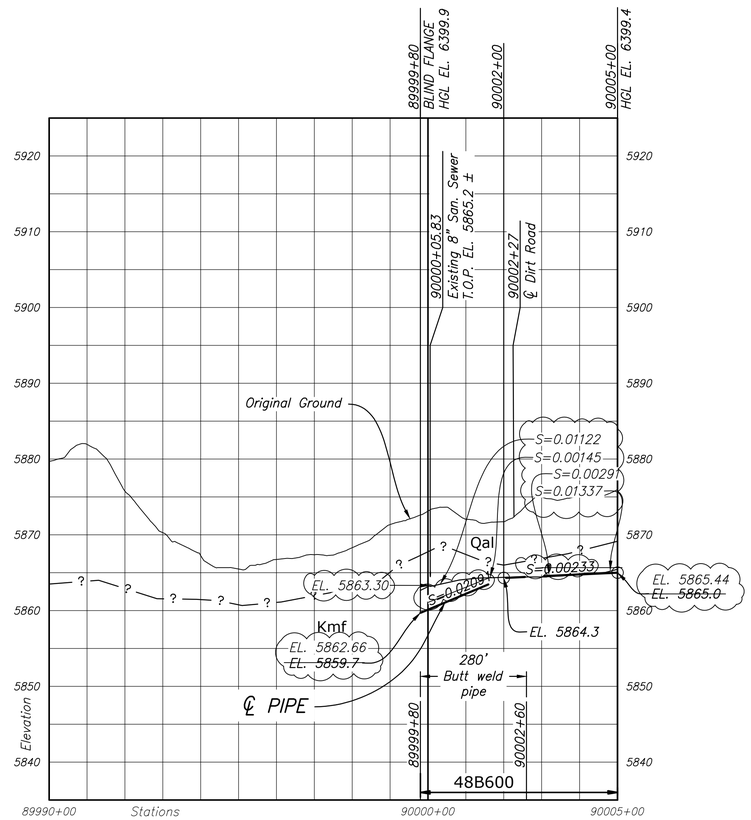
LAST SAVED DATE
2026-05-29
LAST SAVED BY
CUNDERWOOD

CAD SYSTEM
AUTOCAD LDS (LMS TECH)
CAD FILENAME
1695-D-62588 REV 1.DWG



PLAN

OPEN–CUT CROSSINGS		
Station	Road / Wash	% Compaction
90002+02 TO 90002+52	Maintained Road	95%



PROFILE

GENERAL NOTES

1. General Notes, Symbols, Abbreviations, and Pipe Classification apply to all plan and profile drawings.
2. Contractor shall determine the voltage at all power lines and share this information with the COR.
3. At angle points that do not require miter bends, long radius curves (vertical and horizontal) shall be constructed by pulling joints.
4. Stations are approximate and may be varied as directed.
5. Minimum earth cover over the top of the pipe shall be 3.5 feet unless the line pipe is at an existing wash, or at a roadway, and then the minimum cover is 5 feet, unless otherwise noted.
6. All joints are to be restrained under wash crossings between the stations identified in the tables on the plan and profiles.
7. Where the line pipe crosses an existing pipe, a minimum clear distance of 2 feet outside to outside between line pipe and existing pipe shall be provided unless otherwise directed. The backfill between the crossing pipelines shall be controlled low strength material (CLSM) as stated in the specifications.
8. Hydraulic grade line elevations (H.G.L. El.) shown are for the design Q of 47.7 cfs at steady state for Reach 9 and Reach 10, and 40.4 cfs for Reach 11.
9. Unknown elevations and diameters of existing utilities shall be determined by the contractor and reported to the COR in time to allow adjustment of grades and design if required before contractor submits pipe, fittings, and miter bends for approval.
10. For limits on excavation near gas lines, see specifications.
11. Grid based on New Mexico State Coordinate System West Zone NAD 83. Vertical datum based on NAVD 88.
12. Reach 9 Ground to grid conversion factor is 0.999691831997. Reach 10 Ground to grid conversion factor is 0.999703693004. Reach 11 Ground to grid conversion factor is 0.999703150995.
13. Metallic pipe is required, and joints shall be restrained when on slopes greater than 0.20. See (60333) for additional steep pipe details.
14. Open cut roads require CLSM and compacted backfill per specifications. Maintained roads require gravel surfacing per specifications. See (60329) for details.
15. CAUTION: Call New Mexico One Call 811 before you dig.

GEOLOGIC NOTE

For General Geologic Legend Explanation, and Notes, see 1695-529-60076.

ABBREVIATIONS

CR - County Road
HGL - Hydraulic Grade Line
NTUA - Navajo Tribal Utility Authority
OH - Over Head
PNM - Public Service Company of New Mexico
ROW - Right-of-Way
RTE - Route
TP - Test Pit

PIPE CLASSIFICATIONS

The line pipe is designated as 48B600, 42C350 etc. The 48 and 42 indicate the nominal pipe diameter in inches. The letters B, C, and D denote cover class of 10, 15, and 20 feet respectively, of design earth cover over the top of the pipe. The letter K denotes steep pipe sections. The number 600, 350, etc. denotes the maximum allowable hydrostatic pressure head, in feet, measured at the centerline of the pipe.

SYMBOLS

- Air Valve
- Blowoff
- ℄ Line Pipe
- Chain Link Fence
- Construction ROW
- Dirt Road
- Field Fence
- Geologic Unit Contact
- Geologic Unit Approx Contact
- Geologic Unit Queried Where Inferred
- Helium Line(abandoned)
- Manhole
- Manhole with Air Valve
- Manhole with Blowoff
- Natural Gas Pipeline
- Paved Road
- Powerline
- Sectionalizing Valve
- Sewerline
- Telephone
- US Hwy 491 ROW
- Waterline

NOTE

1. For details at 90000+00, see (60325).

GEOLOGY

TECH. APPR. JUSTIN GILBERT
SUBMITTED PAUL GARDNER
JUSTIN GILBERT Supervisory Geologist
APPROVED

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - REACH 9

LINE PIPE

REV.NO	DATE	DESCRIPTION	BY
D	Asbuilts:		
6	2023-08-04	Revised profile and removed sectionalizing valve at 90000+11.25 due to existing sanitary sewer.	Gladden, Ryan L P.E.
5	2019-02-11	Revised Data Reference for alignment to account for wash crossing verifications.	Bowles, Linda M P.E.
4	2018-11-27	Added horizontal and vertical datum information to Note 11.	Marner, Andrew (Andy) J P.E.
3	2018-08-15	Moved sectionalizing valve to station 90000+11.25.	Bowles, Linda M P.E.
2	2017-12-06	Revised Note 13 to require metalic pipe on slopes greater than 0.20.	Bowles, Linda M P.E.
1	2017-06-07		Bowles, Linda M P.E.

Andrew Marner
DESIGNED
Andrew Marner
DRAWN
D.L. Gesundheit
CHECKED
Bowles, Linda M P.E.
TECH. APPR.
Duke, Wylie C (Chris) P.E.
APPROVED Manager - Water Conveyance Group
PEER REVIEWER

DENVER, CO 2017-03-30

PLAN AND PLOT

BEGINNING TO STA 90005+00

1695-D-60281

SHEET 1 OF 26

Released

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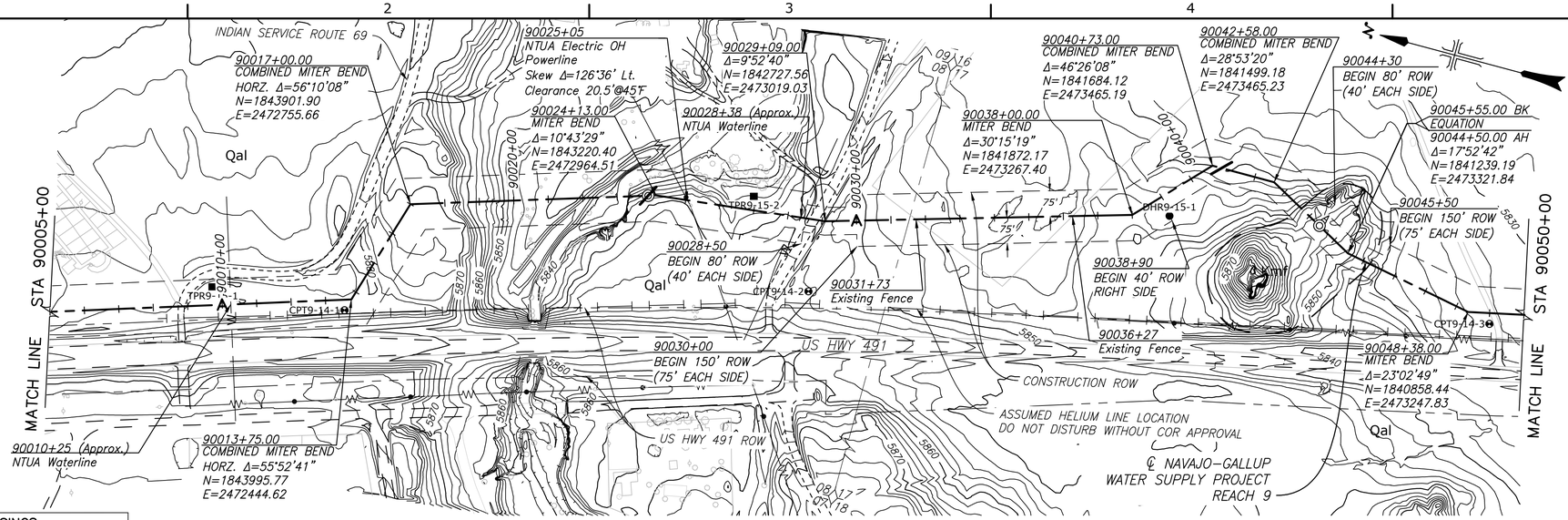
CAD FILENAME
1695-D-60281-RD-1 OF 26.DWG

Released



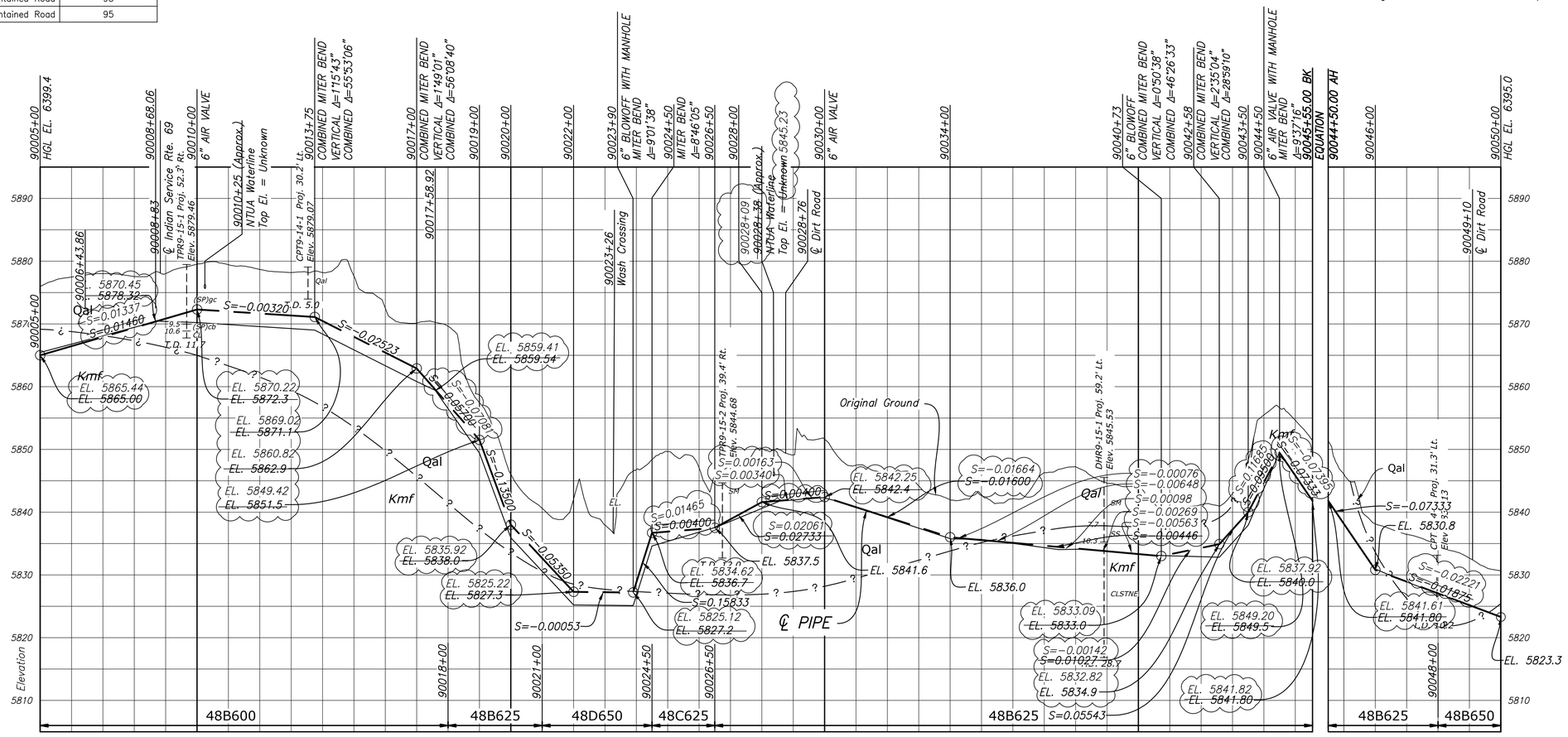
PLAN

SCALE OF FEET: 200 0 200 400 600

OPEN-CUT CROSSINGS		
Station	Road / Wash	% Compaction
90008+73 TO 90008+93	Maintained Road	95
90020+60 TO 90024+02	Wash	85
90028+52 TO 90029+00	Maintained Road	95
90048+90 TO 90049+25	Maintained Road	95

NOTES

1. For General Notes, Symbols, Abbreviations, and Pipe Classification, see (60281).
2. For wash crossing details at 90023+26, see (60337).



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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - REACH 9

LINE PIPE

REV NO	DATE	BY	DESCRIPTION
3	2023-09-04	Gladden, Ryan L P.E.	As-builts.
2	2017-12-07	Bowles, Linda M P.E.	Added Existing NTUA Waterline crossing at Sta. 90018+25 and Sta. 90028+38.
1	2017-05-22	Marner, Andrew (Andy) J P.E.	Added Construction ROW width labels.

Andrew Marner — DESIGNED
Andrew Marner — DRAWN
D. L. Gesundheit — CHECKED
Bowles, Linda M P.E. — TECH. APPR.
Duke, Wylie C (Chris) P.E. — APPROVED
PEER REVIEWER Manager - Water Conveyance Group

DENVER, CO 2017-03-30

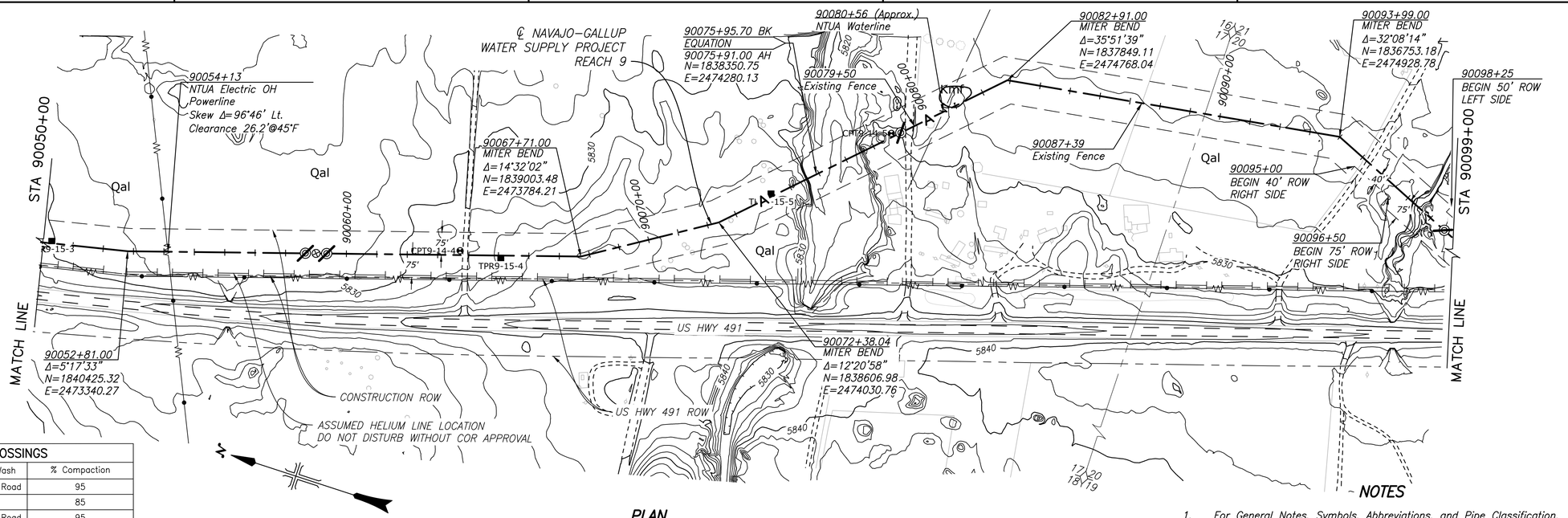
GEOLOGY	
TECH. APPR.	JUSTIN GILBERT
SUBMITTED	PAUL GARDNER
APPROVED	JUSTIN GILBERT Supervisory Geologist

PLAN AND PROFILE

STA. 90005+00
TO
STA. 90050+00

1695-D-60282

SHEET 2 OF 26

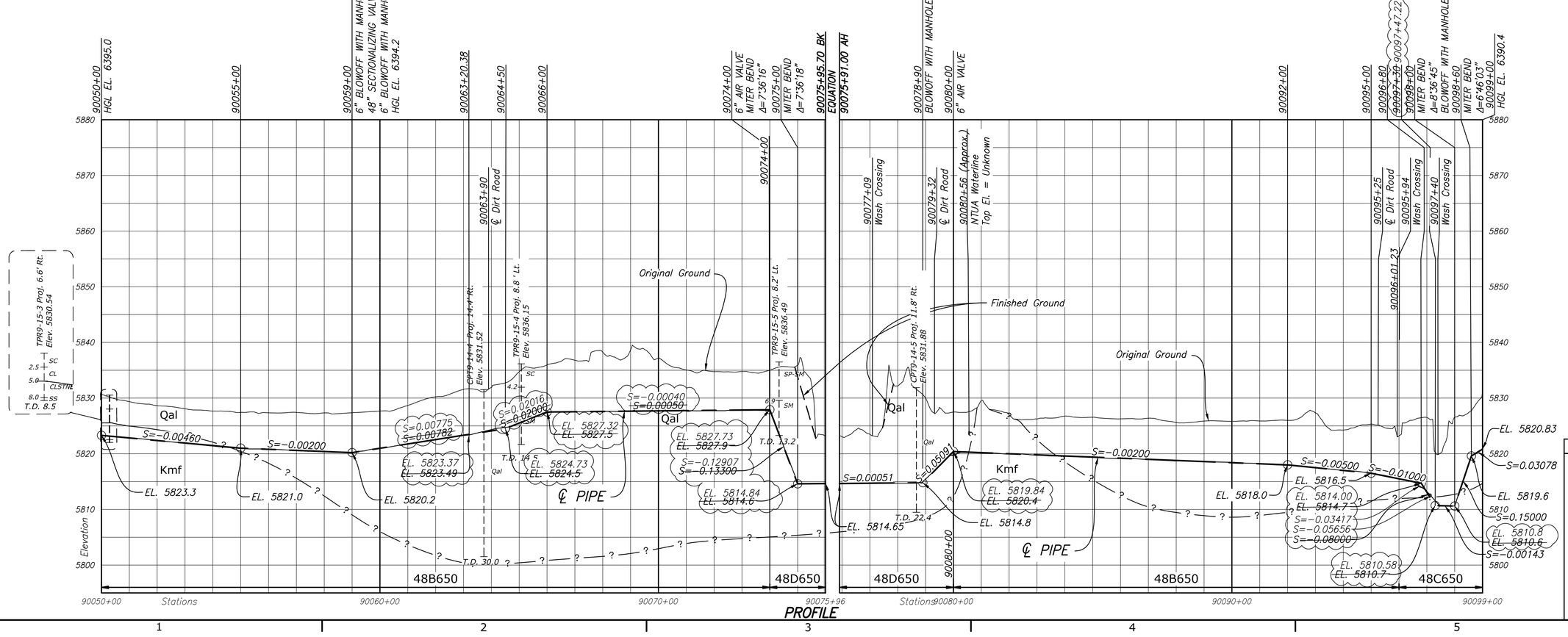


OPEN CUT CROSSINGS		
Station	Road / Wash	% Compaction
90063+70 TO 90064+10	Maintained Road	95
90074+50 TO 90078+00	Wash	85
90079+10 TO 90079+55	Maintained Road	95
90095+10 TO 90095+40	Maintained Road	95
90095+50 TO 90098+00	Wash	85

PLAN

SCALE OF FEET 200 0 200 400 600

1. For General Notes, Symbols, Abbreviations, and Pipe Classification, see (60281).
2. For wash crossing details at 90077+09, see (60338).
3. For wash crossing details at 90097+40, see (60339).



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CAD SYSTEM AutoCAD Civil 3D (LMS TECH)
CAD FILENAME 1695-D-60283-RD-3 OF 26.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - REACH 9

LINE PIPE

REV NO		
3	D	As-builts.
2	D	Added Existing NTUA Waterline crossing at Sta. 90080+56
1	D	Added Construction ROW width labels.

2023-08-04	Gladden, Ryan L P.E.
2017-12-06	Bowles, Linda M P.E.
2017-05-22	Marner, Andrew (Andy) P.E.

Andrew Marner DESIGNED
Andrew Marner DRAWN
D. L. Gesundheit CHECKED
Bowles, Linda M P.E. TECH. APPR.
Duke, Wylie C (Chris) P.E. APPROVED Manager - Water Conveyance Group PEER REVIEWER

DENVER, CO 2017-03-30

GEOLOGY

TECH. APPR.	JUSTIN GILBERT
SUBMITTED	PAUL GARDNER
APPROVED	JUSTIN GILBERT Supervisory Geologist

PLAN AND PLOT

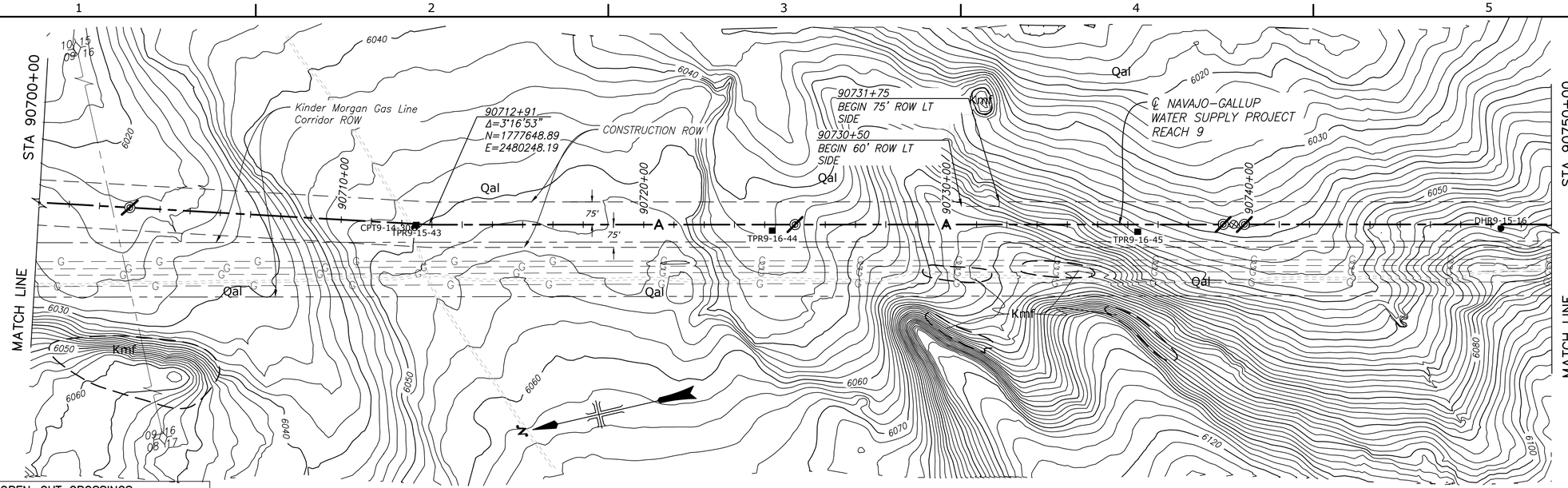
STA. 90050+00
TO
STA. 90099+00

1695-D-60283

SHEET 3 OF 26

Released

Released



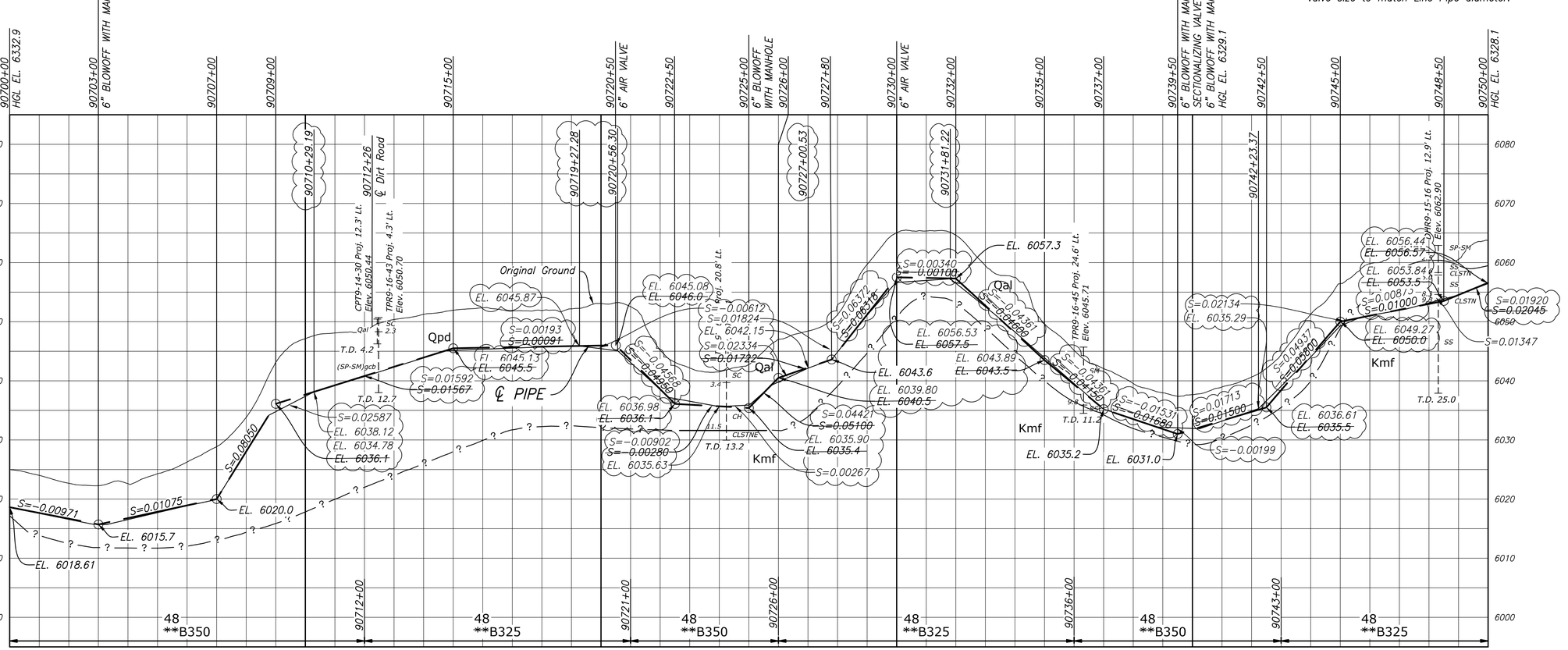
PLAN

SCALE OF FEET 200 0 200 400 600

OPEN–CUT CROSSINGS		
Station	Road / Wash	% Compaction
90712+12 TO 90712+40	Unmaintained Road	95

NOTES

1. For General Notes, Symbols, Abbreviations, and Pipe Classification, see (60281).
2. Diameters with ** designation are dependent on pipe material type, see Line Pipe Material and Diameter Table (60306). Sectionalizing valve size to match Line Pipe diameter.



PROFILE

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1695-D-60296-RD-16 OF 26.DWG

BUREAU OF RECLAMATION

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - REACH 9

LINE PIPE

REV NO		
1	D	As-builts: 2023-08-04 Gladden, Ryan L P.E.

DESIGNED	Andrew Marner
DRAWN	Andrew Marner
CHECKED	D. L. Gesundheit
TECH. APPR.	Bowles, Linda M P.E.
APPROVED PEER REVIEWER	Duke, Wylie C (Chris) P.E. Manager - Water Conveyance Group
	DENVER, CO 2017-03-30

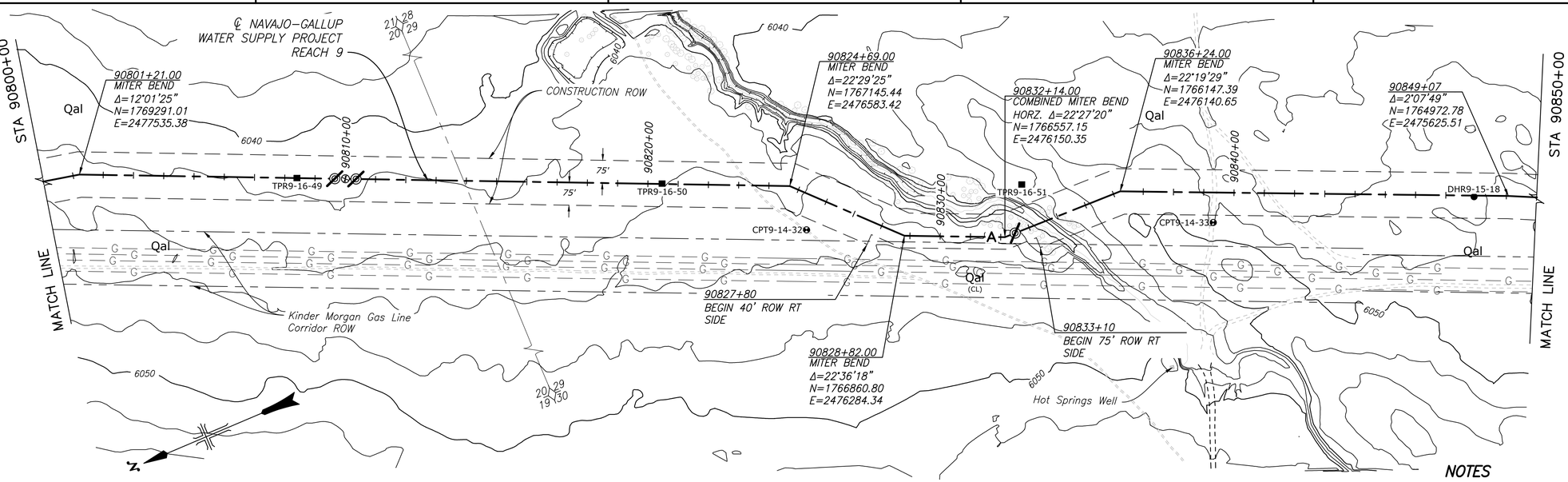
GEOLOGY	
TECH. APPR.	JUSTIN GILBERT
SUBMITTED	PAUL GARDNER
APPROVED	JUSTIN GILBERT Supervisory Geologist

PLAN AND PROFILE

STA. 90700+00
TO
STA. 90750+00

1695-D-60296

SHEET 16 OF 26



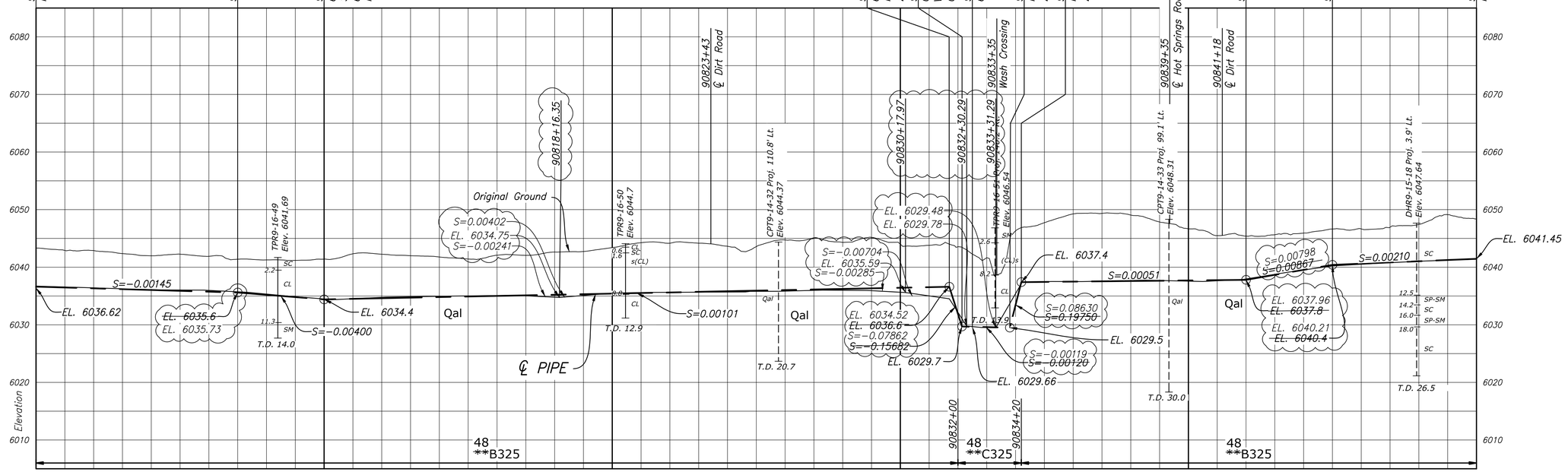
PLAN

200 0 200 400 600
SCALE OF FEET

OPEN–CUT CROSSINGS		
Station	Road / Wash	% Compaction
90823+30 TO 90823+55	Unmaintained Road	95
90832+25 TO 90833+80	Wash	85
90839+20 TO 90839+50	Maintained Road	95
90841+05 TO 90841+30	Unmaintained Road	95

NOTES

1. For General Notes, Symbols, Abbreviations, and Pipe Classification, see (60281).
2. For wash crossing details at 90833+35, see (60355). Base flow is anticipated at this location, contractor to submit dewatering plan to COR.
3. Diameters with ** designation are dependent on pipe material type, see Line Pipe Material and Diameter Table (60306). Sectionalizing valve size to match Line Pipe diameter.



PROFILE

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CAD FILENAME 1695-D-60298-RD-18 OF 26.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - REACH 9

LINE PIPE

REV NO		
1	D	As-builts.
	2023-08-04	Gladden, Ryan L P.E.

DESIGNED	Andrew Marner
DRAWN	Andrew Marner
CHECKED	D. L. Gesundheit
TECH. APPR.	Bowles, Linda M P.E.
APPROVED	Duke, Wylie C (Chris) P.E. Manager - Water Conveyance Group
PEER REVIEWER	
DENVER, CO	2017-03-30

GEOLOGY	
TECH. APPR.	JUSTIN GILBERT
SUBMITTED	PAUL GARDNER
APPROVED	JUSTIN GILBERT Supervisory Geologist

PLAN AND PROFILE

STA. 90800+00
TO
STA. 90850+00

1695-D-60298

SHEET 18 OF 26

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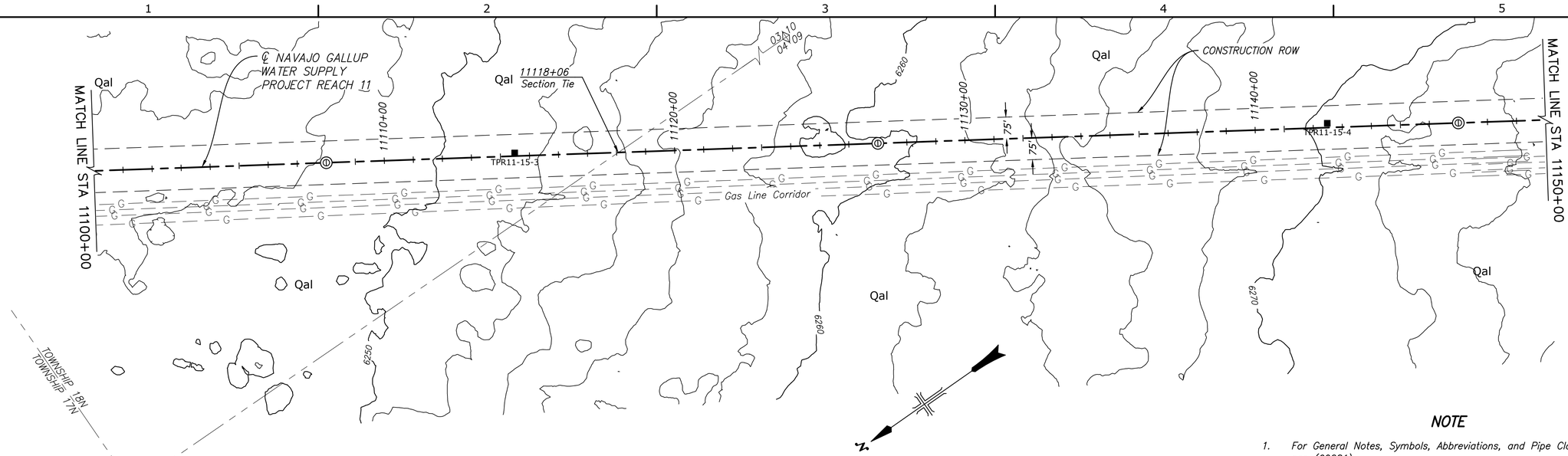
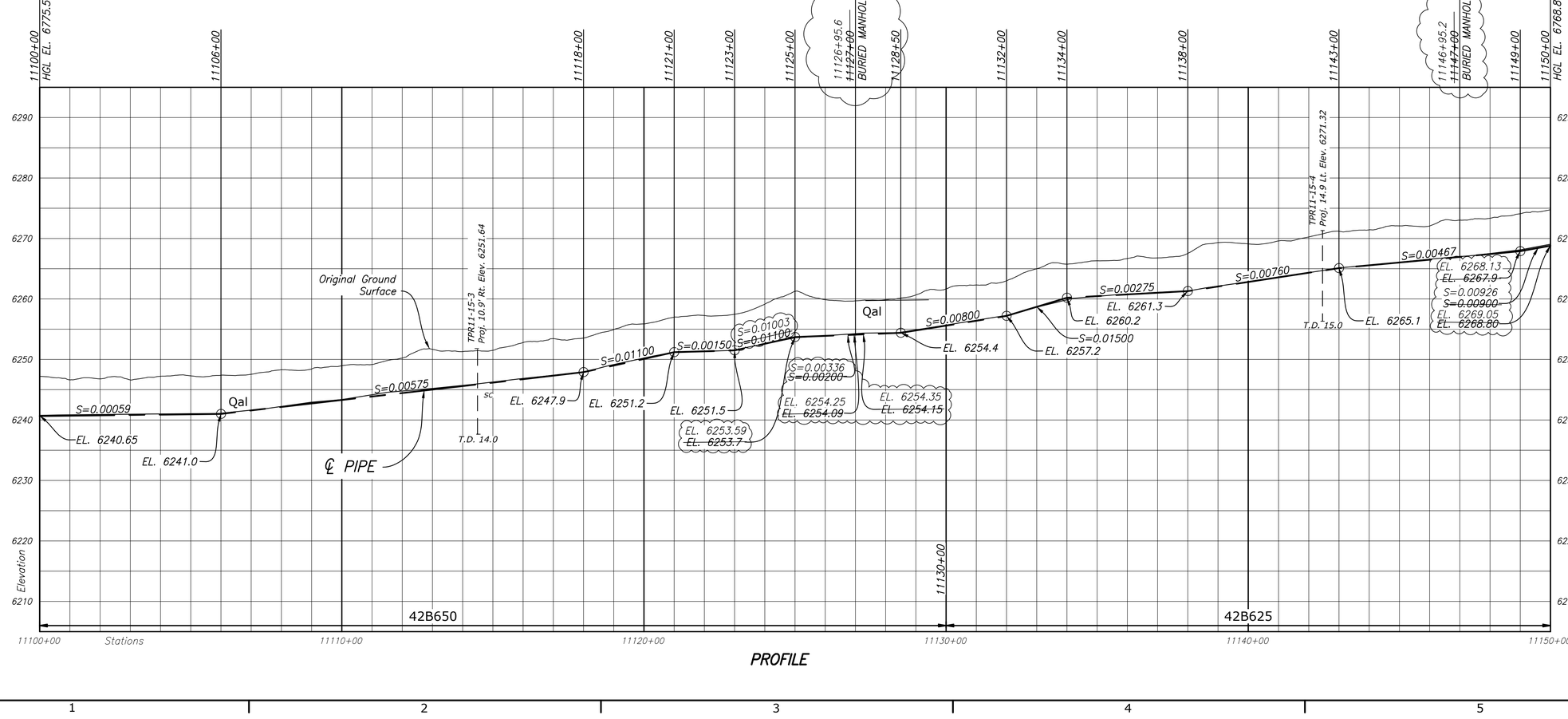


NOTE

1. For General Notes, Symbols, Abbreviations, and Pipe Classification, see (60281).

PLAN

SCALE OF FEET: 200 0 200 400 600



PROFILE

GEOLOGY	
TECH. APPR.	JUSTIN GILBERT
SUBMITTED	PAUL GARDNER
APPROVED	JUSTIN GILBERT Supervisory Geologist

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - REACH 11

LINE PIPE

REVNO		
1	D	As-builts
	2023-08-04	Gladden, Ryan L P.E.

Andrew Marner	DESIGNED
Andrew Marner	DRAWN
D. L. Gesundheit	CHECKED
Bowles, Linda M P.E.	TECH. APPR.
Duke, Wylie C (Chis) P.E.	APPROVED Manager - Water Conveyance Group ADMIN APPROVAL
DENVER, CO	2017-03-30

PLAN AND PROFILE

STA. 11100+00
TO
STA. 11150+00

1695-D-60310

SHEET 2 OF 5

Released

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CAD FILENAME
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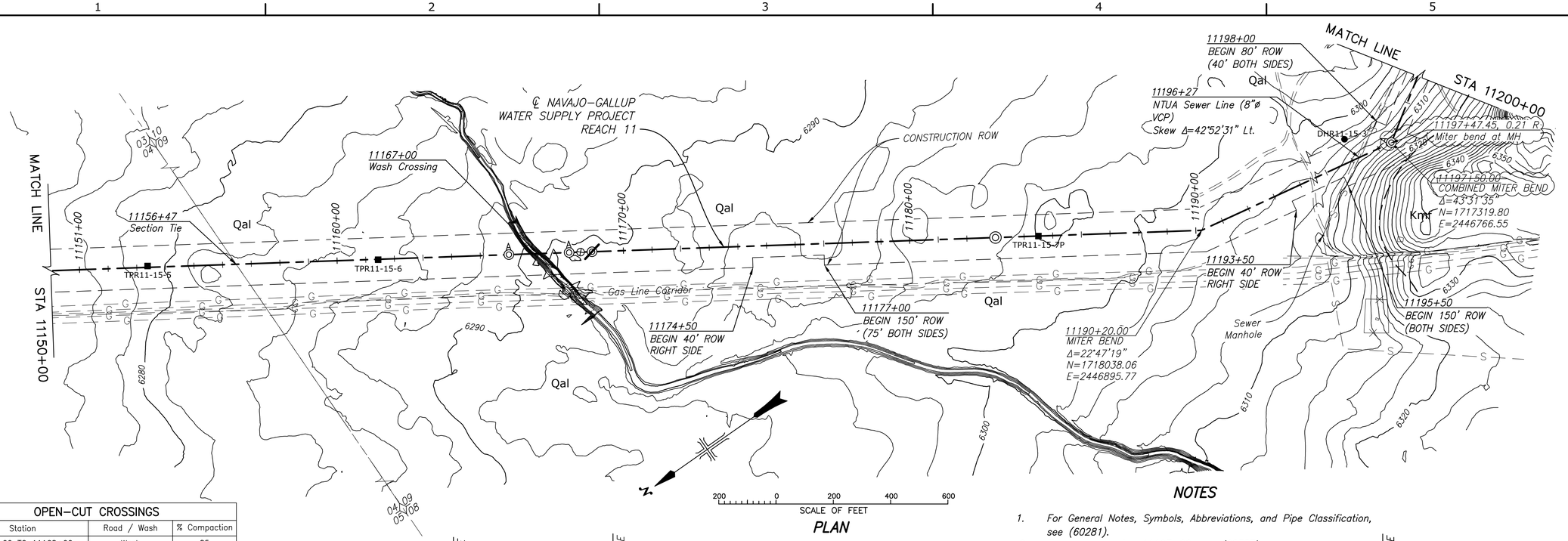
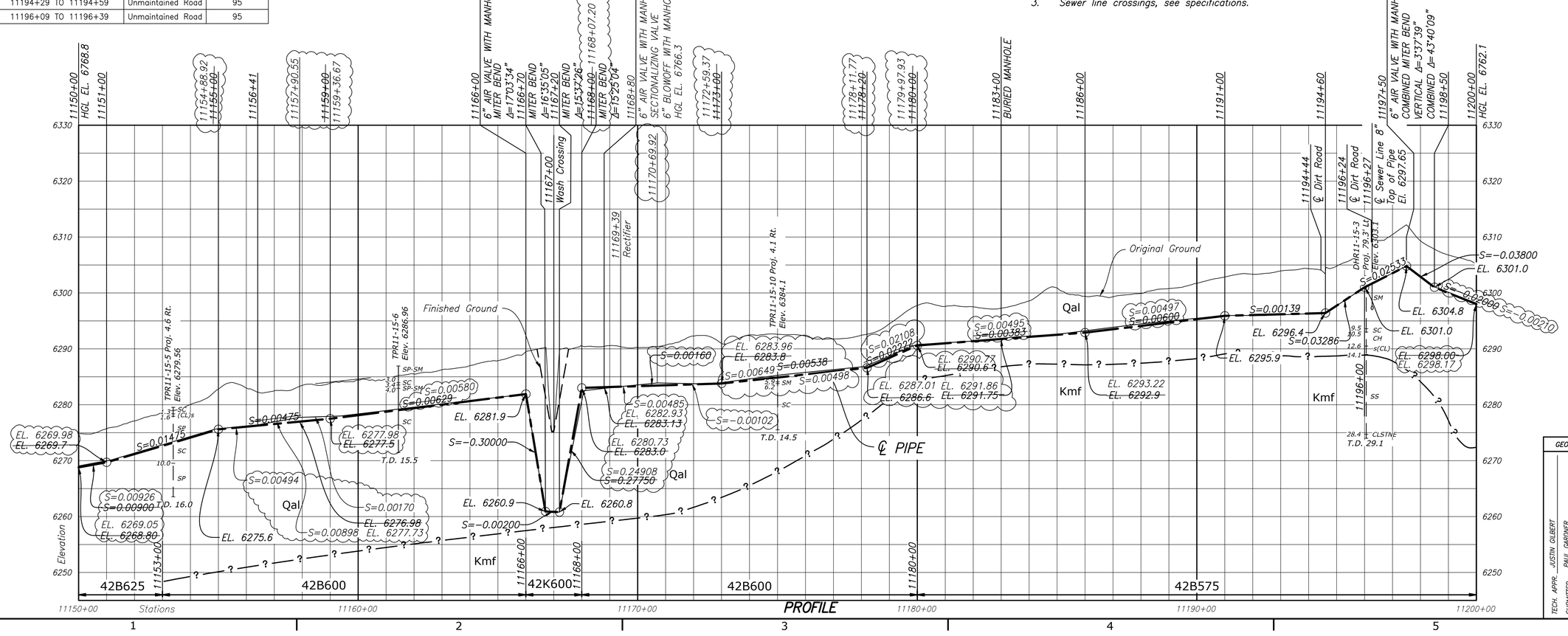


PLAN

OPEN–CUT CROSSINGS		
Station	Road / Wash	% Compaction
11166+00 TO 11168+00	Wash	85
11194+29 TO 11194+59	Unmaintained Road	95
11196+09 TO 11196+39	Unmaintained Road	95

NOTES

1. For General Notes, Symbols, Abbreviations, and Pipe Classification, see (60281).
2. For wash crossing 11167+00, see (60357).
3. Sewer line crossings, see specifications.



PROFILE

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - REACH 11

LINE PIPE

REV NO	DATE	BY	
4	2023-08-04	Gladden, Ryan L P.E.	As-builts.
3	2018-06-21	Bowles, Linda M P.E.	Increased Construction ROW width to 80' at Sta. 11196+00.
2	2018-02-23	Bowles, Linda M P.E.	Reduced Construction ROW width to 50' at Sta. 11198+75.
1	2017-05-19	Bowles, Linda M P.E.	Revised Construction ROW widths:

Andrew Marner
DESIGNED
Andrew Marner
DRAWN
D. L. Gesundheit
CHECKED
Bowles, Linda M P.E.
TECH. APPR.
Duke, Wylie C (Chris) P.E.
APPROVED Manager - Water Conveyance Group
ADMIN APPROVAL

DENVER, CO 2017-03-30

GEOLOGY

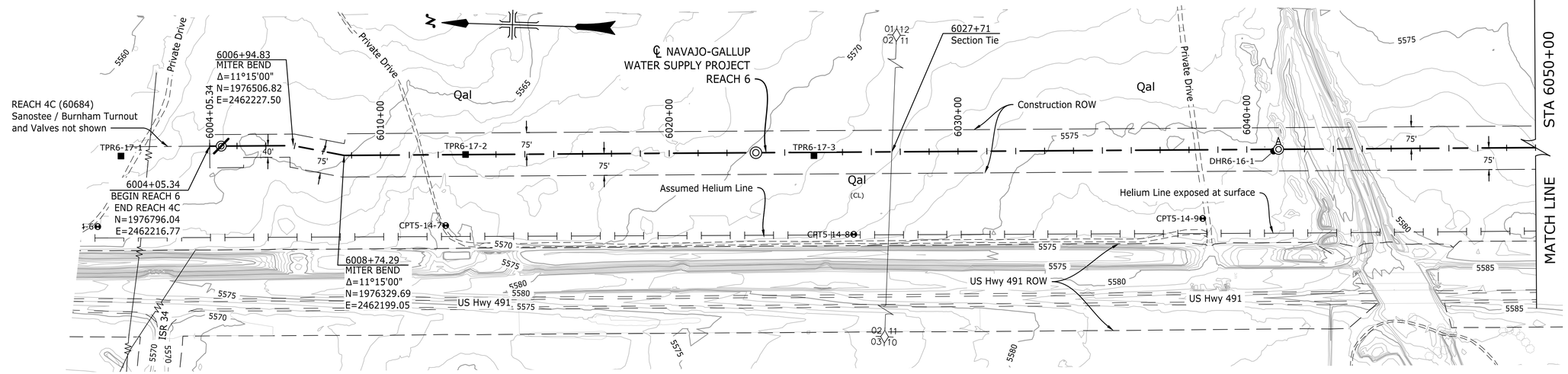
TECH. APPR. JUSTIN GILBERT
SUBMITTED PAUL GARDNER
APPROVED JUSTIN GILBERT Supervisory Geologist

PLAN AND PROFILE

STA. 11150+00
TO
STA. 11200+00

1695-D-60311

SHEET 3 OF 5



OPEN-CUT ROAD CROSSINGS	
Station	Surfacing
6011+39 to 6011+48	Unmaintained Road
6038+21 to 6038+37	Unmaintained Road

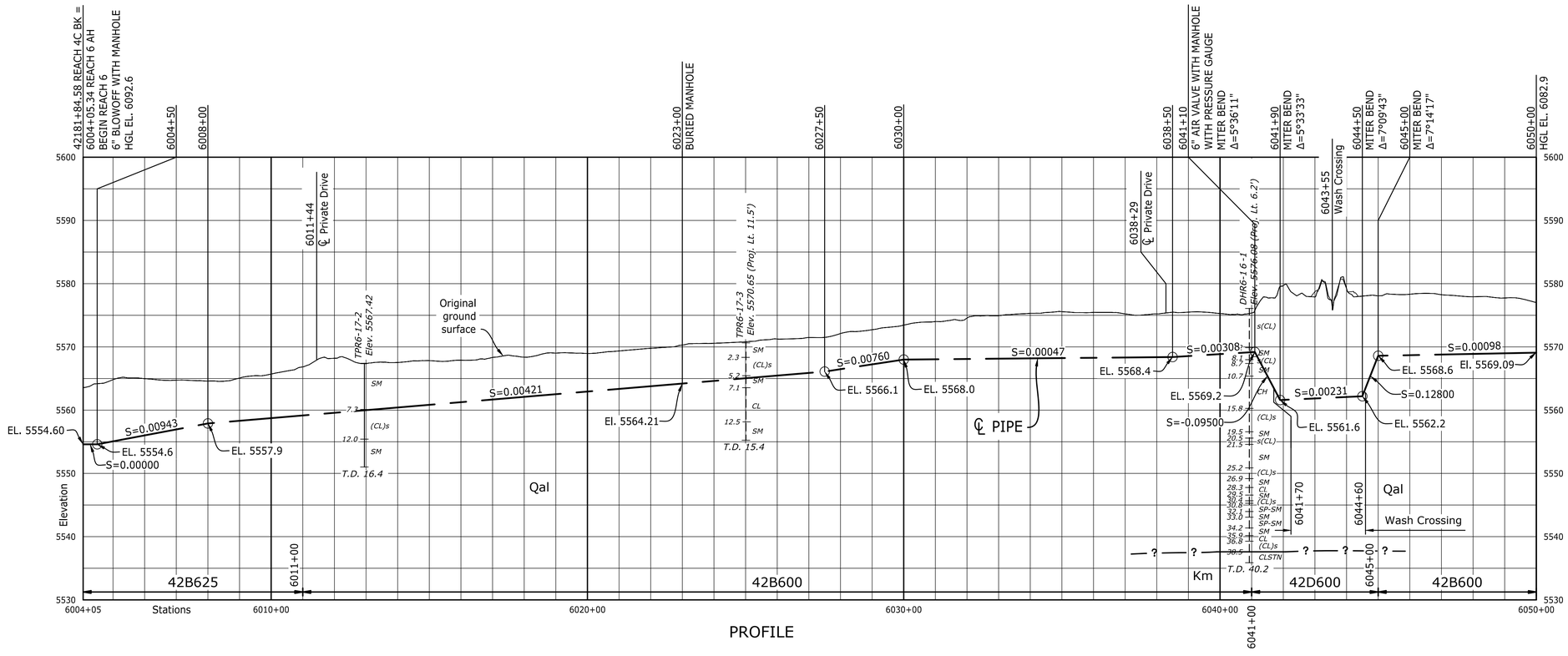
CONSTRUCTION ROW			
Station		Width (Ft.)	
Start	End	Left	Right
6004+05.34	6005+22.4	40	75
6005+22.4	6006+83.2	40	40
6006+83.2	6008+66.9	40	75
6008+66.9	6050+00.0	75	75

PLAN

0 200' 400'

NOTES

1. For General Notes, Symbols, Abbreviations, Pipe Classification, and Geology Note, see (60677).
2. For wash crossing at Sta. 6043+55, see (60738).



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BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

REACH 6 LINE PIPE

REV NO			
4	D	2020-05-15 Marner, Andrew (Andy) J. PE	REVISED DATA REFERENCES TO REVISE VERTICAL ALIGNMENT FOR WASH CROSSING CHANGES AND POWERLINE BORE AND JACK.
3	D	2019-09-19 Marner, Andrew (Andy) J. PE	REVISED DATA REFERENCES TO REVISE ALIGNMENT ON 1695-D-60695.
2	D	2019-09-06 NICHOLAS W. CLOUGH PE	REVISED DATA REFERENCES FOR POWERLINE CROSSING ON 1695-D-60686.
1	D	2019-05-17 NICHOLAS W. CLOUGH PE	MINOR REVISIONS.

M. LEAVITT
DESIGNED
M. LEAVITT
DRAWN
D. L. GESUNDHEIT
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer
DENVER, CO 2019-03-12

GEOLOGY

TECH. APPR. Chris Beyer
SUBMITTED Debby Thornburg
APPROVED Paul Gardner *Supervisory Geologist*

PLAN AND PROFILE

BEGINNING
TO
STA. 6050+00

1695-D-60685

SHEET 1OF11

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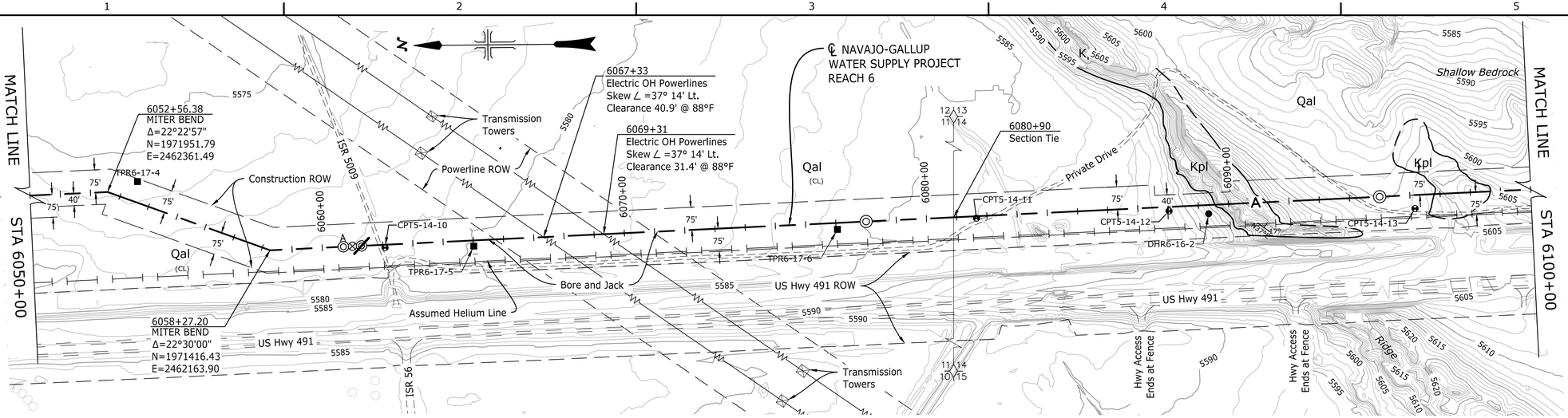


PLAN

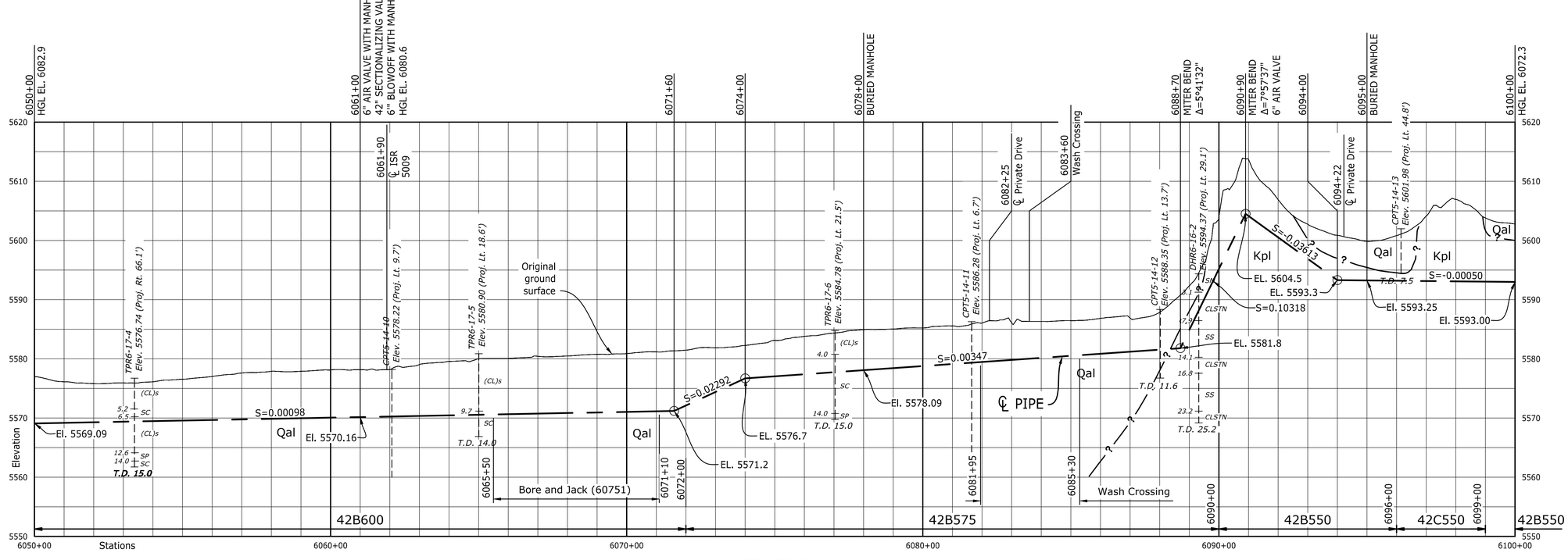
0 200' 400'

OPEN-CUT ROAD CROSSINGS	
Station	Surfacing
6061+85 to 6061+96	Maintained Road
6082+11 to 6082+39	Unmaintained Road
6094+15 to 6094+30	Unmaintained Road

CONSTRUCTION ROW			
Station		Width (Ft.)	
Start	End	Left	Right
6050+00.0	6050+91.7	75	75
6050+91.7	6052+64.3	75	40
6052+64.3	6087+57.7	75	75
6087+57.7	6090+07.4	40	75
6090+07.4	6100+00.0	75	75

NOTES

1. For General Notes, Symbols, Abbreviations, Pipe Classification, and Geology Note, see (60677).
2. For wash crossing, see (60736).
3. For bore and jack, see (60751).



PROFILE

REV NO	DATE	DESCRIPTION
3	2020-05-15 Mamer, Andrew (Andy) J P.E.	D BORE AND JACK ADDED AT POWERLINE CROSSING. REVISED PROFILE AT POWERLINE CROSSING.
2	2019-09-06 Clough, Nicholas W P.E.	D LOWERED PIPE FOR POWERLINE CROSSING.
1	2019-05-17 Clough, Nicholas W P.E.	D

M. LEAVITT — DESIGNED
M. LEAVITT — DRAWN
D. L. GESUNDHEIT — CHECKED
Clough, Nicholas W P.E. — TECH. APPR.
Bowles, Linda M P.E. — APPROVED (PEER REVIEWER · Peer Reviewer)

GEOLOGY
TECH. APPR. Chris Beyer
SUBMITTED Debby Thornburg
APPROVED Paul Gardner Supervisory Geologist

DENVER, CO 2019-03-11

PLAN AND PROFILE

STA. 6050+00
TO
STA. 6100+00

1695-D-60686

SHEET 2 OF 11

RECLAMATION
Managing Water in the West

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NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

REACH 6 LINE PIPE

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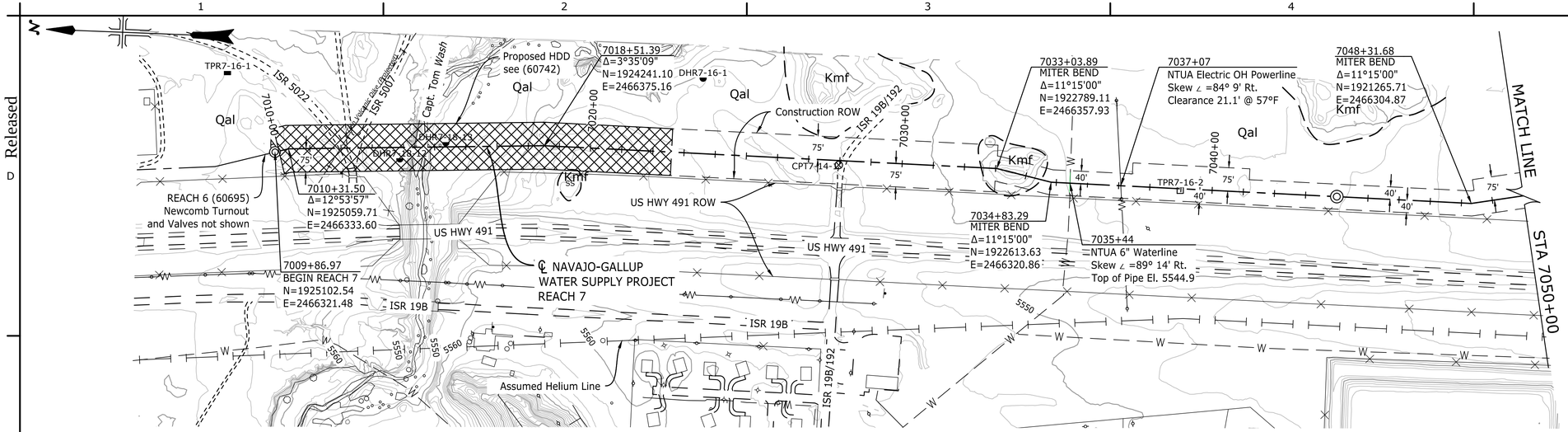


PLAN

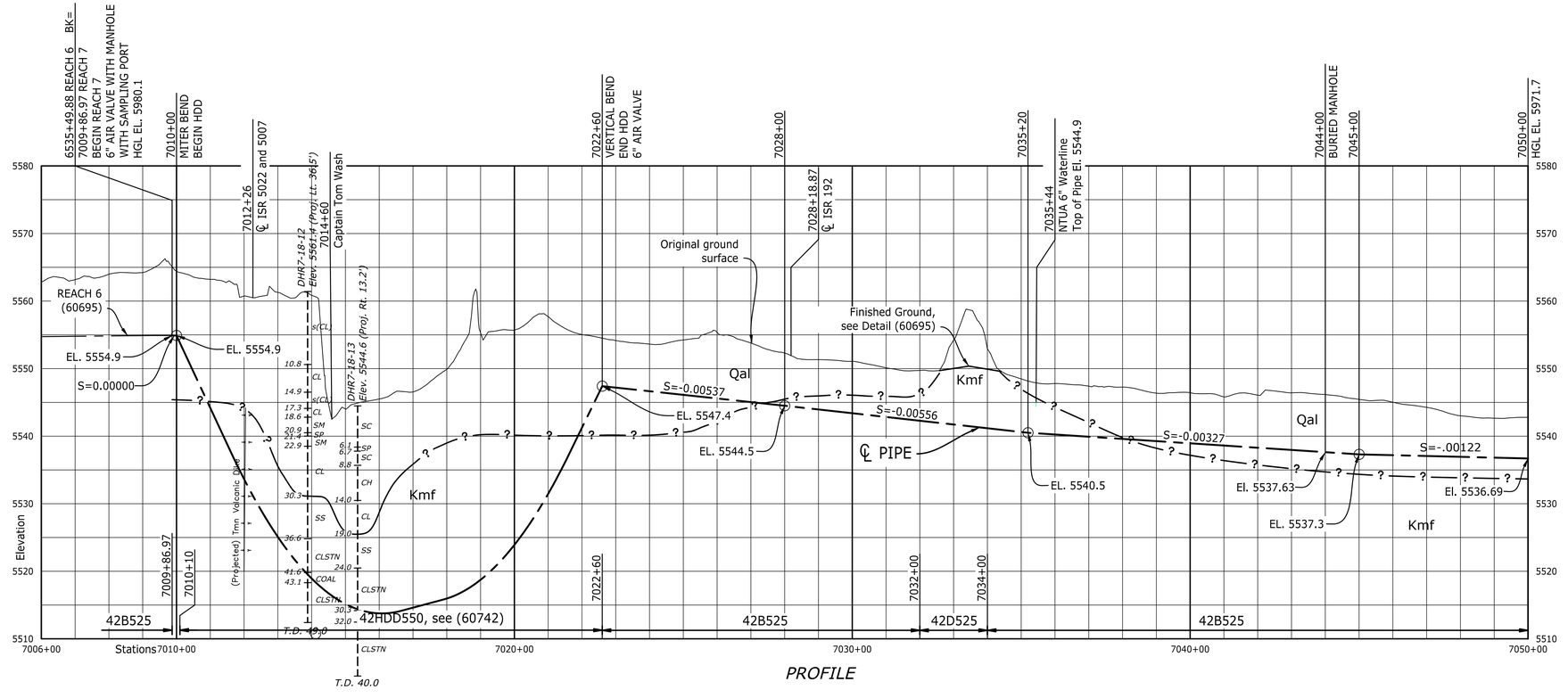
0 200' 400'

OPEN-CUT ROAD CROSSINGS	
Station	Surfacing
7027+98 TO 7028+40	Maintained Road

CONSTRUCTION ROW			
Station		Width (Ft.)	
Start	End	Left	Right
7009+87	7033+04	75	75
7033+04	7037+42	40	40
7037+42	7041+17	75	40
7041+17	7048+47	40	40
7048+47	7050+00	75	40

NOTES

1. For General Notes, Symbols, Abbreviations, Pipe Classification, and Geology Note, see (60677).
2. For Captain Tom HDD plan, see (60742)



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CAD FILENAME
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NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

REACH 7 LINE PIPE

REV NO	DATE	BY	DESCRIPTION
3	2020-09-02	D Nicholas W. Clough, P.E.	Revised HDD stations.
2	2019-09-19	D Nicholas W. Clough, P.E.	Updated DH-18 labels. Contains revised alignment for edraws.
1	2019-05-17	D Nicholas W. Clough, P.E.	Corrected slope callouts in profile.

M. Gemperline
DESIGNED
D. Kosakowski
DRAWN
Andrew Marner
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer

GEOLOGY

TECH. APPR. Chris Beyer
SUBMITTED Debbie Thornburg
APPROVED Paul Gardner Supervisory Geologist

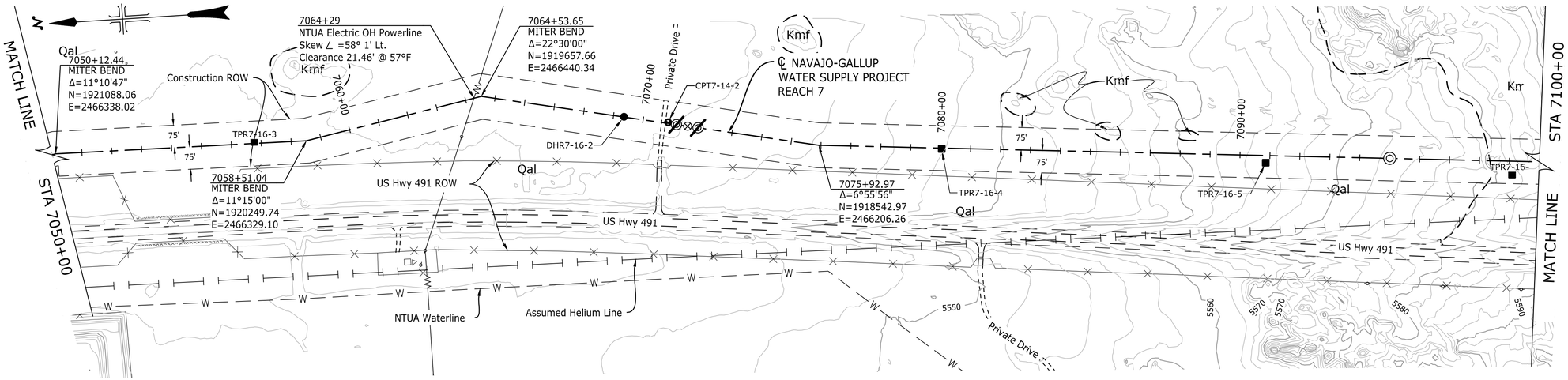
DENVER, CO 2019-03-11

PLAN AND PROFILE

BEGINNING
TO
STA. 7050+00

1695-D-60696

SHEET 1OF8



PLAN

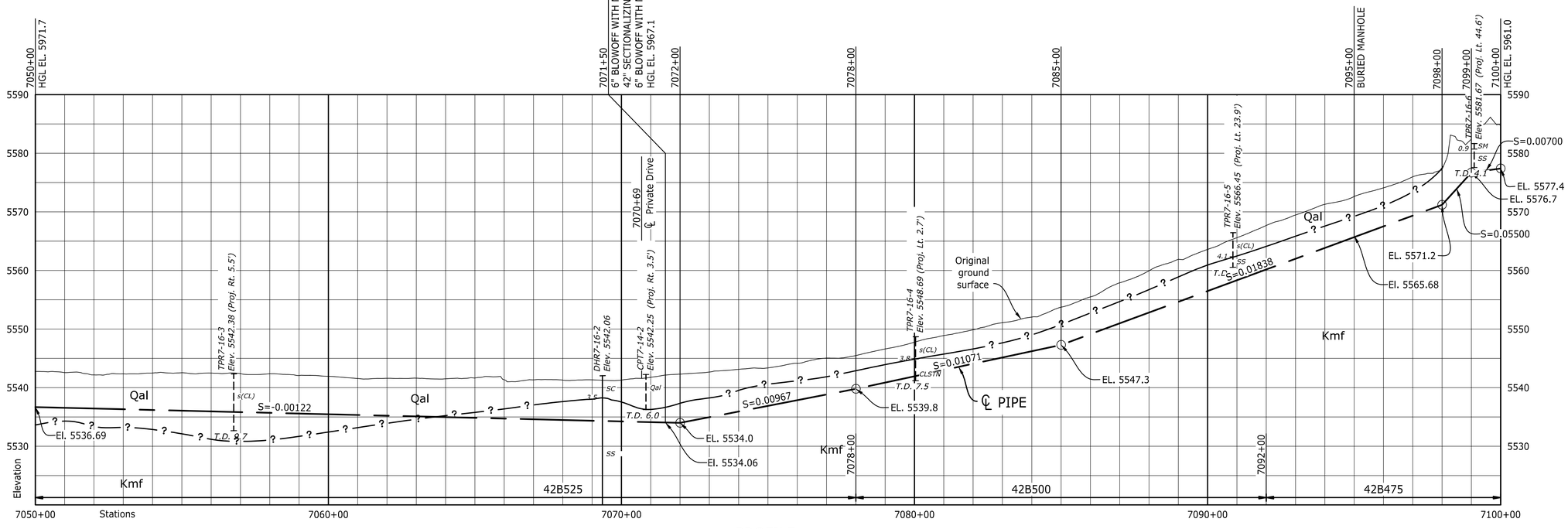
0 200' 400'

OPEN-CUT ROAD CROSSINGS	
Station	Surfacing
7070+52 TO 7070+86	Maintained Road

CONSTRUCTION ROW			
Station		Width (Ft.)	
Start	End	Left	Right
7050+05	7100+00	75	75

NOTE

1. For General Notes, Symbols, Abbreviations, Pipe Classification, and Geology Note, see (60677).



PROFILE

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2019-05-17
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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

REACH 7 LINE PIPE

REV NO	DATE	DESCRIPTION
1	2019-05-17	D Revise depths for minimum cover requirements. Clough, Nicholas W P.E.

M. Gemperline
DESIGNED
D. Kosakowski
DRAWN
Andrew Marner
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer

GEOLOGY
DENVER, CO *2019-03-11*

TECH. APPR. *Chris Beyer*
SUBMITTED *Debbie Thornburg*
APPROVED *Paul Gardner* *Supervisory Geologist*

PLAN AND PROFILE

STA. 7050+00
TO
STA. 7100+00

1695-D-60697

SHEET *2 OF 8*